UCLA Department of Ecology and Evolutionary Biology

Self-Review

2014-2024

Section A. Introduction

The Department of Ecology and Evolutionary Biology is excited to present its 2023-2024 self-review as part of an 8-year review process. The Department's last review was written in 2014. Since that time, we have made significant progress with faculty hires and diversifying our graduate student cohorts. For this review, we began our self-review process with faculty retreats in the Fall of 2022 and 2023 where we discussed the mission of the department, its current direction, major challenges from staff, faculty, and student perspectives, and our strengths and weaknesses. We followed up with discussion in faculty meetings and a series of "Blue Sky thinking sessions" where we encouraged our faculty to think broadly about the mission of the department over the next 10 years. We met with undergraduate student leadership and surveyed students from all three majors. We also met regularly with grad student leadership, held town halls for the graduate students, and surveyed current students and alums. We also held town halls and distributed surveys to the staff. As part of the review process the Vice Chairs of the Graduate (Morgan Tingley) and Undergraduate Programs (Paul Barber) and the Vice Chair of Community (Nathan Kraft) met frequently to discuss aspects of the review with support from current Chief Administrative Officer, Shannon McClenton, former CAO Kimberly Linsley, and Director of Academic Programs and Student Services Eileen Mansoorian. VC Tingley drafted the graduate program section, VC Barber drafted the undergraduate program section, and VC Kraft drafted the diversity section. CAO McClenton drafted the section on staff resources and support. Joselyn Castellanos (chairs assistant) and Sheena Sanchez (former chairs assistant) assisted with drafting sections of the general information as well as managing the process.
Committee members were charged with sharing drafts of the review with the faculty as they developed their individual reports. A partial draft report was discussed on Monday, November 4th 2024. Modifications were made and the report presented at the Monday, November 18th faculty meeting. The formal vote occurred on November 22nd-November 26th via email ballot. The final vote to approve the review was: 18 in favor 0 opposed 0 abstain of 21 faculty eligible to vote.

Section B. General Information

1. Roadmap to Self-Review

Roadmap Overview for 8-Year Review

This self-review highlights the Department of Ecology and Evolutionary Biology's (EEB) remarkable strengths, recent challenges, and opportunities for growth since 2014. EEB is home to an exceptional faculty, spanning early-career researchers achieving significant success and demonstrating immense potential, to mid-career and senior faculty with vibrant research programs that shape the fields of ecology, evolutionary biology, and behavior. These faculty are influential leaders in their disciplines, driving innovation and making critical contributions to science and society.

The department is navigating a period of transition, shaped by retirements, faculty losses, administrative turnover, and the challenges of reestablishing community after the pandemic. Despite these hurdles, EEB remains well-positioned to capitalize on the extraordinary creativity, talent, and research innovation of its faculty, graduate students, and undergraduate programs. With strategic support from the university, the department can unlock its full potential and further its mission of advancing knowledge in conservation, global change science, and quantitative biology, while maintaining core strengths in ecology, evolution, and behavior.

1. **Departmental Mission and Faculty Transition (Section B & Faculty Section of General Information)**
   EEB continues to lead in areas of traditional strength for the department, including conservation genomics, behavior, marine biology, tropical biology, community ecology, quantitative biology, and other key areas of ecological and evolutionary sciences. However, upcoming faculty retirements and current faculty losses highlight the urgent need for strategic hiring to sustain leadership in these fields. Building on the momentum of recent hires, the department sees an unprecedented opportunity to grow in both size and expertise, addressing key gaps in marine biology, conservation genomics, evolutionary medicine, and behavior while expanding its strengths in quantitative and predictive biology. Here we outline a strategic plan for addressing these challenges through targeted faculty recruitment. This includes a critical request for the university's support to not only replace retiring faculty but to expand the department's FTE to 35, a level necessary to meet the growing demands of its majors, enhance research capacity, and maintain EEB's leadership in ecological and evolutionary sciences.

2. **Administrative Turnover and Need for Strategic Planning (Section B, Faculty and Administrative Resources)**
   In recent years, administrative turnover, including multiple changes in department chairs and Chief Administrative Officers (CAOs), has posed challenges to continuity and long-term planning. With a renewed focus on stability and strategic vision, EEB can

address these gaps and strengthen its administrative foundation to better support its faculty, students, and staff.

3. **Staffing Needs and Department Culture (Section B, Administrative Resources)**
   Staffing challenges have been a significant issue for the Department of Ecology and Evolutionary Biology (EEB), stemming from two primary factors. EEB is a large and highly complex department, with faculty research programs that span international fieldwork, diverse funding streams (e.g., NSF, NIH, NGOs, state and local agencies), and training programs that connect to Historically Black Colleges and Universities (HBCUs). Faculty also engage in innovative classroom pedagogy and research. The breadth and variability of these activities far exceed those of a similarly-sized department with a narrower focus. Managing such complexity requires experienced, mid and senior level staff capable of navigating the department's diverse needs. With support from the university to address these staffing issues, EEB can unlock the full potential of its faculty, improve faculty and staff morale, and increase research productivity. This investment in staffing would not only alleviate current challenges but also lead to greater grant activity, improve recruitment and retention efforts and enhance support for both faculty and students.

4. **Undergraduate Program Dynamics (Section D)**
   The undergraduate program in EEB continues to experience significant growth, particularly in capstone majors like Ecology, Behavior, and Evolution (EBE) and Marine Biology. However, this growth coupled with a drop from a high of 31 EEB FTE during the review period to 25 current faculty has strained the department's ability to meet rising student demand. Limited course enrollment capacities and unit assignments for many courses often fail to account for the actual contact hours required for lectures, labs, and discussions, creating bottlenecks that impact both students and faculty. These challenges are further compounded by the poor faculty-to-student ratio in EEB, which is among the worst in the Life Sciences Division. In EEB, this disparity is even more acute, as the department remains below pre-pandemic faculty FTE levels despite significant growth in its undergraduate majors. Curricular adjustments and interdisciplinary Life Sciences courses have helped address evolving student needs, but they also underscore the urgent need for modernized teaching facilities and increased instructional support. The growing size of EEB's undergraduate program places additional stress on outdated infrastructure and further highlights the necessity for investment in teaching spaces that can accommodate high-demand courses effectively. To truly support its students and uphold its educational mission, EEB must not only restore pre-pandemic FTE levels but also secure additional resources to keep pace with student growth. This investment is vital to maintaining the quality of education that has defined EEB's reputation and to meeting the expectations of its increasingly diverse and ambitious student body.

5. **Graduate Program and Mentorship (Section E)**
   Graduate training remains central, especially with an increased emphasis on interdisciplinary and quantitative skills. Limited administrative capacity and staff turnover has strained graduate student support, leading to delays and reduced service quality.

Planned enhancements to graduate advising and postdoctoral mentoring are part of EEB's strategy to uphold its commitment to high-quality training and diversity in the field. Increased faculty FTEs would help expand the graduate curriculum to provide advanced classes in areas of faculty speciality, further enhancing the graduate training program.

6. **Diversity and Community Engagement (Section G)**
   EEB has advanced its diversity initiatives through efforts led by the Committee for Anti-Racism and Equity (CARE). Funding for these initiatives would further the department's goals for a cohesive, inclusive culture as EEB navigates this transition. Enhancing community-building and professional development is essential to support both students and faculty during this period.

7. **Comparative Review and Strategic Planning (Section I)**
   Comparing this cycle to previous reviews, EEB has experienced significant changes in leadership, infrastructure needs, and educational demands. Administrative turnover, coupled with increasing reliance on limited facilities, underscores the urgent need for strategic planning. This initiative will guide the department's approach to adapting its programs and aligning with UCLA's broader mission of teaching and research excellence.

8. **Resource and Facility Needs (Section J)**
   The ability to expand research and teaching is constrained by infrastructure limitations, particularly outdated lab spaces in the Life Sciences Building (LSB). Additional lab space and facility improvements are critical for faculty recruitment and for training in ecology and evolutionary biology. Enhancing facilities for marine research and dive training will also ensure EEB's competitive edge in field-specific research.

Strategic Requests The department's requests in Section K reflect these challenges. With targeted investments in replacement hires for retiring faculty, additional FTE to support the departments research and education mission, expanded and improved lab and teaching space, enhanced staffing to restore effective service, and support for strategic planning, the department is poised to emerge stronger from this transitional phase and strengthen its leadership position in research, training, and education in Ecology, Behavior, and Evolutionary Biology.

2. Program Overview

The Department of Ecology and Evolutionary Biology traces its lineage back to the original Departments of Zoology and Botany and the College of Agriculture, which were established when UCLA emerged as the southern branch of the University of California. The Department of Zoology was relatively stable through the 1940s and 1950s, but by the late 1950s, the urbanization of Los Angeles led to the reduction of the College of Agriculture, whose few remaining members joined their plant colleagues in a department curiously called the Department of Botany and Plant Sciences, soon to be renamed the Department of Botanical

Sciences. Discussions in the 1960s caused this relatively small department to merge with Zoology, forming the Department of Biology. During the 1970's, the rather large Department of Biology was divided into four sections, eventually leading to independent sections—one focusing at the molecular level and the other at the organismic level. By the 1980's, they evolved into two departments, one retaining the name of Biology to reflect its broad mission and the other one now known as the Department of Molecular, Cell, and Developmental Biology. To help clarify the professional areas covered, the Department of Biology was renamed the Department of Organismic Biology, Ecology, and Evolution in the late 1990s, and has now become the Department of Ecology and Evolutionary Biology. Since the last review in 2014 the department has retained this title and followed a strategic vision focused broadly on quantitative biology, conservation biology with a local and global focus, and behavior. In many respects EEB has achieved a great deal of success in moving in this direction (which arose following discussion and retreats following the last review cycle). However the events since the pandemic coupled with retirements, deaths, and faculty departures have created a strong sense that the department is once again at a transitional stage. This self review will highlight the many recent challenges that the department has weathered as well as the opportunities for growth that have been identified in a wide ranging series of discussion that include faculty meetings, retreat sessions, cohort-focused "blue sky" sessions, surveys, and town halls.

3. Research Mission of the Department

The mission of the Department of Ecology and Evolutionary Biology is to achieve excellence in research and teaching across the fundamental domains of ecology, evolution, and behavior. We are committed to advancing knowledge and developing solutions that address the critical challenges of global change in the Anthropocene era. Our historic strengths have been in conservation biology and genomics, plant physiology, population and community ecology, behavioral ecology, evolutionary biology, and marine and tropical biology. The department's strategic vision seeks to maintain these strengths while building expertise in emerging fields of data science and predictive biology. With regards to taxonomic groups, EEB currently has a strong cadre of faculty working on plants and vertebrates (especially birds). However one of the most essential aspects of the department is its strong conceptual and quantitative focus that is highly interdisciplinary. Faculty have world-leading expertise in ecological and evolutionary modeling, landscape genomics, and analysis of both large scale and small-batch, richly curated data sets. EEB is also remarkable for the local and global reach of its field research and for disseminating this work to society. Many faculty have links to local, state, and federal agencies, NGOs, and governmental agencies in other nations.

Perhaps the most significant challenge for the department going forward is how to navigate the recent and impending loss of faculty that have been leaders in crafting many of the strengths noted above. The death of Professor Robert Wayne in December 2022 marked the tragic loss of a leader and co-founder of the fields of conservation biology and conservation genomics. Four other faculty who represent the senior core of conservation studies in the department have either recently retired (Professor Tom Smith) or are very likely to retire within the timeframe of the next review. These faculty all lead or have led substantial research units with important ties to EEB

including the Congo Basin Institute, the Center for Tropical Studies, La Kretz Center for California Conservation Science, and the Mathias Botanical Garden. And while leadership transition plans have been developed for CBI and CTR (with Professors Ordway and Zapata stepping into directorship roles) continued support for the changing direction of these units is needed. EEBs strength in marine biology is even more at risk. Since the last review three marine faculty have retired, and our most recent hire was recruited away leaving just three FTEs with a marine focus, all of whom are senior faculty. The recent hire of Professor Monica Medina will help to address this shortage of marine faculty but we anticipate the need for additional hires to continue to serve the major and to maintain our strength in this area.

4. Teaching Mission of the Department

The Department of Ecology and Evolutionary Biology has a large undergraduate program consisting of approximately 1800 students in across three majors: Biology (1600 students) Ecology, Behavior, and Evolution (100 students), and Marine Biology (70 students) (3 majors, 2 minors, ~1800 students). EEB offers minors in Conservation Biology and Evolutionary Medicine. The department strongly prioritizes undergraduate research through independent study, a departmental research symposium in the spring, and intensive field and marine biology quarters (Appendix D6 includes a list of undergraduate publications during the review period). Graduates from our program have diverse career trajectories that include (but are certainly not limited to) graduate school in the biological and biomedical sciences, professional schools in medicine, health, law, as well as advanced training in teaching. Those that do not immediately seek advanced training often get jobs working for state and federal agencies as well as jobs working in private industry.

The graduate program historically focused on training PhDs but since the last review the Masters Program has grown to become a significant part of our training. There are currently 71 PhD students in the program, 7 EEB MS students and 4 Masters students in the Ecology and Evolution of Medicine Program. By design, we seek to enroll a variety of students with different career trajectories. Some students follow academic paths that begin with additional postdoctoral training that then lead to jobs in a variety of colleges and universities. Others seek work in museums or zoos, and others work in state and federal laboratories, for NGOs and for private research organizations. We are proud to report that almost all of our students obtain work in desired fields upon graduation. The department has made diversifying the graduate program a significant priority–since the last review EEB's revisions to the graduate admission program include the elimination of GRE scores, a change in application prompts to allow for the assessment of accomplishment given the challenges and resources available to the applicant, and a sweeping change to SOPs of ranking and selecting students that has been vetted in faculty discussion and, most recently, by the CARE committee. However, the graduate program is also a source of significant challenge for EEB. The core graduate curriculum has undergone significant redesign following graduate student outcry over course content in 2020 and 2021. The graduate student strike in 2023 elevated tensions and introduced new financial stresses on the department. The violence on campus in May 2024 following protests and counter protests over

the war in Gaza has left some of our graduates feeling unsafe on campus and also created uncertainty around the possibility of another UAW strike.

5. Faculty

The Department has 29 faculty members, 25 of whom have FTE and voting rights. *Over the review period EEB grew to 31 FTE and thus EEBs current size marks a decrease in the size of the department driven by faculty passings, retirements, and departures to other universities.* In addition, we have 8 adjunct faculty members. Of the 25 with FTE, 3 are Assistant Professors, 4 are Associate Professors, 18 are Full Professors, and 2 are Distinguished Professors. Of the 25 with FTE, 10 are women, and 2 come from underrepresented minority groups. Faculty Research Interests are listed in Appendix B1 and Faculty CVs are available asAppendix B2. Faculty CVs.pdf Further information on trends in faculty diversity is given in section I5: Improving Faculty Diversity.

Faculty Diversity.

	ETHNICITY	GENDER	2015-16	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24
PROFESSOR OF TEACHING - SOE (210)	White	Female									1	1
PROFESSORIAL-NON-TENURE (011)	Asian	Other/Unknown						1	1	1	1	1
	Hispanic/Latino	Male			1	1	1	1	1	1	1	
	Multiple	Female	1	1	1	1	1					
		Male					1	1	1			
	White	Male	1	1	1	1			1	1	1	
		Female			1	1				1	1	1
PROFESSORIAL-RECALL (012)	White	Male	1	1	2	1	1	1	1	1	1	
PROFESSORIAL-TENURE (010)	Asian	Female	1	1	1	1	1	1	1	1		1
	Hispanic/Latino	Male	2	2	2	2	2	2	2	2	2	3
	Multiple	Female						1	1	1	1	1
	Other/Unknown	Male	1	1	1	1	1	1	1	1	1	1
	White	Male	9	10	10	9	10	9	9	9	8	7
		Other/Unknown	5	6	6	6	3	4	5	5	4	4
		Female	4	4	4	5	3	3	3	3	3	3
TOTAL			25	27	30	29	24	25	27	27	25	23

Faculty Promotion and Mentorship

EEB is strongly supportive of its junior faculty and all assistant professors that have come up for tenure have been successfully promoted during the review period. Junior faculty are assigned mentors and meet with them at least twice annually and also receive advice and support from their other colleagues. Junior faculty are given teaching and administrative relief and are encouraged to apply for university grants to provide additional teaching relief or research support. Junior faculty are also encouraged to apply for (and some have received) both in-house named professorships (De Logi Chair in Biological Sciences), and extramural fellowships and grants (e.g., Searle, Sloan, etc.) that help them launch their careers. At the Divisional level, there

are diversity committees and junior faculty support groups in which a number of faculty have elected to participate.

Faculty excellence EEB faculty have achieved widespread recognition for the quality and impact of their research. Faculty distinctions during this review period include: election to the American Academy of Arts and Sciences (**Shaffer, Smith, Sork**), election as a Fellow of the California Academy of Sciences (**Sears, Sork**), recognition as a McDonnell Complex Systems Scholar (**Savage**) and as Highly Cited Researchers (**Kraft, Sack**), Quest Award for Outstanding Contribution in Animal Behavior (**Pinter-Wollman**), appointment to the Lancet/PPATS Commission on the Prevention of Viral Spillover (**Lloyd-Smith**), External advisor to World Health Organization on 'Prioritization of Pathogens' (**Lloyd-Smith**), NSF CAREER Award (**Garud, Barber**), Paul Allen Distinguished Investigator designation (**Garud**), Scialog! Microbiome Neurobiology and Disease Fellowship (**Garud**), Outstanding Ecological Theory Paper Award (**Song**), Hellman Award (**Zapata, Ordway**), Naturalist of the Year (**Barber**), External Expert Advisory Board (EEAB) of the Biodiversity Genomics Europe (BGE) Project (**Shaffer**), UCLA Public Impact Research Award (**Smith**), W.K. Kellogg Foundation Community Engagement Scholarship Award (**Smith**), American Ornithological Society Elliott Coues Award (**Smith**), Faculty Gold Shield Award (**Barber**, **Blumstein**), Quest Award from the Animal Behavior Society (**Blumstein**), and the Richard Buchholz Conservation Behavior Award from the Animal Behavior Society (**Blumstein**).

Leadership in Professional Societies Our faculty play key leadership roles in professional societies. **Morgan Tingley** has recently been named President-Elect of the American Ornithological Society, **Karen Sears** served as President of the PanAmerican Society for Evolutionary Developmental Biology, and **Brad Shaffer** has served on the Board of Governors and as President for the American Society of Ichthyologists and Herpetologists. **Dan Blumstein** is the President of the Board of the Rocky Mountain Biological Laboratories. **Greg Grether** was nominated for President of the American Society of Naturalists. Faculty also serve in editorial leadership roles for prestigious journals. **Editors** include: **Greg Grether** (*Ecology Letters*), **Karen Sears** (*Biological Journal of the Linnean Society*), **Lawren Sack** (*Journal of Plant Hydraulics*, *Frontiers in Functional Plant Ecology*, and *PrometheusWiki*), **Victoria L. Sork** (*Molecular Ecology*), **Nandita Garud** (*PLOS Genetics* and *mSystems*), **Dan Blumstein** (founding Chief Editor of *Frontiers in Conservation Science*), and **Kirk Lohmueller** (*Journal of Heredity*, special issue honoring Bob Wayne). **Associate Editors** include: **Karen Sears** (*EvoDevo*, *Proceedings of the Royal Society B*, and *Development, Genes, and Evolution*), **Greg Grether** (*The American Naturalist*), **Nathan Kraft** (*Ecology* and *Ecological Monographs*), **Michael Alfaro** (*Evolution* and *Systematic Biology*), **Dan Blumstein** (*Quarterly Review of Biology* and *Evolution, Medicine and Public Health*), and **Kirk Lohmueller** (*PLOS Genetics*, *PLOS Computational Biology*, *Genome Biology and Evolution*, and *Evolution*), **Felipe Zapata** (*Quarterly Review of Biology)*.

Leadership in Diversity and Inclusive Excellence Faculty in our department have played a significant role in promoting Diversity, Equity and Inclusion on campus and in professional societies. Since last review five faculty (**Alfaro, Barber, Fong, Kennison, Sork**) have received the DEI Award from the UCLA senate in recognition for outstanding commitment and leadership

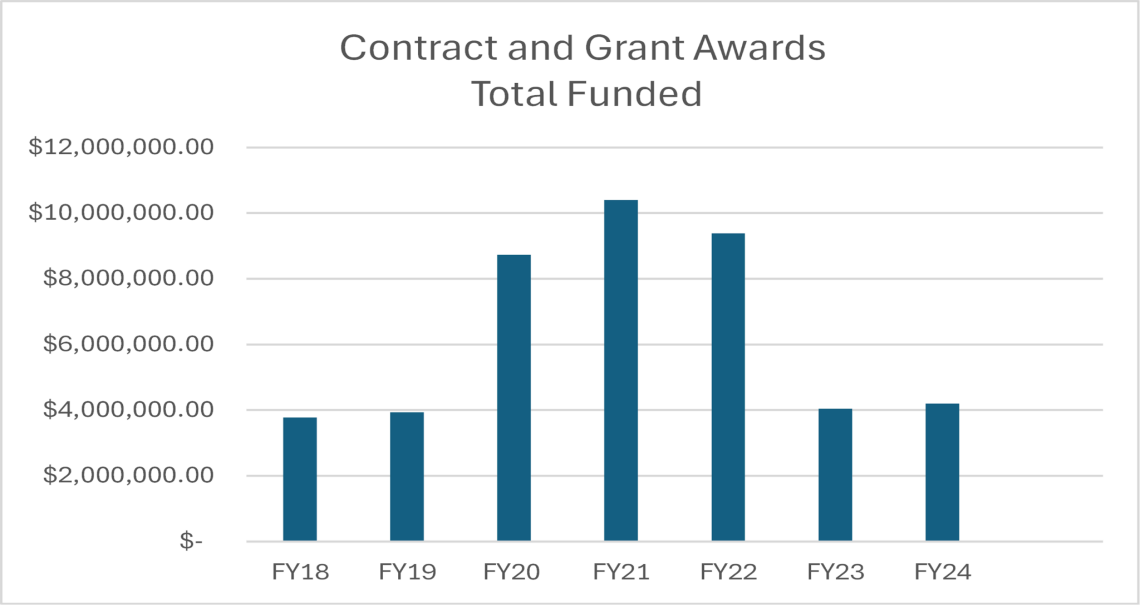
to campus DEI. This unprecedented concentration of recipients in our department speaks to the long history of effective and impactful change EEB faculty members have helped to bring to UCLA. In addition, EEB faculty chair the SSE Committee on Diversity (**Yeh**), have received the DEI Award to IRADCA (**Kennison**), the SACNAS Distinguished Student Mentor Award (**Barber**) and the UCLA Life Sciences Award for Inclusive Excellence (**Kraft**).

Teaching Excellence The Department of Ecology and Evolutionary Biology (EEB) at UCLA is distinguished by its faculty's exceptional commitment to teaching and mentorship. During the review period, several faculty members have been honored for their contributions underscoring the department's dedication to fostering a culture of teaching excellence and mentorship, supporting the academic and professional success of UCLA students. **Paul Barber** received the 2019 UCLA Gold Shield Faculty Prize, the university's highest faculty honor, recognizing his distinguished record in undergraduate teaching, research, and university service. **Greg Grether** and **Jamie Lloyd-Smith** were awarded the UCLA Distinguished Teaching Award for outstanding contributions to undergraduate education and mentorship. **Paul Barber** received the UCLA Undergraduate Research Center Faculty Mentor Award. **Noa Pinter-Wollman, Morgan Tingley,** and **Rachel Kennison** each earned the Life Sciences Excellence in Teaching Award acknowledging innovative and inclusive teaching practices. **Peter Nonacs** received the **UCLA** Undergraduate Mentorship Award for his dedication to fostering student research and academic growth. **Brad Shaffer** was recognized with the Meritorious Teaching Award in Herpetology, celebrating his expertise and influence in the classroom. **Kirk Lohmueller** received the Department of Ecology & Evolutionary Biology Faculty Teaching Award.

Campus Leadership Faculty in the Department of Ecology and Evolutionary Biology (EEB) have held significant leadership roles at UCLA, reflecting their dedication to advancing the university's academic mission. A selection of these leadership positions includes:

- Dean of Life Sciences (**Sork**)
- EEB Chair: **Alfaro**, **Sears**, **Nonacs, Grether, Blumstein**
- Vice Chair for Graduate Studies: **Tingley**, **Kraft**, **Nonacs**
- Vice Chair for Undergraduate Affairs: **Barber**, **Sears**, **Alfaro, Sack**
- Vice Chair for Academic Personnel: **Yeh**, **Lohmueller, Grether**
- Vice Chair for Community: **Kraft** (2022– present)
- Life Sciences Equity Advisor (**Barber**)
- Director, UCLA La Kretz Center for California Conservation Science: **Shaffer**
- Faculty Director, Center for Education Innovation and Learning in the Sciences (CEILS): **Kennison**
- Institutional Leader UCLA's Center for the Integration of Research, Teaching, and Learning: **Kennison**
- Chair of Computational and Systems Biology: **Savage**
- Vice Chair of graduate admissions in Biomathematics/Computational Medicine: **Savage**

Figure B1: Contract and Grant Post Award Data



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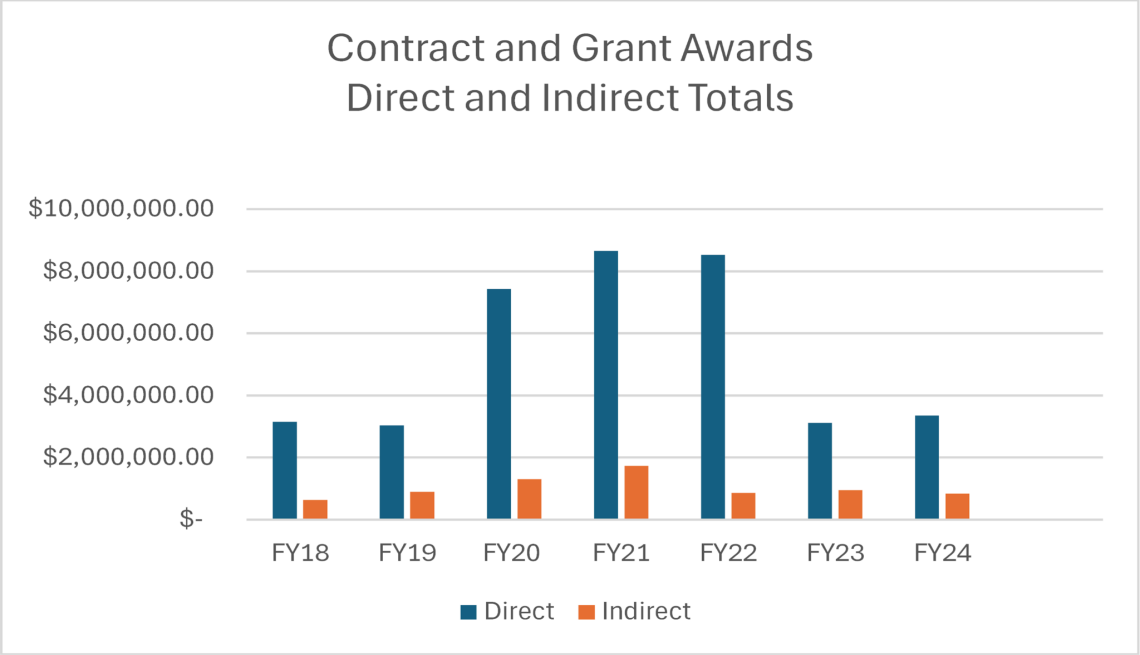


Figure B2: Direct and Indirect Totals

Faculty Teaching

We are delighted to report that we have award-winning faculty who are highly motivated to teach at the undergraduate and graduate level. The normal workload for a ladder-rank faculty is one ‘large’ undergraduate course, one ‘smaller’ undergraduate course, and one graduate course annually. This is a guideline and we are somewhat flexible in how this is administered. For instance, team-teaching half a large (200+ person undergraduate course along with an associated lab) may count for a full course, or teaching the large course plus the lab may count for a full load. Teaching in the labor-intensive field or marine biology quarters may also count for a full load. Teaching relief is associated with major administrative loads including leadership positions within EEB. Many faculty opt to over-teach, thus masking our need for additional instructional staff. Since the last review EEB has hired Rachel Kennison as a Teaching Professor. Teaching Professors are ladder rank faculty who specialize in pedagogy. Professor Kennison has had a transformative impact on developing teaching best practices and inclusive pedagogy through her role as the Director of the Center for Education Innovation and Learning (CEILS) and is currently advertising a search for a second Teaching Professor that will focus on teaching EEB content in the Life Sciences Core curriculum as well as EEB specific courses.

Use of non-ladder faculty We currently employ four lecturers who teach a wide range of topics in the major.

LERYN GORLITSKY,: has been integral in co-teaching EEB 100 & 100L; has co-taught multiple field trip courses (EEB 136 & 154); had consistently taught two courses (EEB 111 & 151A) that meet diverse needs for our undergraduates: one is a stand-alone lab course and the other a lecture.

TONYA KANE, : has taught numerous sections of LIFESCI 7B each year, has consistently taught sections of EEB 109 & 101 each year, has co-led a Local FMBQ in Bodega Bay every other year; has stepped in to co-lead an international MBQ.

ALISON LIPMAN, : has been integral in co-teaching EEB 100 & 100L; has taught EEB 116 (Conservation Biology) consistently since we haven’t had a ladder faculty member available to teach; has co-taught multiple field trip courses (EEB 136 & 154)

JAMES DINES : has consistently taught numerous vertebrate lab courses (EEB 110, 115, & 170), two of which in lieu of faculty retiring (110 & 170) and has been the sole instructor for the courses. Additionally, he has developed a marine biology course for our Marine Biology majors (EEB 140).

Academic staffing priorities for the future

Viewed solely through the lens of graduation rates and student time to degree, our department is largely meeting the needs of our undergraduate teaching mission (see section D. Undergraduate Programs). However, this success is only being accomplished by employing a substantial roster of lecturers (three continuing and one pre-six), and by focusing on a smaller group of core classes. This combination has led to both frustration and dissatisfaction among lecturers who desire to teach a greater diversity of classes, as well as frustration among students, as many

classes in our department have not been offered in many years or are offered so infrequently that there are substantial bottlenecks for enrollment when they are offered. It has also led to the stagnation in growth of our capstone majors, as there are insufficient research faculty to teach these offerings. These problems are exacerbated by the large number of EEB faculty who are engaged in substantial departmental, divisional and UCLA service that come with teaching release (at least 9 faculty hold leadership positions with teaching release). Within this landscape, UCLA administration is asking for more teaching by EEB ladder-rank faculty, particularly within the Life Sciences core.

Maintaining success in the EEB education mission while fulfilling the teaching expectations of UCLA administration requires additional faculty. In particular, specialized organismal biology courses (mammalogy, herpetology, ornithology, ichthyology, etc) require faculty with specialized expertise. Also central to strengthening these and other course offerings, is increasing the number of Lecturer with Potential for Security of Employment (LPSOE) positions in the department. For instance, adding an LPSOE to develop and teach core courses such as EEB 109, 116, and 120 would not only provide continuity and oversight for these key classes but also create opportunities for lecturers to rotate teaching responsibilities as needed, and allow ladder rank faculty to offer more specialized courses in our majors. Similarly, a dedicated LPSOE focused solely on developing and teaching stand-alone lab courses (e.g., EEB 110, 111, 115, 170) would ensure consistent curriculum development and support lecturers in updating content to maintain pedagogical excellence.

It is also important to address recent constraints imposed by the Division of Graduate Education (DGE) on hiring teaching staff for undergraduate courses. Until Fall 2023, our department was able to hire Non-student Tutors (NSTs) in Academic Student Employee (ASE) positions for courses requiring specific teaching expertise—for example, the Plant Physiology course (EEB 111), which enrolls the largest number of lab students in the department and necessitates specialized teaching assistants with expertise in plant lab techniques. These NSTs, often alumni of our undergraduate or Master's programs, had prior experience either taking or TA'ing the course or its equivalent. Their involvement allowed us to hire the most qualified individuals for each course without compromising teaching quality or pedagogy. Unfortunately, the DGE and UCLA's Labor Relations office have since prohibited hiring NSTs in ASE positions, citing a violation of the ASE collective bargaining agreement. This decision has introduced significant challenges for our department, including: (1) the need to hire and extensively train less experienced graduate students; (2) the cancellation of some or all sections of certain courses due to a lack of qualified teaching staff; and (3) concerns from instructors that the pedagogy of the affected courses has been compromised, resulting in a diminished ability to maintain teaching excellence.

6. Administrative Resources

The department's administrative team consists of the Chief Administrative Officer, Director, Academic Programs and Student Services, Manager, Graduate Program and Student Services,

(2) Undergraduate Student Services Advisors, (1) Graduate Program Coordinator, Director, Financial Services, (1) Purchasing Coordinator, (3) Fund Managers, (1) Academic Personnel Analyst, (1) HR Generalist and the (1) Chair's Assistant. The department's administrative functional areas include Finance, Operations, Research Administration, Procurement, Human Resources, Academic Personnel, Academic Program Coordination, Under-graduate & Graduate Student Services, and Facilities Management (see sections Building Operations and Physical Plant for details). The CAO oversees these functional areas both directly and indirectly.

Administration/Operations

The CAO is responsible for strategic planning, development, implementation and management of administrative structures and procedures in support of the research and instructional activities of the department. The CAO oversees the financial well being of the department by managing operational budgets, gift and endowments, as well as indirect oversight of the contract and grant administration.

Office Administration

The Chair's Assistant oversees numerous items that can be categorized as General Office Administration, including scheduling, calendaring; managing EEB Seminar Series and department events; creating, developing and maintaining the departments external website and internal intranet; handling the initial phase of academic recruitments (job posting and administration of applications); managing the department committee assignments.

Financial Services/Research Administration

The finance team oversees financial services, research administration, fund management, and procurement services. Financial services includes, but is not limited to, management of Dean and Chancellors commitments to faculty recruitment and retention, gift fund management and state funds. Research administration includes pre and post award support. The fund management team is trained in full cycle fund management, this includes budget preparation, proposal submission, post award monitoring, monthly reconciliation, and award closeouts. Fund management also includes oversight of expenses related to staff and graduate student and postdoctoral expenses related to the PI funds.

Procurement

Procurement services include purchase services for administration, student and laboratory services, processing reimbursements and travel services.

Human Resources/Payroll

The current structure consists of two career staff, a HR Generalists responsible for personnel matters and payroll transactions related to non-academic staff and ASE populations and an Academic Personnel HR Generalist responsible for all actions, merits, promotions for ladder and non-ladder faculty.

Academic Programs & Student Affairs

Academic Program Coordination, Undergraduate Student Services, Graduate Student Services
The student affairs and academic program areas are managed by the Director of Academic Programming and Student Services. The Director manages a team of one Manager of Graduate Programs and Student Services, two Undergraduate Student Services Advisors, and one Graduate Program Officer. The members of this team work both independently and at times work together to meet the department's instructional, academic program and student services needs. Working closely with the CAO, the Director is responsible for creation and implementation of the department's annual instructional plan; creation and implementation of the department's course schedule; faculty course assignments and teaching assistant course assignments. The team oversees graduate student recruitment; undergraduate student advising; graduate student advising; academic programming creation development and implementation.

IT services

Computing support is provided by Life Sciences Computing. This is a shared resource provided to the department by the College of Life Sciences. In addition to desktop support and networking, services include application and web development services and the network/systems services.

7. Building and Operations

A. Building and Operations

EEB's space is managed by the Life Sciences Building and Operations team. This is a shared resource with other LS departments. The team is managed by Joel Latimer, with assistance from Humberto Barba and Guillermo Castillo. This includes approximately 221,000 square feet of building space including faculty research laboratories, academic services, electrical shop, and the research facilities, including the Plant Growth Center, the Drosophila Media Facility, the Mildred E. Mathias Botanical Garden, and Stunt Ranch located in the Santa Monica Mountains Reserve. Mr. Latimer is also responsible for managing renovations of classrooms, faculty laboratories, and faculty and staff offices, and provides coordination for the major capital projects that affect the Life Sciences departments.

Faculty of the Department of Ecology and Evolutionary Biology are currently located across three buildings: Life Sciences, La Kretz Botany Building, and Terasaki Life Sciences. Space assigned to a faculty member includes a faculty office, lab office, and research lab space. Graduate and undergraduate students advised by a faculty member must be accommodated within the assigned space. While La Kretz and Terasaki are newer and renovated spaces, the Life Sciences building which was constructed in 1954 is in need of major renovations.

Related Units

The UCLA Mathias Botanical Garden and Herbarium

The Mission of the Garden is to promote botanical knowledge and appreciation of nature to the UCLA community and the public at large. As a living museum, Its diverse plant collections are designed to assist the teaching and research mission at UCLA while creating a tranquil environment within the urban surroundings. The Garden acts as a community gathering place for a wide variety of groups and through public events promotes plant conservation, botanical research, and horticulture.

The Garden and Herbarium were established when the University of California's southern campus was moved to its current Westwood location. Due to the land-grant mission of the University of California, a 30-acre experimental station was maintained until the land-grant function of the UCLA campus was transferred to UC Riverside. At that time, the Garden retained its living collections but the focus shifted to teaching and research collections and the species composition changed. A major tree planting effort began almost immediately, with a special focus on subtropical tree species from Australia, Southeast Asia, southern Africa, and Central and South America, which thrived in the mild Southern California climate. By the late 1940s, more than 1500 different species of trees were being cultivated, a number which has grown to ~4000 trees, shrubs, and herbaceous plants. During that time, the herbarium collections also grew to more than 150,000 vascular plant specimens, largely due to the expertise of several prominent plant systematists and plant evolutionary biologists. The recently renovated UCLA Herbarium is located in a building adjacent to the Garden available to researchers, students, and the public. In 2017, the Garden opened an entrance building, the award-winning La Kretz Garden Pavilion, which is used for university classes, public programs, K-12 school visits, and privately-rented events. The Garden has just completed a major water conservation project by implementing a water-efficient irrigation system with an integrated beautiful stream and capture pond.

The Garden, which serves as an educational and research facility for the entire campus community, has a special relationship with the Department of Ecology and Evolutionary Biology (EEB) and its predecessor department, Biology. The Director of the Botanical Garden has almost always been a professor of Biology or EEB and the faculty in EEB serve its advisory boards. In fact, the Garden is named after Dr. Mildred Mathias, who was a professor of Biology and director from 1956-1974, whose vision was to promote the Garden as a highly diverse teaching garden for students and the public. In line with that tradition, many of the faculty use the Garden and Herbarium for teaching classes such as introductory biology, introduction to Ecology and Behavior, Plant Physiology, and Plant Evolutionary Biology. Many undergrads associated with EEB majors use the Garden and Herbarium for class projects and independent research projects. Moreover, the department uses the Garden for formal and informal gatherings that promote a positive culture for the department

LaKretz Center for California Conservation Science - Stunt Ranch Reserve and La Kretz Center Field Station

EEB is an equal partner with the Institute of Environment and Sustainability in the center. The mission of the UCLA/La Kretz Center for California Conservation Science is to conserve California's biodiversity and ecosystems through research, education and public programs.

La Kretz Center Field Station

La Kretz Center works with a broad set of partners to fund scientific research and educational activities that inform and enable conservation management through university-agency cooperation to meet the conservation challenges facing 21st century California. The goal is to leverage UCLA's world-class environmental scholars to achieve applied conservation solutions based on cutting edge research.

The La Kretz Center is affiliated with a diverse network of UCLA faculty, postdocs and students, and we work closely with our partners, including the U.S. National Park Service, the Museum of Natural History of Los Angeles County, The Nature Conservancy, the U.S. Geological Survey, the U.S. Fish and Wildlife Service, the California Department of Fish and Wildlife, the U.S. Bureau of Land Management, and the Natural Communities Coalition. Our collective goal is to protect and restore California's biodiversity, in ecosystems ranging from urban LA landscapes to pristine national parks.

Stunt Ranch Reserve

Research
Stunt Ranch Reserve supports a variety of short-term and long-term research projects managed by university-level students and faculty researchers. The reserve welcomes researchers from universities across California and beyond to utilize the 310-acres of biologically diverse terrain and state-of-the-art field station to study biology, botany, anthropology, archaeology, geology, earth sciences, environmental engineering, physical geography, public health, and resource management.

The reserve also works in tandem with UCLA's La Kretz Center for California Conservation to create and fund research opportunities for university students, leading to dozens of published papers annually.

Education
Stunt Ranch Reserve is an exciting setting for students to learn about the history of the Santa Monica Mountains, local plant and animal ecology, and ongoing conservation research conducted at the reserve. In particular, the reserve is an ideal space for students to gain hands-on experience in data collection, which can be applied to their future research endeavors. In addition to supporting university-level courses, the reserve partners with local environmental organizations to lead field trips for K-12 students. Since its establishment, thousands of inner-city students have explored the reserve's trails, gaining valuable environmental knowledge and observing wildlife in the natural setting of the Santa Monica Mountains.

Outreach
Stunt Ranch serves as a unique setting for outreach events that support California conservation and community engagement.

The reserve can accommodate large group events and provides an intimate setting to promote your cause. We encourage university clubs and organizations to consider the reserve for events relevant to California environmentalism, broadly considered. Residents of the surrounding area may also apply to use the reserve for community meetings and events.

Donald R. Dickey Bird and Mammal Collection
The UCLA Dickey Bird and Mammal Collection consists of over 70,000 skins and skeletons of birds and mammals from the United States, Canada, Mexico, Central America, and islands in the Pacific. The collection was developed by Donald R. Dickey of Pasadena and his associates between 1910 and 1932. Donated to UCLA in 1940 by his widow, Florence Van Vechten Dickey, it is one of the world's best collections of birds and mammals of the American Southwest and Central America. Also encompassing a broad teaching collection of bird and mammal skins, skulls, and skeletons developed by Loye Miller, the UCLA Dickey Collection serves the campus's biological curricula and the research efforts of faculty, students and visiting researchers. The Dickey Collection and the LaKretz Center are the two largest endowments associated with EEB.

The La Kretz Garden Pavilion
The La Kretz Garden Pavilion, located in the UCLA Mildred E. Mathias Botanical Gardens, is a multi-functioning facility that serves as a visitor's center, classroom, and meeting room, as well as a display space for garden specimens. The facility was made possible by a generous donation from Morton La Kretz and spans 1,815 square feet.

La Kretz Botany Building
Thanks to a generous grant from Morton La Kretz, the Botany building underwent a complete renovation in 2022, resulting in the La Kretz Botany Building. The building now boasts a refurbished entrance and lobby; upgraded HVAC systems; modern labs for teaching and research; and two new technologically enabled class labs. These renovations have significantly enhanced office, research, and teaching spaces for EEB, showcasing a notable advancement in the department's facilities.

EEB Seawater Facility
The department has long maintained a seawater facility to serve instructional and research needs in the department and for other campus users. This facility has been replaced with entirely new equipment in a new location in the Life Sciences Building as of 2024. This replacement (due to structural impacts of the old facility) includes a new storage tank, temperature controlled culture tanks for both tropical and California marine biota, facilities for experimental manipulation of invertebrates, and for decanting seawater for instruction and research use elsewhere. The new facility has also been designed for potential expansion to facilitate new instructional lab space and faculty research space should that be desired (Please see appended Marine Science Lab Plan for status and potential expansion)

B. Physical Plant

Needed Improvements

Space renovation and improvements are needed throughout the Life Sciences Building. The Heating, Ventilation, and Air Conditioning (HVAC) systems are dated and need to be replaced. HVAC is insufficient and costly in many of the LSB spaces, resorting to window air conditioning units due to the lack of a central system. These units do not provide the proper air exchanges required by UCLA standards, placing the maintenance burden on the department. Research lab rooms, lab benches, and controlled temperature rooms are inadequately configured, impacting the department's recruitment capabilities. Additionally, several EEB classrooms require renovations, including new paint, updated flooring, new lighting and replacement ceiling tiles. The department would be financially responsible for these updates. While important for overall recruitment and retention of faculty as well as the best student experience, there needs to be some shared expense for the renovations.

8. Staff

Since the last review, the Ecology and Evolutionary Biology (EEB) department has faced ongoing staffing difficulties, including retention, frequent turnovers and subsequent shortages that have significantly impacted the ability to meet the needs of faculty, students, and researchers. These shortages, compounded by the 2018 reorganization of the Life Sciences South Administration (LSSA), have created substantial challenges in maintaining effective administrative support. The reorganization, which aimed to streamline services by decentralizing administrative functions, resulted in increased workloads and lack of cross coverage for the staff assigned to EEB. The ongoing staffing issues have resulted in a strained departmental culture, particularly between faculty and staff. Faculty have felt the quality of services has left critical gaps in support for our department’s academic mission. Staff morale has been impacted by increased workloads and have reported feeling overwhelmed and undervalued.

Life Sciences South Administration (LSSA) Reorganization

In 2018 a decision was made to reorganize LSSA. The impact of the reorganization to EEB was not fully realized at the time. Prior to the reorganization, key administrative roles were centralized, providing specialized support and cross coverage across departments. Post reorganization staff were redistributed, and many service units were downsized while responsibilities expanded. Staff experienced a higher volume of daily duties without sufficient resources or preparation, leading to a decline in the quality of support services.

Research Administration/Finance

The Research Administration team consists of three fund managers, with one working at the senior level and the Director of Finance. Since the reorganization in 2018, this team has also experienced excessive turnover and staffing shortages. From 2015 - present there have been four (4) Directors of Finance, with the most recent Director being in the position since June 2023. From 2014 - present, there have been 12 fund managers. With the most recent 3 hires occurring between November 2022 - July 2024. While contract and grant management is an extremely important factor in the research mission, as well as recruitment and retention, EEB has struggled with finding and retaining an experienced finance team.

Procurement

EEB has one procurement specialist who handles a high volume of daily work. From 2018-present, there have been 5 specialists in this role. The current purchaser has been in place since July 2023. In the 2022-23 academic year this team processed 1,334 purchase orders, 84 travel booking requests, 493 travel reimbursements, 572 ticketing system inquiries, and 76 stipend and honorarium requests. Current volumes average 120 Zendesk tickets per month, most of which are assigned to and completed by a single purchasing assistant. This involves an average of $118,000 worth of requisitions per month in BruinBuyPlus and an additional $10,000 purchasing card transactions per month in Concur. With increase in the number of faculty and redistribution of duties, as well as the implementation of new platforms (Concur/Bruinbuy Plus) the procurement specialist has experienced increased volume of work with no additional administrative support.

Student Services

Per the reorganization of LSSA, the Computational and Systems Biology IDP was added to the Biology Group's purview. This resulted in the oversight of an additional staff member plus management of a separate undergraduate program. The Manager of Undergraduate Advising now has to split focus between two instructional areas, resulting in a more managerial role and reducing time left for student-facing academic advising. The timeline of changes is as follows:

> Summer 2018 – Fall 2021, the Director of Student Affairs (LSSA) was promoted to CAO of the Biology Group and subsequently left that role. This turnover left little to no continuity between the Graduate and Undergraduate offices, resulting in miscommunication and a disjointed team.
>
> June 2021- UCLA campus underwent the Career Tracks changes, which was meant to align job titles and compensation with external organizations. All student affairs officers were renamed student services advisors.
>
> October 2021 - the Interim CAO and Chair restructured the Student Services team, promoting the Manager of Undergraduate Advising to Director of Academic Programs and Student Services to provide more oversight of all Graduate and Undergraduate Student Services Advisors in the department.
>
> June 2023 - Director of Academic Programs and Student Services restructured the team, reclassifying the Student Services Advisor 4 to Manager, Graduate Programs and Student Services for EEB and CaSB, now overseeing the Student Services Advisor 2 for EEB Graduate Studies and Student Services Advisor 2 CaSB IDP. This created a better distribution of work, with the goal of retaining staff who are looking to grow.

Student Services has had 8 staff members leave their positions within the past 2.5 years. This has been incredibly difficult to help maintain continuity of work and student services, resulting in a greater demand on existing staff members. With such high turnover in select positions

(specifically the Student Services Advisor 2 position in the EEB Grad Office), it has been problematic to create a consistent flow of work. This, coupled with university and UCOP-level changes to timelines has created a high level of stress.

Undergraduate Program

Since the reorganization, staff in the graduate and undergraduate offices have experienced an unusually high turnover rate, with eight staff members leaving positions in the last two years. Staff turnover and leadership changes exacerbated issues, leading to a loss of institutional knowledge and continuity in service. A year by year recent summary is as follows:

> 21-22 AY: In the 2021-22 academic year, 5,700 students enrolled in EEB courses and provided advising for 1,740 majors and 54 minors. These advisors processed over 4,800 student messages and conducted nearly 1,550 advising appointments, all while handling a range of administrative tasks such as petition approvals and contract course setups.
>
> 22-23 AY: In the 2022-23 academic year, 4,700 students enrolled in EEB courses and provided advising for 1,804 majors and 90 minors. These advisors processed over 4,500 student messages and conducted nearly 1,900 advising appointments, all while handling a range of administrative tasks such as petition approvals and contract course setups.
>
> 23-24 AY: In the 2023-24 academic year, 4,900 students enrolled in EEB courses and provided advising for over 2,000 majors and 70 minors. These advisors processed over 4,600 student messages and conducted over 1,800 advising appointments, all while handling a range of administrative tasks such as petition approvals and contract course setups.

In addition to advising duties, student advisors are tasked with coordinating the annual events, including the graduation ceremony, 20 new student orientation sessions, the Undergraduate Research Symposium, and community outreach fairs. They also support curriculum updates, maintain the undergraduate section of the department website, and work closely with the Faculty Executive Committee. With only two advisors, it has become increasingly difficult to provide timely and personalized support to students, resulting in delays and a reduction in service quality.

Graduate Program

The EEB Graduate Program is managed by two staff members: a Graduate Program Coordinator and the Supervisor. Together, they handle all aspects of graduate advising, from committee constitution to funding management and teaching assistant assignments. Additional responsibilities include hiring ASEs and GSRs each quarter, along with planning recruitment events and managing graduate admissions, has stretched the capacity of the team. The graduate program has experienced increased delays in responding to students' needs and managing critical processes.

Personnel/Payroll

The Personnel and Payroll team consists of two career staff reporting to the CAO. The subject matter involved in this area involves understanding UCLA policies, University of California (UC) policies, State labor laws and Federal laws, as well as keeping abreast of the human resource practices and laws related to the Visa process. There are also numerous contracts that UC has secured with unions that govern many staff members' employment. Knowledge of and adherence to these policies, laws and union contracts is critical. Following the policies, laws and union contract stipulations can be time consuming and requires skill and efficiency.

The Academic Personnel coordinator processes academic matters for the Academic Senate Faculty, Lecturers (governed by a union contract), Postdocs (governed by a union contract), Research Scientists, Project Scientists and Adjunct Professor appointments. The work conducted in this area is specialized and extremely important to the department. The work involves submitting high quality and timely faculty merit and promotion cases for review by the Dean and Senate Committee on Academic Personnel. The subject matter involved in this area is also complex and requires specialized training and experience necessary to administer faculty and academic personnel and payroll functions for federal, state, private and other funding sources, as well as academic dossiers. This is another area where there has been significant turnover. Since the last review there have been 3 academic personnel coordinators and at the time of writing this review the position is currently vacant.

Section C. Bylaws

BYLAWS

UCLA DEPARTMENT OF ECOLOGY AND EVOLUTIONARY BIOLOGY

PREAMBLE

This document is written to be in conformity with both the Standing Rules of the Regents of the University of California and Bylaw 55 of the State-wide Academic Senate of the University (UC SB 55). It was adopted by two-thirds majority vote of Departmental members of the UCLA Academic Senate following a regularly scheduled faculty meeting held on November 16, 2015 (20 of 29 eligible faculty voted; 22 yes, 0 no, 0 abstain).

ORGANIZATION

1.1. The chief Departmental administrative officer is the Chair. The Chair receives assistance and advice from the Standing Committees of the Department, any Ad hoc Committees that may be appointed from time to time, and from the Faculty as a whole. Non-academic Staff assist the Chair in operational matters. The duties and responsibilities of the Chair are spelled out in the Announcement of the President entitled "Duties of Department Chairs (or Equivalent Officers)"

1.2. One or more Vice-Chairs may be appointed by the Chair with the advice and consent of the Department and the approval of the Divisional Dean of Life Sciences. The normal term of service of a Vice-Chair will be three years with the appointment being renewed annually. Vice-Chairs will be responsible for duties assigned to them by the Chair.

1.3. There are five Standing Committees. Composition of committees is determined after consultation with faculty, who are asked for their preferences and allowed to review the list of assignments before it is finalized. Composition of committees is determined each year by the Chair, taking into account the following: (1) areas of expertise and interest of each faculty member; (2) discussions between Chair and Vice Chairs to achieve some balance of subject matter fields in a given year or on a rotating basis; (3) availability of faculty during the academic year for service on departmental committees; and (4) fairness of load, so that certain faculty members do not consistently carry an unfair burden. The standing committees are:

- Personnel Committee

- Curriculum Committee
- Graduate Admissions and Support Committee
- Seminar Committee
- Departmental Awards Committee

Excepting only the Personnel Committee, Standing Committees may have student members, said students having full participatory and voting rights. The Chair works with appropriate Undergraduate and Graduate Student Organizations to identify student members. Students will normally serve for terms of one year. Except as explicitly provided below faculty appointments to standing committees are normally for terms of one or more years. The Chair ensures that terms are staggered in time.

1.4. Ad hoc Committees (e.g., Search Committees for new faculty members) may be appointed by the Chair as needed for the operations of the Department. The Chair determines the charges for these committees as is appropriate.

1.5. Special Functions of the Department may be assigned by the Chair as responsibilities of individual members of the faculty. These Functions may include, but are not restricted to:

- Biomedical Library Liaison(s)
- Departmental Webmaster
- DWQE Coordinators
- Director, Botanical Garden/Herbarium
- Graduate Advisor
- Undergraduate Advisor
- MBQ Coordinator
- FBQ Coordinator
- Dickey Collection Advisor to the Chair

2. OPERATIONS

2.1. Parliamentary Procedures and Voting

2.1.1 The Department operates according to the rules of both the UC state-wide Academic Senate and the UCLA divisional Academic Senate. These bylaws apply to all Departmental actions on “substantial departmental questions” [UC SB 55(A)(1)] and all academic personnel actions [UC SB 55(B)(1-7)]. Except as specified below, standard parliamentary procedures are followed. When needed A. Sturgis' The Standard Code of Parliamentary Procedures (4th edition, 2001) is used for guidance.

2.1.2 Faculty members in the Department who are members of the Academic Senate and have active appointments are eligible to vote on the matters specified in sec. 2.1.1. Only one series of titles is presently represented: Regular Series (19900, Ladder; Tenured and Tenure-track). Changes in this governance document can only be made if the proposed action is a listed agenda item for a meeting of Departmental faculty that has been scheduled and publicized in advance according to normal Department

procedure. A 2/3 majority vote of all Senate members present at such meetings is required for approval of changes. This document will be studied and revised by an ad hoc Bylaws Committee every 4 years; one of these revisions must occur during the year of Departmental self-review prior to the Department's 8-year review.

2.1.3 All faculty members with Regular series titles and active appointments have equal voting rights in all Departmental matters, including personnel actions. Personnel voting rights on Appointments were extended to Assistant Professors by a two-thirds majority of Associate and Full Professors (16 yes, 0 no) following the November 16, 2015 faculty meeting. Voting rights on Full Professor Promotions and Merits were extended to Associate and Assistant Professors by a two-thirds majority of Full Professors (15 yes, 0 no) following the November 16, 2015 faculty meeting. Voting rights on Associate Professor Promotions and Merits were extended to Assistant Professors by a two-thirds majority of Associate and Full Professors (16 yes, 0 no) following the November 16, 2015 faculty meeting. Emeriti can participate in Departmental meetings; their voting privileges on personnel cases are determined at 4-year intervals at the time of revision of these Bylaws. Currently, emeriti do not have voting privileges on personnel cases, whether they are recalled, not recalled, or appointed as Research Professors. Recalled emeriti have voting privileges on all non-personnel substantial department questions. Adjunct faculty members can participate in Departmental meetings not involving personnel actions, but do not have voting privileges. Temporary faculty members may attend Departmental faculty meetings, but do not have voting privileges. Faculty members with joint appointments in other units have voting rights as agreed upon within the terms of their appointments.

2.1.4 Faculty members may vote by absentee ballot. Faculty members on leave may vote, but their votes will not be solicited. Secure electronic voting is available when feasible.

2.1.5 Majorities are determined on the basis of pro and con votes only; votes to abstain do not count.

2.1.6 Departmental meetings are normally called by the Chair or the Chair's designee. Personnel actions and other substantial matters for consideration at Departmental meetings must appear on written agendas distributed to all faculty members at least three working days prior to the meetings. Exceptions to this rule require unanimous approval.

2.1.7 A quorum consists of 50% of the regular faculty members not on leave.

2.1.8 Recruitment of new faculty is done by *ad hoc* search committees appointed by the Chair. Search committee recommendations are made to the department at scheduled faculty meetings as agenda items. Recommendations are discussed by all eligible faculty in attendance at those meetings. Faculty with active Regular series appointments will vote on proposed appointments by secret written ballots for periods of two business days, beginning immediately after the presentation meeting.

3. CHARGES TO COMMITTEES

3.1.RESPONSIBILITIES OF STANDING COMMITTEES

3.1.1. PERSONNEL COMMITTEE

All discussions of personnel actions are confidential. Committee membership consists of a minimum of four Regular series faculty. Ad hoc members may be added as needed, as in cases involving faculty with joint appointments with other units.

The Personnel Committee is primarily responsible for reviewing and recommending to the Chair and the Department the full range of personnel actions involving Regular series members of the Academic Senate, also appointees in the non-Senate Adjunct Professor series. There are two special categories of actions for which the Committee is designated as the final reviewing body, making recommendations directly to the Chair:

1) Actions involving non-Senate appointees in the Adjunct Professor, Professional Researcher and Project Scientist series.

2) 5-year performance reviews of Senate members when such reviews are required under terms of the APM.

The Chair has discretion to bring any of these actions to the full Department if conditions arise in which Departmental review is considered essential.

Personnel Actions - Promotions

1) The Personnel Committee reviews and analyzes all aspects of a candidate's dossier, and makes a recommendation for action to the Department. Extramural references for candidates, when required, will be selected by the Department Chair after the candidate has been given the opportunity to provide suggestions.

2) The Personnel Committee presents its findings at a regularly scheduled Departmental meeting. After hearing the review and recommendation of the Committee, all regular members of the Department present discuss and consider the recommendation, then vote on it. A secret written ballot is held two working days after the discussion. The LSSA Office of Academic Personnel is responsible for conducting the ballot, the result of which is reported to the Chair. It is the responsibility of the Chair to inform the candidate of the exact vote.

3) The Personnel Committee then advises the Chair in the preparation of the Departmental letter.

4) The Chair forwards the Departmental recommendation to the Dean. At the Chair's discretion a second ("Chair's") letter may be included as described in University policy.

Personnel Actions - Merit Increases

Merit increases, including fourth-year appraisals for Assistant Professors, are handled in the same manner as promotions except that external reviews are not solicited. Voting eligibility for merit increases is the same as for promotions.

Personnel Actions - Adjunct Professor Series

1) Any member of the regular faculty may nominate a candidate for appointment in the Adjunct series. Fully documented nominations are referred by the Chair to the Personnel Committee for review and recommendation. Appointments, advancements in rank, and merit increases are handled in the same manner as for regular series faculty. Each initial nomination and all recommendations for subsequent actions must include statements specifically describing the ways in which the candidates will contribute to departmental teaching programs at either or both the undergraduate or graduate levels during the duration of the initial appointment or renewal appointment. Those statements must be developed jointly by the faculty sponsor and the candidate.

2) Votes on reappointments must be conducted by the Department every two years for appointees at the Assistant and Associate Professor levels, and every three years for appointees at the Full Professor level. The results are reported to the Divisional Dean.

Personnel Actions – Research Professor Series

1) Appendix 38: Research Professor was added to the UCLA CALL on April 6, 2012. The operative sections of that document state:

> "Professors Emeriti from UCLA who have continued a high level of research scholarship since their retirement may be eligible to utilize the title Research Professor as a working title. Appointees to Research Professor status must be able to demonstrate that their research activities since retirement have led to one or more of the following outcomes: an active external grant, a proposal to obtain external grant funding, or a recent history of external grant funding, internal research funds, publication of scholarly books, monographs, or journal articles, scholarly presentations at professional meetings
>
> II. TERMS OF SERVICE
>
> Appointments as Research Professor shall be made for three years, or in cases where a Research Professor is Principal Investigator of an active grant, for three years or the duration of the grant, whichever is longer.
>
> III. PROCEDURES
>
> Appointment authority is delegated to academic deans and may not be further delegated. Requests for appointment as Research Professor are normally initiated through the department chair or ORU Director by submission of a letter by the applicant that documents his or her continuation of a high level of research scholarship since retirement. In accordance with department or ORU procedures, the chair or director should consult appropriately with the faculty. If the unit concurs, the chair or director shall recommend the appointment to the appropriate dean. The dean will notify the applicant of the decision in writing with a copy to the recommending unit.

Renewal of an appointment is possible, but not automatic. The procedures for reappointment are the same as described above in this section."

2) Departmental implementation of this policy will be done as follows:

a) Professors Emeriti wishing to be appointed as Research Professor will submit a letter to the Department Chair making that request. The letter must document their post retirement research activity in accordance with the terms of Appendix 38. It must also specify the requested duration of the appointment per the Appendix.

b) The Chair will refer the letter to the Personnel Committee for its determination as to whether or not the request meets the standards of the Appendix. If it does meet the standards the Committee will recommend that the Chair forward the request to the Dean of Life Sciences. If it does not, the Committee will return the letter to the Chair with its recommendation as to what additional information may be needed to satisfy the standards. A revised letter will be reviewed as a new submission.

c) This policy and procedure will go into effect July 1, 2012.

3.1.2 CURRICULUM COMMITTEE

Committee membership consists of a minimum of three regular faculty members, one of whom is committee Chair, the Lead Undergraduate Student Affairs Officer, and the Faculty Undergraduate Advisor.

Committee responsibilities are:

1) In general, to keep the catalog and the schedule of classes up to date. To consider proposals for new courses or deletion or modification of existing courses by faculty members. The Committee presents its recommendations for new courses, changes in course descriptions, and deletions of courses to the Chair for final approval.

2) Proposed changes in core curriculum and the preparation for the major must be approved by the Department.

3.1.3 GRADUATE ADMISSIONS AND SUPPORT COMMITTEE

Committee membership consists of five regular faculty members, one of whom is committee Chair, plus the Graduate Student Affairs Officer, and the Faculty Graduate Advisor. The latter two people are both ex officio members.

Committee responsibilities are:

1) The committee provides for the acceptance and admission of new graduate students. The committee monitors the calendar for admissions and works to ensure that offers of admission are as competitive as

resources will permit. Through the Graduate Affairs Office it handles correspondence with prospective graduate students, manages student files, provides student files for faculty review, and designates provisional advisors.

2) The committee advises on policies and procedures for the selection, appointment and assignment of teaching assistants. It is responsible for collection of data for evaluation of the performance of teaching assistants. It generates and develops materials for orientation of new appointees. It also provides advice on the administration of graduate fellowships such as Traineeships, Fellowships, and others which may be sought by graduate students in the Department. It administers Departmental (e.g., travel and dissertation research support) and Graduate Division funds that support graduate students.

3.1.4 SEMINAR COMMITTEE

The committee is responsible for organizing, scheduling, and overseeing the Departmental seminar program. The committee solicits suggestions from the faculty, graduate students, and postdoctoral fellows for desired speakers. Committee structure and operations vary at the discretion of the Chair.

3.1.5 DEPARTMENTAL AWARDS COMMITTEE

The committee is composed of at least two faculty members. The committee oversees the Departmental awards program with the advice and assistance of the Lead Undergraduate Student Affairs Officer and the Graduate Student Affairs Officer. The committee assists the Chair in preparing and selecting nominations for the numerous extra-departmental faculty awards and prizes (e.g., Faculty Research Lecturer, Distinguished Teaching Award, Woman Scientist of the Year, etc.) for which the Department is asked to submit names and dossiers each year.

4. APPENDICES

A. Policy on Adjunct Professors

1) An Adjunct appointment will be made only when the Department will be clearly well served by such an appointment.

2) All such appointees must meet the hiring standards that we impose on any candidate for a regular tenure track position.

3) The candidate must demonstrate a commitment to teach on a regular basis. This will include participation in a graduate or advanced undergraduate course, and may also

involve lower division or core course teaching. While there is no commitment to a salary for such participation, it is not precluded for lower division and core teaching. Salaries for such teaching must be negotiated with the Department Chair. Appointees are encouraged to serve as members of advisory committees for graduate students.

4) All extramural funding for research administered through UCLA must be "signed off" by the University, although overhead may not necessarily go to UCLA unless the University is the sole underwriter of the grant.

5) Space must be negotiated with the Department Chair.

6) The candidate will be eligible to co-sponsor graduate students.

7) Advancement will follow the same procedures as for regular faculty. Evidence of continuing academic excellence and productivity, similar to our active faculty, will be a condition of continued appointment.

B. Policy on Acceleration Requests

1) All acceleration requests must be accompanied by a letter to the Chair from the candidate justifying the request. Regardless of the justification, all acceleration requests made by May 1 will be reviewed by the Personnel Committee, presented to the Department, and submitted to the Dean of Life Sciences, unless withdrawn by the candidate. Late acceleration requests, defined as those made after May 1, must be based on a significant achievement, such as a national award, that occurred or was announced after May 1. Whether this criterion has been met will be evaluated by the Personnel Committee.

2) If the Personnel Committee determines that a late acceleration request is justified, the case will proceed as usual, unless an extension of the Dean's deadline is needed. If an extension of the Dean's deadline is needed, the Department will submit a request for an extension to the Dean, along with the candidate's justification.

3) If the Personnel Committee finds that a late acceleration request is not justified, the candidate will be informed of the basis for this decision in writing. If the candidate chooses to appeal the decision, their justification for the acceleration request and the Personnel Committee's decision will be submitted to the Dean. The Department will proceed with the case, if asked to do so by the Dean.

C. Departmental Equipment and Services

1) All equipment designated for instructional use and purchased with Departmental funds must be made available for regularly scheduled courses.

2) When such equipment is not in use in courses, it may be used for research with the understanding that it must be maintained in proper order by the faculty member concerned and that it must be relinquished promptly when needed for a course.

3) Whenever appropriate, major pieces of equipment purchased with Departmental funds and requiring special technical knowledge for their use will be assigned by the Chair to a specific faculty member who will assume responsibility for overall maintenance.

4) A current list of all pieces of Departmental equipment purchased for courses and their location will be available in the Stockroom. This will include a listing by courses.

5) The Facilities Manager will be responsible for maintaining a listing of all technical services (photographic, illustration, workshop, etc.) available in the Department.

Section D. Undergraduate Programs

1. Overview

The overall goals of the undergraduate programs of the Department of Ecology and Evolutionary Biology are: 1) to provide a broad education to students interested in fields where biological training serves as a stepping stone to professional schools (medicine, optometry, dentistry, pharmacy and veterinary), or careers in fields such as education, industry, environmental and conservation sciences, and public health; and 2) to provide a more focused research-oriented capstone experience for students exploring careers as research scientists and academics in ecology, evolution, behavior, and marine science.

2. Rationale

The Department of Ecology and Evolutionary Biology is dedicated to attracting and retaining a wide diversity of undergraduate students of the highest caliber, providing them with an outstanding educational experience, and connecting that education with their career goals and the needs of the community.

For many years, the majority of incoming Life Sciences majors at UCLA have been interested in careers in the health professions, and this has remained unchanged over the last 10 years considered in this review. However, students have come to understand that their major need not be directly associated with their professional training, resulting in pre-professional students pursuing a broader range of academic majors at UCLA. While such a shift might normally result in a reduction in the numbers of majors within EEB, there was substantial growth in EEB undergraduate majors from 2014-15 until 2016-17 when enrollments began to decline (Figure D-1). Subsequent declines in EEB undergraduate majors are associated with the launch of the Human Biology and Society major in the Institute for Society and Genetics, which grew from ~50 majors in 2012 to >400 in 2023, and to a lesser extent the Computational and Systems Biology interdepartmental program, which grew from 3 majors in 2016 to >90 in 2023. Although current enrollment numbers reflect a decline in total majors from the 2016-17 high, current enrollment reflects an overall 3.3% increase over 2014 levels, demonstrating a strong, sustained demand for the three majors serviced by EEB.

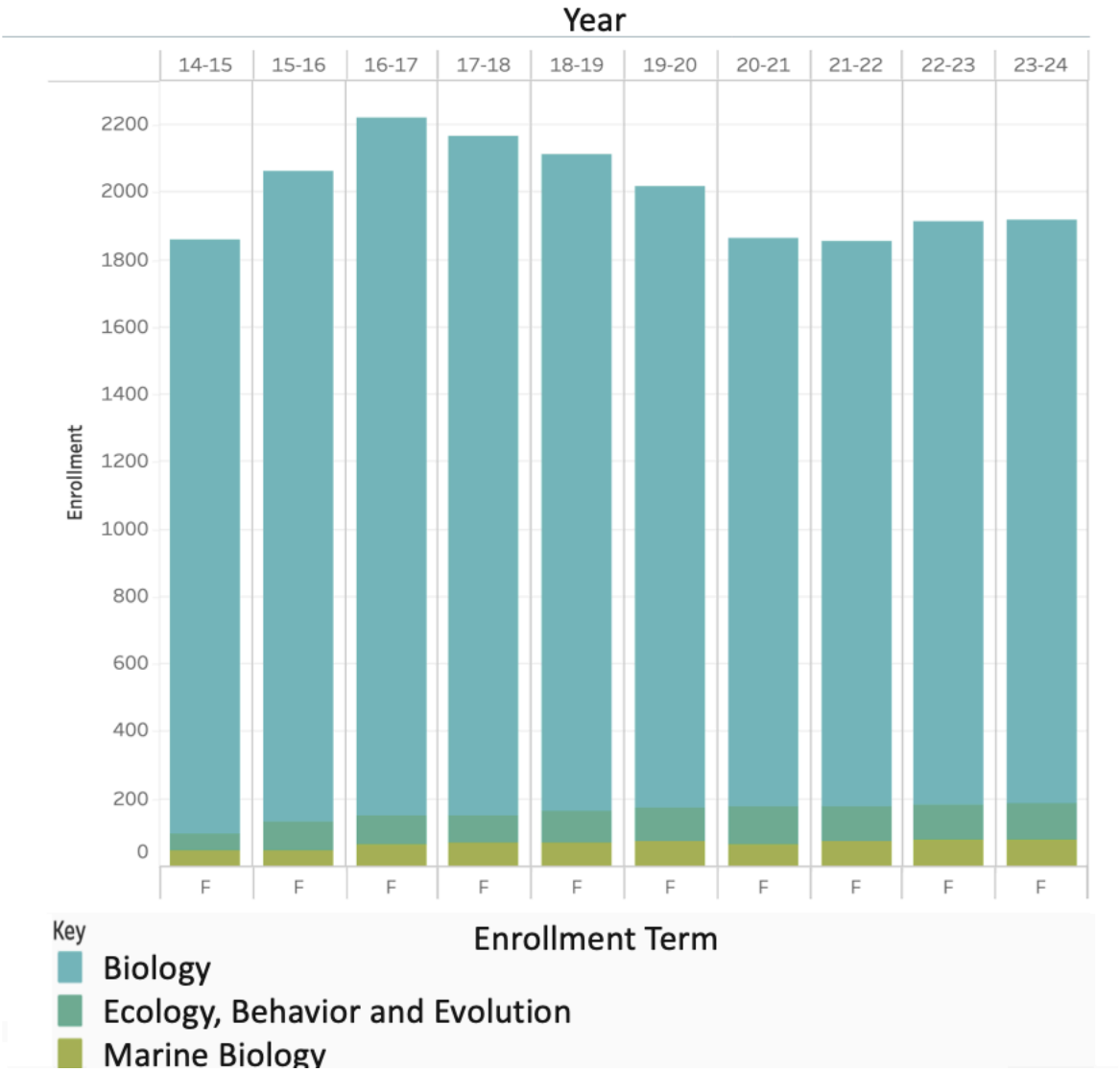


Figure D-1. Undergraduate enrollment in EEB majors (Biology, Ecology Behavior and Evolution, and Marine Biology from 2014-2024

3. Structure

The Department of Ecology and Evolutionary Biology is the home of 3 majors (Biology, Marine Biology, and Ecology, Behavior, and Evolution) and 2 minors (Conservation Biology and Evolutionary Medicine). Biology is by far the largest EEB departmental major, serving more than 90% of our students. This major is, by design, very flexible, allowing students to draw from courses within our department as well as from other Life Science departments in order to meet the entrance requirements of various professional schools, graduate programs, and other careers in biology. It is an especially attractive major for students who wish to focus on breadth of knowledge and for those students whose interests overlap with other departments in the life and physical sciences. The Department of Ecology and Evolutionary Biology is the home of the

Biology major because we offer the broadest training in biology, ranging from molecules to ecosystems and from contemporary processes to deep evolutionary timescales. We also offer a centralized home for administration and counseling for this large number of majors, two important mechanisms to ensure timely graduation and smooth transitions to future careers (see advising below).

In addition to the Biology major, Marine Biology and Ecology, Behavior, and Evolution are two capstone majors and continue to be the route that students take when they want to explore a more specialized research-based and experiential learning environment, particularly for students interested in organismal biology. Students with these majors frequently enter PhD programs in these fields. We also offer two departmental minors. Students in our Conservation Biology minor come from a wide variety of backgrounds, including other majors across the Division of Life Sciences, and often have an environmental career in mind, and enrollment in this minor continues to grow (Figure D-2). Similarly, our Evolutionary Medicine minor, introduced in 2013, has experienced significant growth, particularly in the last 4 years through COVID (Figure D-2).

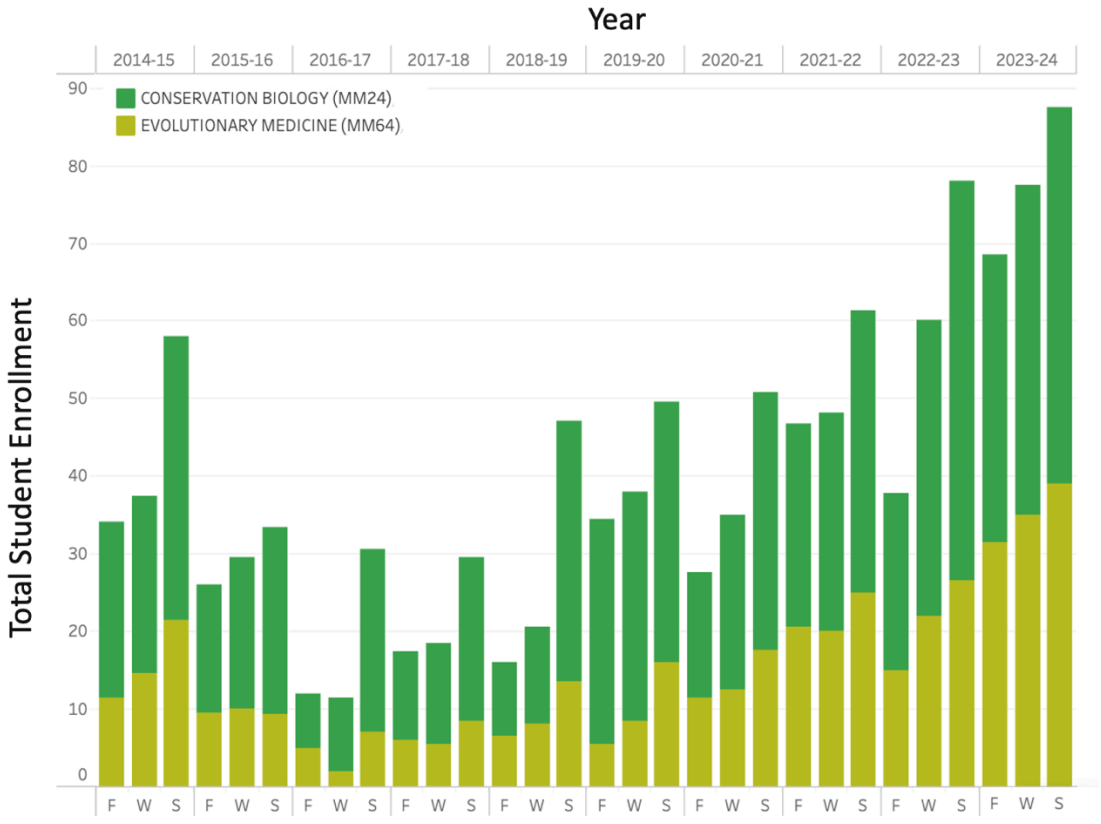


Figure D-2 Growth in the Evolutionary Medicine and Conservation Biology Minors from 2014-2024

We are constantly updating and refining our major requirements and elective choices to reflect the changes in courses taught in our and other departments and the more focused research interests of our current Ecology and Evolutionary Biology faculty. The department gained approval and began the newest revised Biology major in Spring 2020, which retains the advantage of containing elective courses in several life sciences areas, but is more focused on Ecology and Evolutionary Biology courses. The revisions for the Biology major were particularly noteworthy in allowing flexibility within the program, but with an increased focus on departmental courses. Changes to the Marine Biology major were also made in order to focus on current departmental strengths in ecology and evolution. The revised guidelines for Biology and Marine Biology majors are listed in Appendix D1. The most significant changes in our majors revolve around changes to the Life Sciences Core curriculum, which prepares our students for their upper division coursework.

A. The Life Sciences Core

All UCLA life science majors have a unified lower division curriculum (the Life Sciences Core Curriculum), including general and organic chemistry, math, and physics, and a year-long series of life sciences courses (Life Sciences 7a: Cell and Molecular Biology; LS 7b: Genetics, Evolution and Ecology; LS 7c: Physiology and Human Biology. In 2017, this new year-long Life Sciences course series replaced the former 4-course series, with significant pedagogical innovations focused on active learning, and a streamlined curriculum.

The Life Science Core provides the introductory curriculum for the three majors that our department administers and for the life science majors administered by other departments. The core has its own department chair and a largely non-ladder instructional staff and coordinates the efforts of departmental faculty who teach in the core. Faculty from Ecology and Evolutionary Biology primarily teach in LS 7b (Genetics, Evolution and Ecology) and LS 40 (Statistics for Life Sciences). Life Sciences 7b is taught during every academic quarter and in summer sessions A and C. Approximately 500-700 students enroll in LS 7b every academic quarter, and about 100 students take LS 7b in each summer session. LS 40 is taught in winter and spring quarters, with enrollments between 100-250 students. EEB faculty also teach LS110, focused on exploring and preparing students for future careers in the life sciences.
Another significant change in the core curriculum is the shift from the year-long Math 3 calculus series to a two-course series, Life Sciences 30a and 30b (Mathematics for Life Scientists). This course was developed to modernize how our students learn math, teaching it in the context of Life Science, and incorporating the use of python computer programming to teach students modern concepts in computational biology. Importantly, this course series leads to better outcomes in downstream courses like Physics, compared to traditional, pure mathematical focused calculus courses. However, for the minority of students who desire a pure calculus course, Math 3 is still accepted for the mathematics requirement of our major.

B. Biology BS

The Biology major offers a broad curriculum that allows students to study biology across a range of sub-disciplines. All EEB majors begins with a core curriculum in Life Sciences (LS 7 series), Math/Computational Biology (LS30 series, LS40), Chemistry (Chem 14 series), and Physics (Phys 5 series), followed by introductory core of courses specific to our department, Introduction to Evolution and Behavior (EEB 100), and/or Introduction to Marine Science (EEB 109). Students then build on this foundational knowledge through additional upper division courses and research-based courses. Appendix D2 outlines the requirements for completion of this major.

C. Ecology, Behavior and Evolution BS - capstone major

The EBE major is appropriate for students preparing for graduate study in ecology, behavior, and evolution, or for employment in areas such as environmental biology, conservation biology, teaching, museum/curatorial work, and governmental positions focused on wildlife/ecosystem management, and environmental issues. Students in this major are required to complete either the research-oriented Field Biology or Marine Biology Quarter, which are discussed below. Appendix D3 outlines the requirements for completion of this major.

D. Marine Biology BS - capstone major

This major is designed for students wishing to specialize in marine sciences. Students gain an excellent background in biology and take specialized courses in oceanography, marine ecology, and physiology of marine organisms. Marine Biology majors are required to complete one quarter of field research and study in the research-oriented Marine Biology Quarter, discussed below. Appendix D4 outlines the requirements for completion of this major.

E. Comparison with other University of California major curricula

Majors in EEB are similar in intent to other UC campuses, although they vary in plan. All of UC campuses have a major similar to Biology. Davis, San Diego, Santa Barbara, and Santa Cruz have majors analogous to our EBE major. However, our EBE program is the only one that requires a capstone field quarter experience. Similar marine biology majors are offered by UC Santa Barbara and UC Santa Cruz, and UC Davis has the Marine Ecology and Organismal Biology major. UC Berkeley offers a marine science major, administered jointly by the Department of Earth and Planetary Science and the Department of Integrative Biology, but lacks the biological focus of a marine biology major.

F. Addition of courses

Since the last review, the department of Ecology and Evolutionary Biology has created a large number of new course offerings, and modified several others to better meet the pedagogical needs of our students. Of particular note are courses that expand our offerings in capstone majors (e.g. 140, 169) and expand lab offerings (e.g. 144L, 161L, 113AL). In total, EEB added 35 new courses during the review period, but only 24 of these new courses are still active; some

are no longer taught due to loss of instructors and some were designed as single offerings associated with a graduate course focused on pedagogical development. In addition to adding courses, EEB also eliminated 30 courses from our catalog, representing courses that will no longer be taught due to shifts in pedagogy, loss of appropriate instructors, or lack of student interest. In particular, four non-majors courses were removed from our catalog as the market for general education courses is heavily saturated at UCLA, and non-STEM majors tend to not take STEM courses for GE requirements.
All new courses are listed in. Table D-1. More detailed descriptions of their development can be found in Appendix D5.

Table D-1: New Courses Developed During Review Period

Course Number	Course Title	Term First Offered	Developed By	Active
EE BIOL 169	Biology of Bats: The True Superheroes	24W	Sears, Karen	Yes
EE BIOL 167	Natural History Collections in Biological Sciences	23W	Baniaga, Anthony Marcot, Jonathan	Yes
EE BIOL 167L	Natural History Collections in Biological Sciences Laboratory	23W	Baniaga, Anthony Marcot, Jonathan	Yes
EE BIOL 103	Plant Diversity and Evolution	17S	Zapata, Felipe	Yes
EE BIOL 104	Plants and People	23F	Baniaga, Anthony	Yes
EE BIOL 143	Viral Ecology and Evolution	22W	Knowles, Benjamin	Yes
EE BIOL 144L	Prehistoric California Laboratory	22W	Marcot, Jonathan	Yes
EE BIOL C146/C246	Conservation Genetics	22W	Sork, Victoria	Yes
EE BIOL 168	Global Change Ecology	22S	Ordway, Elsa	Yes
EE BIOL 88S	Urban Ecology: Coexistence in Urban Landscapes	22S	Seminar	Yes
EE BIOL 149	Evolutionary Genomics	21S	Garud, Nandita	Yes
EE BIOL 88S	Animal Consciousness and Ethics	21S	Seminar	Yes
EE BIOL 140	Biology of Marine Mammals	20F	Dines, James	Yes
EE BIOL 183	Finding Ecological Solutions to Environmental Problems	19W	Lipman, Alison	Yes

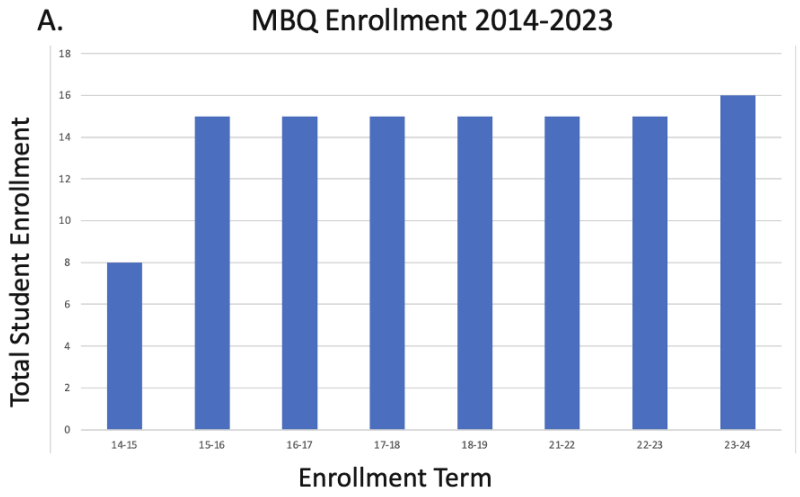
EE BIOL 158	Introduction to Diversity, Health Disparities, and Environment	19F	Barber, Paul	Yes
EE BIOL 184	Evolution, Development, and Disease	19F	Sears, Karen	Yes
EE BIOL 144	Prehistoric California	18W	Marcot, Jonathan	Yes
EE BIOL 113AL	Herpetology Laboratory	18S	Shaffer, H. Bradley	Yes
EE BIOL 156	Biology and Social Justice	18S	Alfaro, Michael	Yes
EE BIOL 88S	Most Extreme: Exploring Extremophiles, Origin of Life, and Search for Life in Universe	18S	Seminar	Yes
EE BIOL C202/C172	Advanced Statistics in Ecology and Evolutionary Biology	17S	Pinter-Wollman, Noa	Yes
EE BIOL 161	Plant Ecology	17F	Kraft, Nathan	Yes
EE BIOL C177/C234	Practical Computing for Evolutionary Biologists and Ecologists	15W	Alfaro, Michael	Yes
EE BIOL 201	Introduction to R for Ecology and Evolutionary Biology	15F	Lloyd-Smith, James	Yes
EE BIOL 98T	Plants and Humans: Ancient Connection	23W	Grad student course	No
EE BIOL 150L	Principles of Genetics Laboratory	22W	Prunier, Rachel	No
EE BIOL 98T	History and Racism of Biology Research: Moving towards Anti-Racist Science Future	22W	Grad student course	No
EE BIOL 161L	Plant Ecology Laboratory	22F	Kraft, Nathan	No
EE BIOL 150	Principles of Genetics	21W	Prunier, Rachel	No
EE BIOL 220	Conservation Science: Theory and Practice	20W	Amarasakare, Priyanga	No
EE BIOL 171	Coming of Age on Planet Earth	20F	Natterson-Horowitz, Barbara	No

EE BIOL M157/SOC GEN M157	Biology of Superheroes: Exploring Limits of Form and Function	19W	Campbell-Staton, Shane	No
EE BIOL 108	Biodiversity in Age of Humans	19S	Wayne, Robert	No
EE BIOL 87	California's DNA: a field course	18F	Wayne, Robert	No
EE BIOL 98T	Flexible Phenotypes and Adaptive Evolution	14W	Grad student course	No

G. The Marine and Field Biology Quarters

The Marine and Field Biology Quarter courses comprise the capstone experience for our Marine Biology and Ecology, Behavior, and Evolution capstone majors. These courses are 10-week long, immersive research experiences. Students carry 16 units of coursework that is spread over two to four courses, each with a significant research component. The site of the Marine Biology Quarter (MBQ) varies, but includes the University of California Richard B. Gump South Pacific Research Station in Moorea, French Polynesia, UC Davis' Bodega Bay Marine lab, and USC's Wrigley Marine Science Center on Catalina Island. For the Field Biology Quarter (FBQ), students travel either internationally to field sites in Australia, Ecuador, Nicaragua, or Thailand, or locally to field sites in California or Mexico. Recently, we added a local FBQ, focused on the urban ecology of Los Angeles

The MBQ currently serves 15-18 students each quarter and is offered once or twice per year. Similarly, the FBQ is offered one or two course options per year, with each option carrying a maximum of 15 students. In some years we also offer a hybrid FMBQ, where students engage in both terrestrial and marine research over the duration of the quarter. There is a very competitive application process for participation in these courses, limiting the total number of students we can serve, resulting in relatively stable enrollment numbers across the years (Figure D-3).



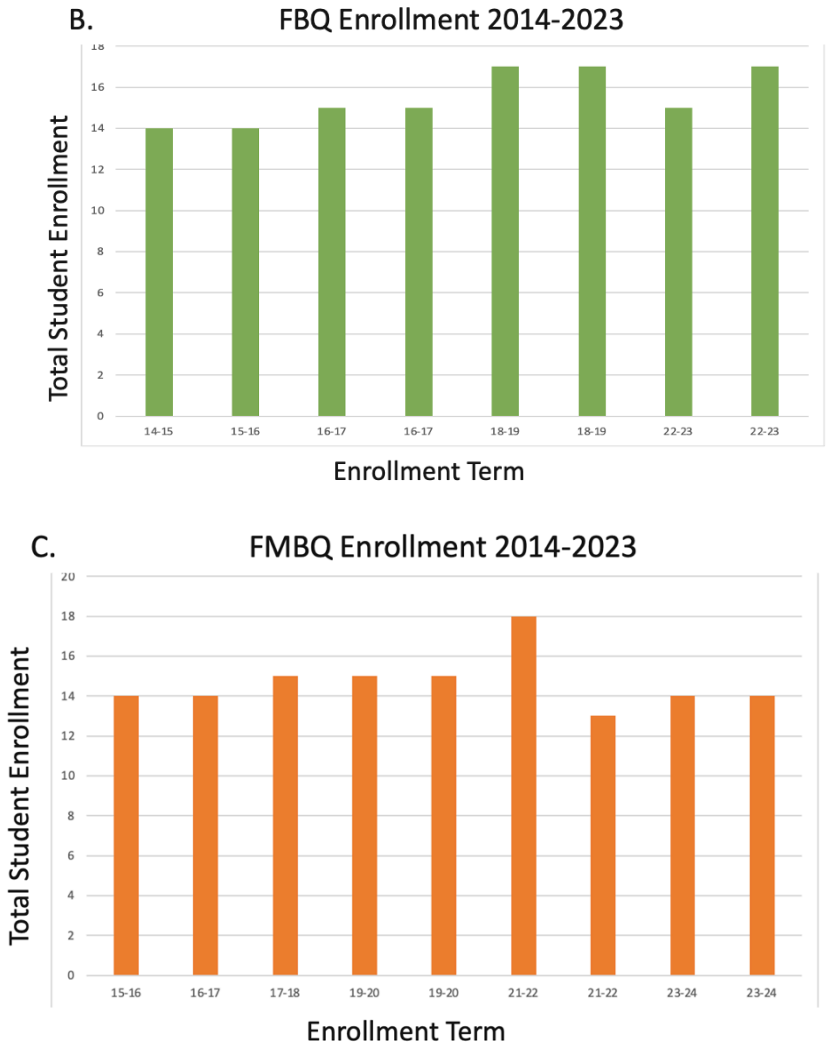


Figure D-3. Enrollment in the A) Marine Biology Quarter (MBQ), B) Field Biology Quarter (FBQ) and C) Field/Marine Biology Quarter (FMBQ) from 2014-2024.

Students evaluate MBQ/FBQ courses extremely highly, and many attest that this experience is what made them decide to become scientists and academics. However, over the past 10 years, the funding of these valuable programs has been an issue of much discussion among the faculty who participate in the Marine and Field Biology Quarters. The instructors are committed to providing regular lower-cost alternatives to the increasing expense of more distant and costly field and marine quarters; on the other hand, many students are primarily interested in the programs that are more exotically located, and the pedagogical needs of the course content are not always best served by more local facilities.

Previously, to help defray student costs, the faculty of the MBQ and FBQ prepared a joint proposal each year to the Office of Instructional Development, helping significantly reduce the

cost to each student. However, UCLA is no longer supporting funding requests for field programs, requiring different strategies. To partially compensate for the loss of these funds, EEB has developed a number of scholarship funds. The Joseph Epperson Memorial Field Biology Scholarship provides need-based scholarships in the range of $500-$1,000 to FBQ participants, and the AR Wallace fund provides ~$500 for students either in the FBQ or FMBQ. Similarly, the Vance-Fohey Fund provides scholarships of ~$500 for students in the MBQ and the Park Nobel Fund provides ~$500 for students in the MBQ or FMBQ . From 2014-2023, these funds provided a total of $53,250 in scholarship funds to 101 unique students. However, the increasing costs of running these capstone field programs, and decreasing levels of institutional support are a significant concern.

Field quarters require a large commitment of teaching faculty, as faculty are with the students full time while in the field. This time commitment constrains the Department's ability to increase either the size of these courses or the frequency of their offerings. In turn, these constraints affect the size of the MB and EBE majors and their potential for growth. We are addressing these limitations in several key ways. 1) Some students take equivalent field coursework through the University of California Education Abroad Program or recognized equivalent from outside the UC, but the extremely high standards that we set for the research aspects of these courses limit the number of options students have outside of UCLA. 2) We recently created an offering local version of the FBQ where students are introduced to field biology in a rapidly changing urban environment and biodiversity hotspot, the city of Los Angeles. As with traditional FBQ/MBQ offerings, the course covers the fundamentals of observing nature, formulating research questions, experimental design, collecting and analyzing data. A central theme of this FBQ is to reflect on the role of humans as ecosystem engineers and the effects of our actions on the local environment as well as the other organisms that inhabit it. Importantly, this FBQ allows students to have the same intensive field research experience without the added expense of travel. 3) Noting that research in EEB is increasingly computational, we are creating a new research-intensive offering, where students receive an in-depth research experience on the UCLA campus doing computationally intensive research, rather than field-based research, the Computational Biology Quarter. In contrast to the typical 10-week intensive experience of the FBQ/MBQ, the CBQ will combine a quarter of preparatory coursework, followed by subsequent mentored computationally focused research. This new model for a capstone research experience will add an important new dimension to our established research-intensive programs, expanding the accessibility and topical focus relative to existing FBQ/MBQ offerings.

Although our department has been creative in ensuring the sustainability of the MBQ and FBQ capstone programs, the new campus-wide Bruin Budget Model, where departmental funding is tied directly to student credit hours, represents a novel challenge. Because of the high demands of leading these capstone courses, teaching these courses fulfills the entire teaching obligations for faculty leading MBQ/FBQ courses. Committing the full teaching load of two faculty for low enrollment MBQ/FBQ courses could become harder to justify for upper administration, given funding implications. Indeed, EEB has already been singled out by administration for providing fewer student credit hours of instruction, relative to other Life Science departments, a direct result of low-enrollment offerings of the MBQ/FBQ, which other departments do not offer.

However, there remains strong support at the departmental level for maintaining these transformative courses.

H. Independent Student Research

Many students in each of the three undergraduate majors and two minors participate in independent research although not all enroll for credit through EEB. Those that do, generally enroll first in EEB 99, the Student Research Program. After a quarter or more of this apprenticeship, many go on to EEB 199 (independent research, upper division) or the EEB 198 series (independent honors research). The number of students in the labs of each faculty member varies, often depending on the nature of the research, but many mentor several each quarter. For the EBE and MB majors, the students conduct independent research projects as part of the FBQ or MBQ, and a substantial number of these projects result in publications. As a result of this combination of research opportunities, several of our faculty members have co-authored a number of publications with undergraduates (over 50 with approximately five in review or submitted, Appendix D6).

General biology majors are encouraged to conduct independent research within our department, as well as other departments in the Division of Life Sciences, Division of Physical Sciences (particularly the Institute of the Environment and Sustainability and Atmospheric and Oceanic Sciences) or biomedical laboratories associated with the David Geffen School of Medicine or Fielding School of Public Health. In addition, many students opt to become paid research assistants in one of the medical research laboratories. The department honors outstanding undergraduate student research at the Biology Research Symposium each spring, and many undergraduate students in our department present their work at Undergraduate Research Week, where Dean's Prizes are awarded for excellence in research. The Undergraduate Science Journal is a student-run academic journal featuring original research articles, review articles, and faculty interviews, sponsored by the Undergraduate Research Center for the Life and Physical Sciences. Thus, the undergraduate research experience opportunities in our department are both varied and extensive.

I. Other Undergraduate Programs

1. Conservation Biology minor

The Conservation Biology minor was established Spring Quarter, 2007, and continues to draw considerable interest from students (Figure 2).This minor provides greater opportunity to foster and strengthen links with the other departments, adding a much-needed curriculum for many on campus who are interested in environmental issues, but do not want to major in the Environmental Science major. In fact, one important function of the Conservation Biology minor was to create an area of concentration for the Environmental Science (ES) majors based in the Institute of the Environment and Sustainability. Enrollments in the Conservation Biology minor have fluctuated over the years, with a peak in Spring 2015 (36 students) and Spring 2023 (51 students), but show an overall pattern of growth over the review period. The growth of the Conservation Biology minor has also resulted in a greater interest in student participation in both the Field Biology Quarter (FBQ) and the Marine Biology Quarter (MBQ). An increasing number

of participants in the field quarters are declared minors because students are allowed to apply the 16 upper division units awarded by the MBQ/FBQ to the minimum 19-unit upper division elective requirement for the minor. The full requirements for the minor are listed in Appendix D7.

2. Evolutionary Medicine minor

Fundamental and established scientific principles within the field of evolutionary biology contain powerful insights for all the biological sciences, and the field of medicine is no exception. Unfortunately physicians, physicians-in-training, and pre-health students receive inadequate education in evolutionary biology, resulting in potentially negative outcomes, such as antibiotic-resistant bacteria. Infusing evolutionary teaching into medical and pre-medical education could significantly improve the quality of scientific thinking of students at all levels of training, resulting in better health outcomes.

To address this need, we created our Evolutionary Medicine Minor which was opened for students in Fall 2013. This minor is designed for students who wish to create new paradigms for investigating and understanding disease, and to foster integrative and creative thinking among future health professionals. Through this program, students develop a diverse tool-kit including a foundational understanding of the core principles of evolutionary biology that prepares them for future discovery as physicians and biomedical researchers.

The Evolutionary Medicine minor has experienced steady growth over the past 10 years, from 12 students in 2014 to 40 students in spring 2024, demonstrating high student interest and demand Student in the Evolutionary Medicine minor come from all three EEB majors, as well as a diversity of other majors, including Astrophysics, Chemistry, Math and Neurobiology. Students must complete the prerequisites before they can be officially accepted to the minor, and the full requirements for the minor are listed in Appendix D8.

4. Enrollments, Course Offerings, and Advising

A. Enrollments and Course Offerings

Outside of Psychology, the Department of Ecology and Evolutionary Biology has consistently housed the most life sciences undergraduate majors at UCLA, with most other life sciences departments serving a much smaller number of majors. Over the last 10 years, this pattern has remained relatively constant, although some programs such as Human Biology and Society, offered through the Institute for Society and Genetics have attracted an increasing number of life science undergraduate students (Figure D-4). Despite growth in other departments, and the launch of two new majors in Life Sciences (Human Biology and Society, and Computational and Systems Biology, EEB majors have grown 3.3% over 2014 levels.

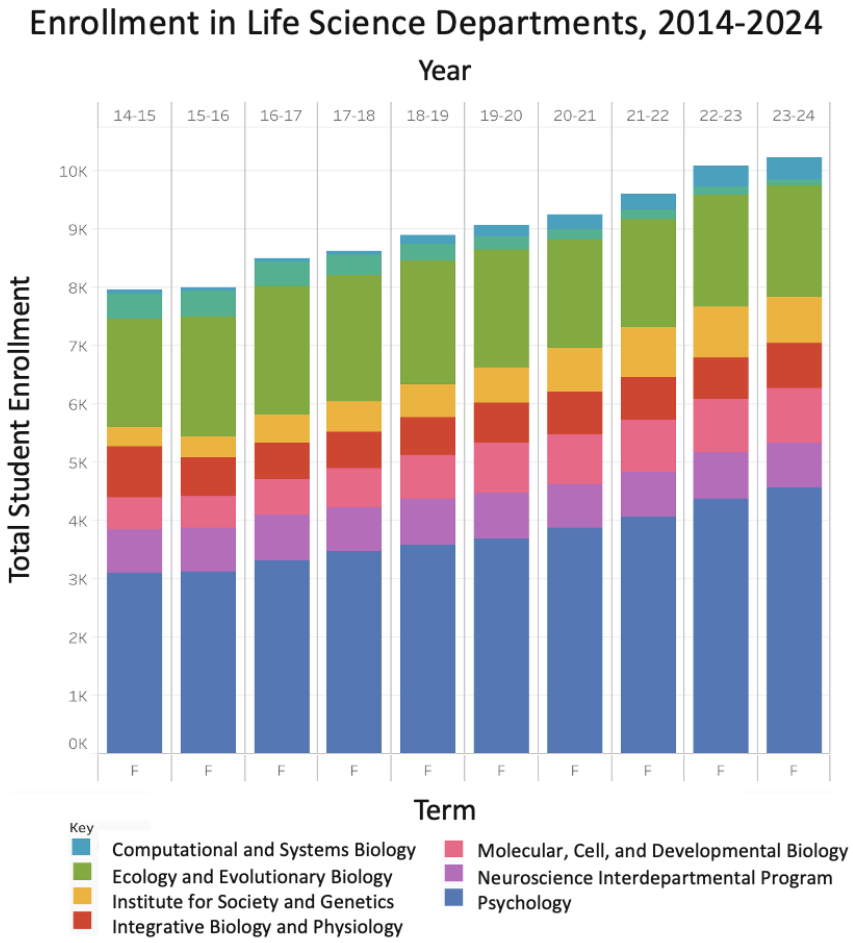


Figure D-4. Total fall student enrollment in undergraduate majors of 7 Life Science units from 2014–2014.

Currently, there are 12 undergraduate majors in the Division of Life Sciences with similar preparatory courses. Many of these majors are highly specialized and therefore subdivide the field of knowledge in life sciences to a degree that does not suit many of our students whose interests may span several majors or who wish to have a broader background in preparation for professional school. For those undergraduates whose interests are not suited to more specialized majors, the Biology major provides the opportunity to bridge all of the life sciences subject areas. (The nine additional Life Sciences majors outside of Ecology and Evolutionary Biology are Human Biology and Society (B.A. and B.S.); Microbiology, Immunology, and Molecular Genetics; Molecular, Cell, and Developmental Biology; Neuroscience; Physiological Science; Computational and Systems Biology; Psychology, Psychobiology, and Cognitive Science).

As of Fall Quarter 2014, there were 1,625 Biology majors, 58 EBE majors, and 54 Marine Biology majors, for a total of 1,737students. Trends since the beginning of each of these programs can be seen in the figure below. Of note is the growth in both the EBE and Marine Biology majors, which has put pressure on enrollment in FBQ/MBQ courses (see above).

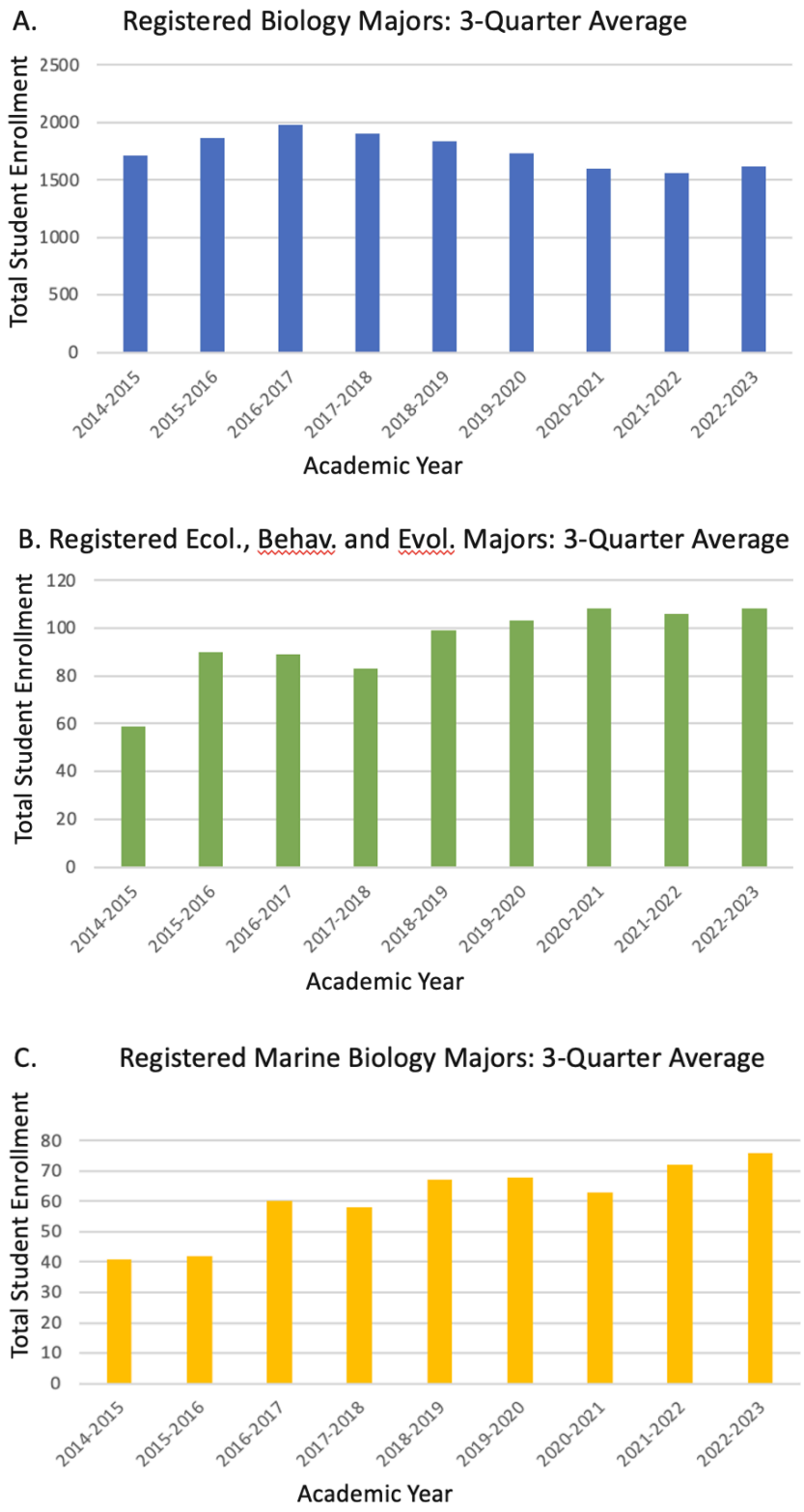


Figure D-5. Total student enrollment in the three EEB majors from 2014-2024 including A) Biology, B) Ecology, Behavior, and Evolution, and C) Marine Biology.

Ecology and Evolutionary Biology offered, on average, 53 lecture and lab courses per year over the time period of 2006 to 2014. Of these courses, an average of 63% were staffed by ladder-rank faculty from Ecology and Evolutionary Biology. The remainder were staffed by a combination of full-time lecturers, temporary lecturers, adjunct faculty, and faculty from other departments. An average of 2600 students were enrolled per year in Ecology and Evolutionary Biology undergraduate courses. In addition, EEB faculty and lecturers offered, on average, 6 courses per year in the Life Sciences Core, serving an average of an additional 1,658 students per year. Appendix D9 shows the classes offered, the number of quarters they were offered, and their average enrollments per quarter over the course of the review period.

High demand for EEB laboratory courses presents a continuing challenge given our present faculty size (since 2014, we have lost 11.5 faculty to retirement, separation, and death), offset by recruitment of 3 new faculty. In addition, highly limited laboratory instructional space limits our ability to offer enough laboratory classes to comfortably cover student needs. Nearly all of our upper division courses, especially the laboratory courses, have more applicants than enrollment slots. While the creation of EEB 100L and 109L went a long way in easing this constraint, adequately serving our students' need for laboratory classes remains an urgent and persistent issue.

In addition to lecture and laboratory courses, Ecology and Evolutionary Biology offers a number of options for independent study for undergraduates. These currently include mentored research experience at various levels of commitment and intensity (e.g., EEB 99, 196, 198A-D, and 199); teaching experience (EEB 192A/B); seminars and directed study (EEB 190, 191, 193 and194A/B); and internships outside of UCLA (EEB 195). Ecology and Evolutionary Biology has averaged 100 undergraduates per year in these independent study courses, with a range of 48 to 156 across the last eight years (Figure D-6). It should be noted also that the Field Biology Quarter and Marine Biology Quarter have a significant research component, serving ~ 30-45 students per year.

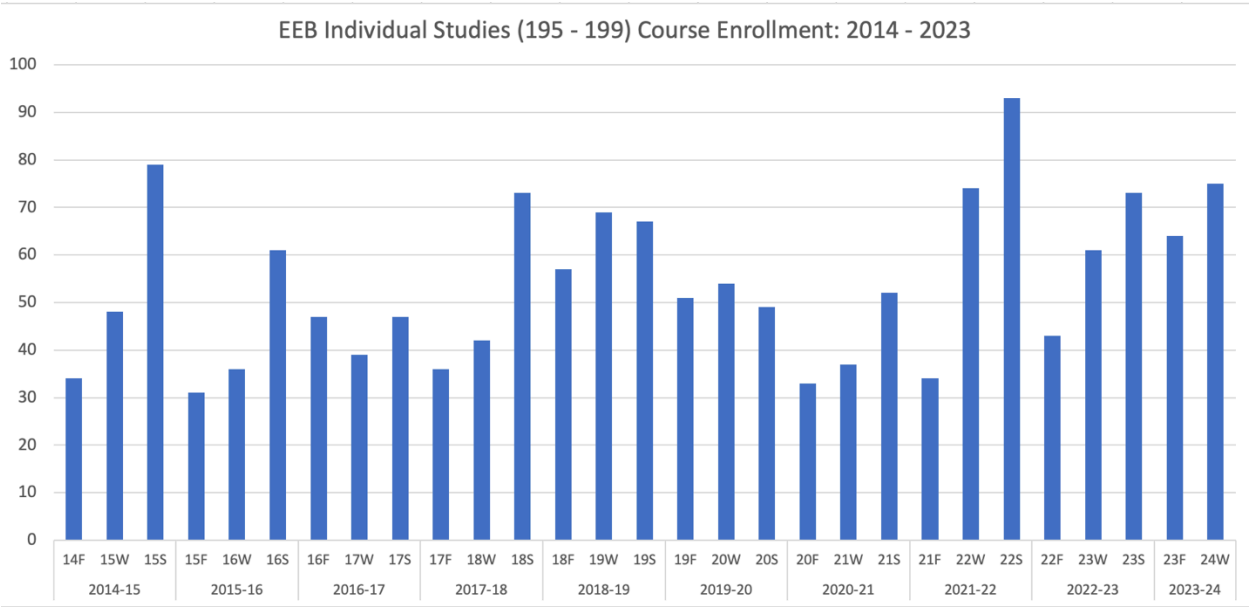


Figure D-6. Enrollment in research-focused individual studies courses (195, 198, 199) in EEB from 2014-2024

B. Advising

One key strength of our department is the advising staff. The undergraduate advisors in the Department of Ecology and Evolutionary Biology provide extensive counseling to assist undergraduates in exploring their interests in the Life Sciences, choosing appropriate majors, finding research opportunities, investigating career paths, and completing their degrees in a timely manner. Because first-year and transfer students interested in the life sciences often enter as Biology majors regardless of their eventual final major, the Ecology and Evolutionary Biology Undergraduate Office serves as the orientation home and initial counseling location for the majority of life science students at UCLA. This pattern results in a larger counseling load than is reflected even by the large number of majors assigned to our department. During summer and fall, the counselors meet with incoming first-year and transfer students. These counseling appointments may be either individual follow-up sessions for students who attended Summer Orientation, or more lengthy sessions for students who were unable to attend the Summer Orientation. In these meetings, counselors cover four-year plans, career planning, research possibilities, and more. This initial effort and the departmental policy of providing individualized four-year plans for each student are labor-intensive, but pay off substantially in students who are better informed, make better decisions, need less remedial counseling later, and have a better educational experience overall.

The Ecology and Evolutionary Biology Undergraduate Office coordinates and organizes three major departmental events each year. During Welcome Week, the undergraduate counselors host a New Student Welcome Event which debuted in Fall Quarter, 2007. All new incoming students, both first-years and transfers, are invited to a presentation where either the Department Chair or Vice Chair for Undergraduate Studies (sometimes both) welcome the new students to the department. Current undergraduates are also invited to give a brief presentation about their UCLA experience, and also share their research activities with these new students. This event also gives new students the opportunity to meet their peers.

The Ecology and Evolutionary Undergraduate Office is also in charge of the Annual Biology Research Symposium and the annual commencement celebration. Until 2024, the Symposium began with a judged poster session (undergraduate and graduate categories), followed by a seminar given by a special guest speaker invited by the Department Chair. The symposium speaker usually had ties to the department, either as an alumnus/alumna, or as a former instructor.

In June 2002, Ecology and Evolutionary Biology held its first departmental graduation ceremony in Royce Hall. Beginning in June 2005 until June 2023, in response to the growth in the numbers of departmental graduates, the ceremony was held in Wilson Plaza. The commencement celebration includes a student speaker, chosen through an audition process, and a keynote speaker, again, someone invited by the Department Chair. One of the graduating students also sings the National Anthem at the beginning of the ceremony and then leads the audience in singing the UCLA Alma Mater at the conclusion of the ceremony. It has meant a

great deal to the students and their families as well as the department's staff and faculty to be able to celebrate the achievements of its students and to recognize them individually.

2014-2019
EEB Commencement Celebration
Wilson Plaza
Traditional ceremony format

Covid
Research Symposium
Due to Covid, there was no in-person Annual Research Symposium in 2020 and 2021. Students were still very eager to participate in the poster session that was part of the Symposium. The EEB Undergraduate Office had a site established via the EEB Undergraduate Counseling list serv (via CCLE) where students could upload their posters. There were the usual categories: Undergraduate and Graduate. The judges, usually EEB alumni/-ae, were given access to view the posters for approximately one (1) week after which they would send their decisions via e-mail to the faculty coordinator or to one of the undergraduate advisors. In 2020, there was not a Symposium Speaker due to the change in the format of the Annual Research Symposium.

In 2021, again, the Annual Research Symposium was conducted similarly to 2020. There was an effort to replicate the in-person experience. Again, the posters were uploaded to the 2021 Research Symposium website where the judges were given the opportunity to view the posters online over the duration of the week. However, on the day that the in-person poster session would usually be held, Wednesday of Undergraduate Research Week, a real-time virtual poster session was held, using the Gather app. In Gather, a virtual conference setting was created with separate conference rooms where the posters were on display. All attendees, including poster presenters, faculty, judges, staff, and guests, would create an avatar which would grant them access to the virtual Research Symposium. Similar to the in-person poster sessions, there was a designated time that the Gather environment was available, e.g., 12:00-4:00PM. Following the poster session, there was a seminar given by the Symposium Speaker, an EEB alumnus/-a, via Zoom. The Research Symposium reverted back to the pre-Covid format In 2022 and 2023.

In 2024, the format of the Annual Research Symposium changed. Instead of being held during Undergraduate Research Week in Spring Quarter, the Symposium became the EEB Research Celebration. In this new format, there would still be an in-person poster session. Instead of an invited Symposium Speaker, there were a series of lightning talks from different members of the EEB community: undergraduate students, graduate students, post-doctoral fellows, and faculty. Another change was that the winners from the poster session were announced after the lightning talks. Initially, the idea was to combine the Research Symposium and the annual end-of-year awards. However, there was only enough time to announce the awards for the poster session.

Another change was that there were other categories for the poster session awards. Previously, there were two (2) main categories, Undergraduate and Graduate, with first, second, and third

place. In 2024, there were a total of eight (8) awards available. The two (2) main categories were renamed the Donald G. Buth Award for Outstanding Poster Presentation, named in honor of Don Buth, the longtime faculty chair of the Research Symposium, which would award potentially six (6) prizes (first, second, and third for undergraduates and graduates). There were two (2) new awards offered, with the winners being either undergraduate or graduate: Robert K. Wayne Award for Outstanding Poster in Conservation Biology and Genomics; Diversity, Equality, and Inclusion Award. The remainder of the event was an informal reception.

EEB Commencement Celebration
For the years 2020 and 2021, there were no in-person Commencement Celebrations due to Covid. Moreover, EEB did not have a real-time, virtual ceremony in these years. On the EEB Commencement page of the EEB website, a virtual EEB Commencement

2020
Recorded Video Messages posted to EEB Commencement website
- · Department Chair
- · EEB Faculty
- · Student Speaker

Graduating students' names were posted to EEB Commencement website

2021
Recorded Video Messages posted to EEB Commencement website
- · EEB Faculty
- · EEB Undergraduate Advisors
- · Student Speaker

Virtual Commencement but not in real time – Commencement Slide Show
- · Students created slides with name, major, UCLA highlights, achievements, future plans, and picture
- · Undergraduate advisors read students' names as slide appeared, with "Pomp and Circumstance" playing in the background

Graduating students' names were posted to EEB Commencement website

2022 & 2023
A return to an in-person Commencement celebration occurred in 2022. The Commencement celebrations in both 2022 and 2023 had the same format prior to Covid.

2024
In-person Commencement celebration but was held at the Los Angeles Tennis Center (LATC) and not in Wilson Plaza. The change of venue was in response to the recent on-campus demonstrations related to the Israel-Hamas war in Gaza. Although both Wilson Plaza and LATC are outdoor venues, LATC has the added security of controlled entrance into the venue. At Wilson Plaza, the general public has access to the venue. Guests are only monitored if they have a ticket for a seat. Moreover, guests who do not have tickets are able to congregate in any public space, e.g., stairs of the Student Activities Center, grassy area behind the seats. This

year, however, all guests had to have tickets in order to enter LATC and attend the EEB Commencement Celebration.

C. Degrees Awarded

Degrees awarded mirror the enrollment figures with a total increase from 412 total degrees awarded in 2014 to 602 in 2019-2020, followed by a decrease to 408 in 2023 (Figure D-7).

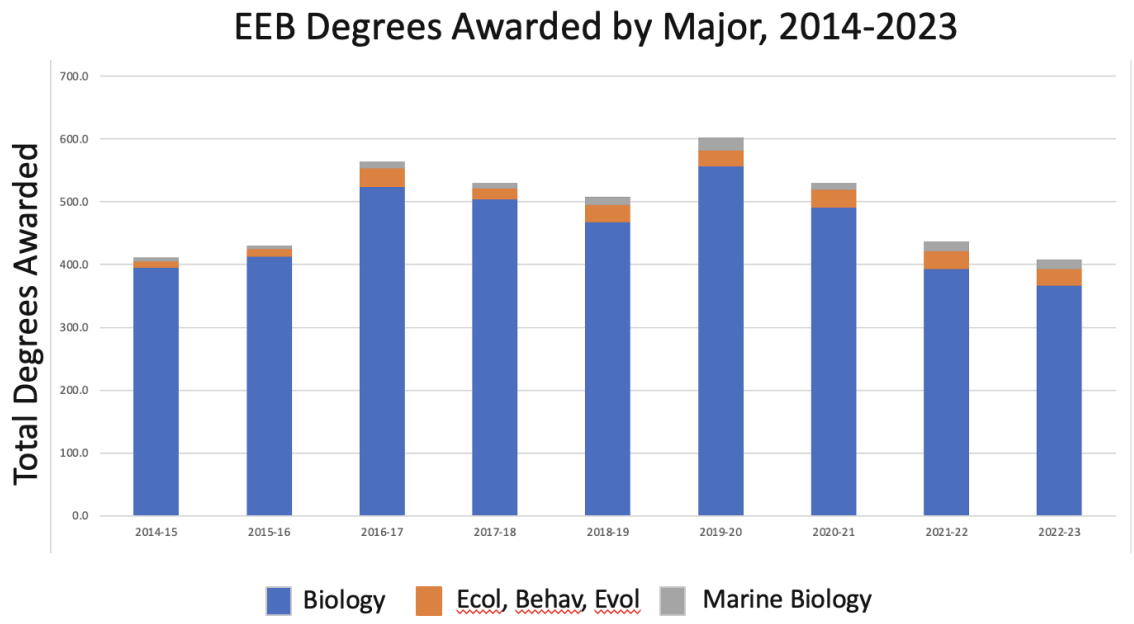


Figure D-7. Total degrees awarded in the three EEB majors from 2014-2024 including i) Biology, ii) Ecology, Behavior, and Evolution, and iii) Marine Biology.

D. Time to degree

For both 4-year admits and transfer students, time to degree has fallen sharply since 2014 (Figure D-8). For 4-year admits, current average time to degree is 11.65 quarters, down from 12.25 quarters is 2014. Similarly the average transfer student completes their EEB degree in 5.92 quarters, down from 6.81 quarters. These gains reflect campus wide trends towards shorter time to degree, but also reflect the success of refining our undergraduate majors, and efforts to offer more laboratory courses, allowing students to complete degree requirements faster.

Overall declines in time to degree across EEB are largely due to declines in degree time for the Biology major and Ecology, Behavior and Evolution major (Figure D-9). In contrast, Marine Biology time to degree has fluctuated greatly since 2014, but has not experienced a downward trend. Slower time to degree for Marine Biology majors may be partly explained by the loss of more than half of EEB marine focused faculty since 2014 (Buth, Zimmer, Zimmer

Faust, Gordon, Kremer), in addition to one of the remaining marine focused faculty (Barber) having other significant service and teaching obligations that precludes teaching in the MBQ.

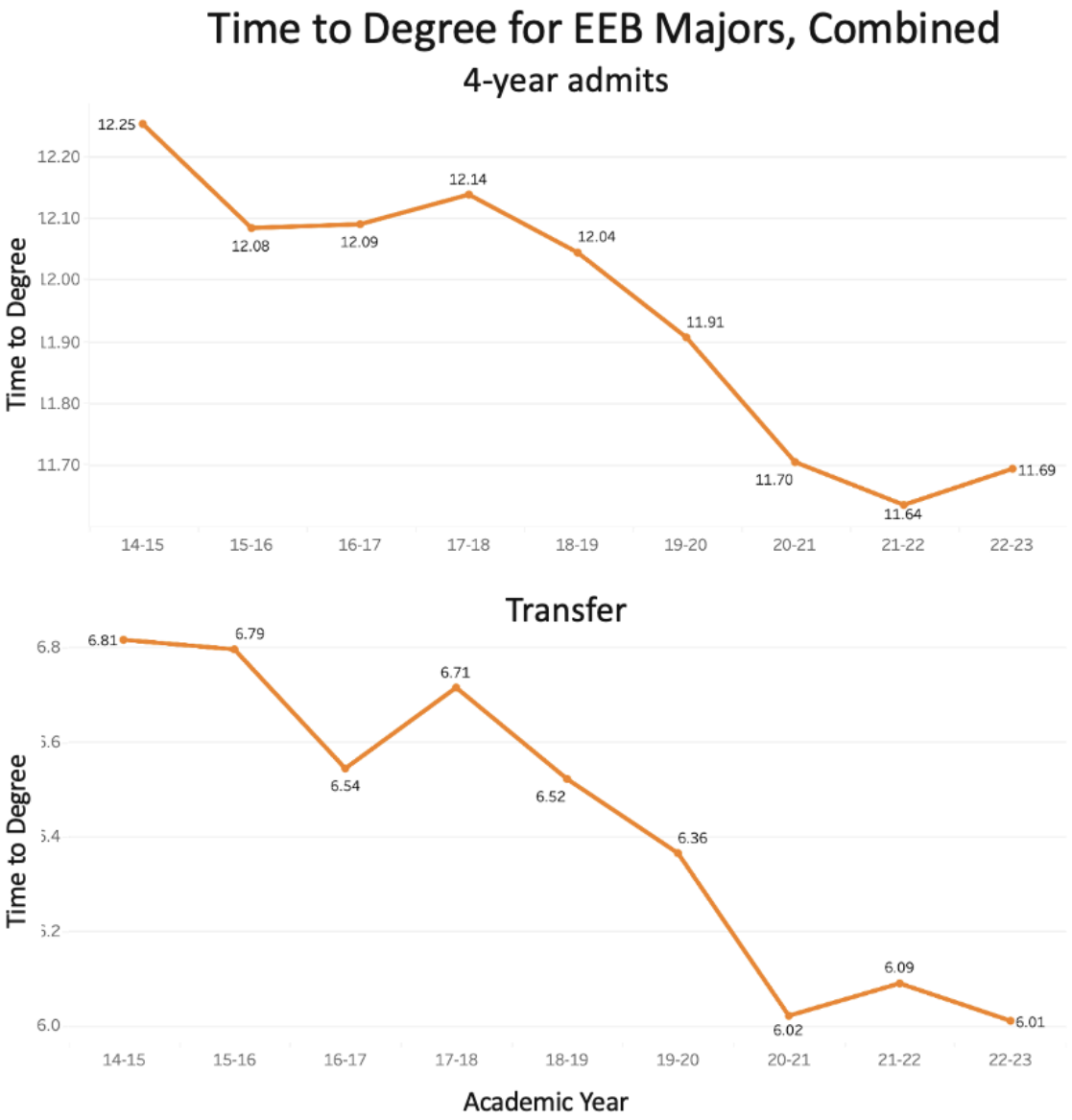


Fig D-8. Time to degree (in quarters) for all three Ecology and Evolutionary Biology majors (Biology, EBE, Marine Biology) over the period 2014–2023 for 4-year admits (top) and 2-year transfer students (bottom).

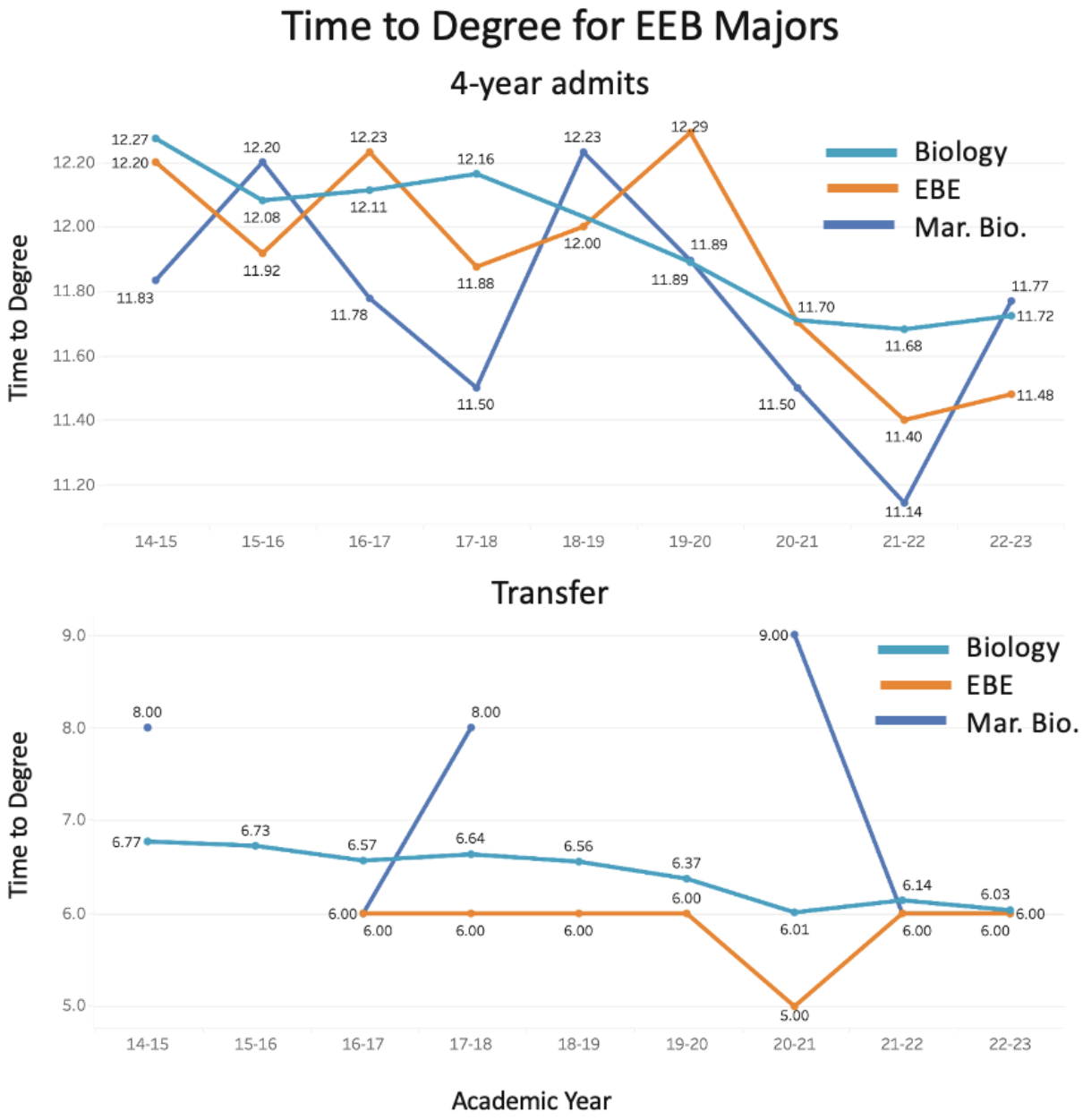


Fig D-9. Disaggregated graduation rates for all three Ecology and Evolutionary Biology majors, Biology, EBE, Marine Biology, from 2014-2023.

E. Departmental graduation rates

4-year graduation rates

As time to degree has decreased, 4-year EEB graduation rates have increased broadly over the review period, spanning cohorts that entered from 2010 to 2019 (Figure D-10). For students that entered in 2019 as one of three declared EEB majors, a total of 88.8% earned a UCLA bachelor's degree (any major) within 4 years, compared to 75.6% in 2010. Similarly, 84.5% of students entering as a declared EEB major completed an undergraduate degree in the College of Letters and Science within 4 years, relative to 69.9% in 2010. This number jumps to 91.5% for the 6-year graduation rate.

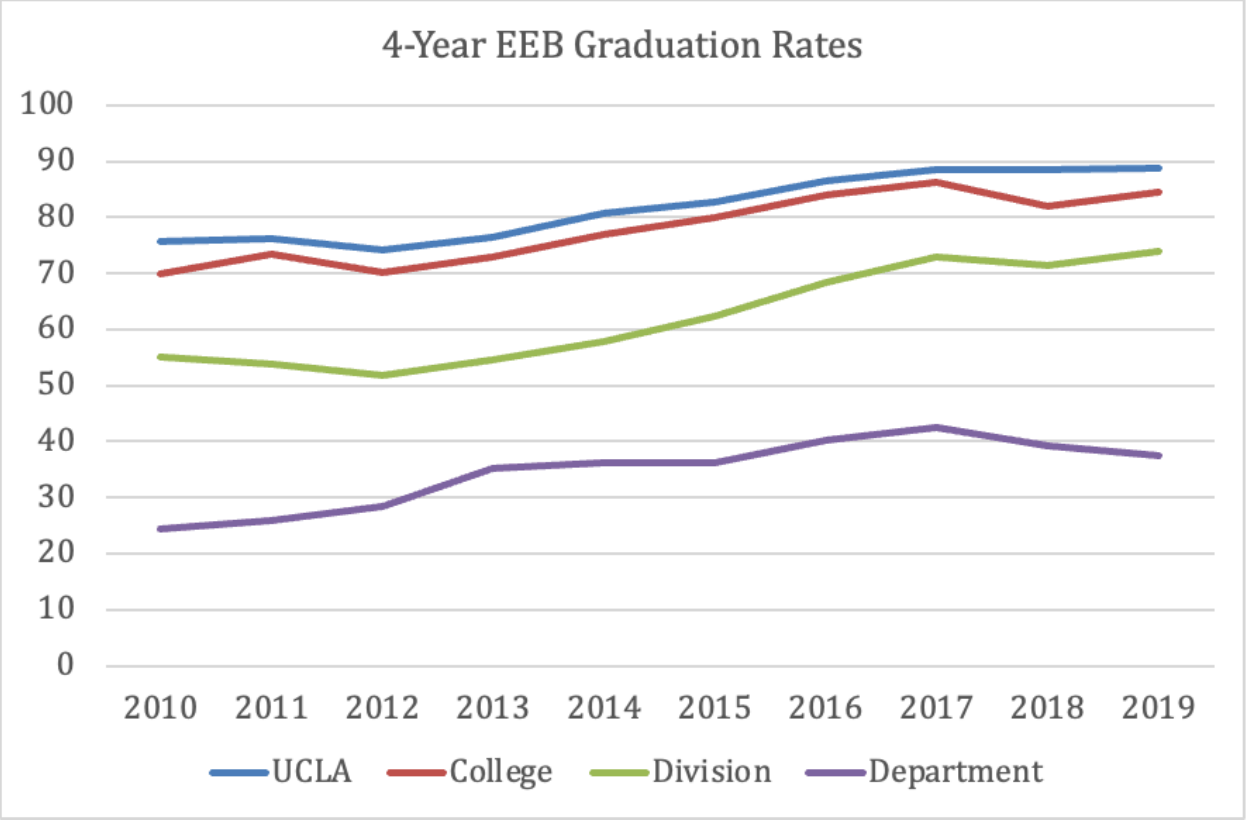


Figure D-10. Four-year graduation rates for students entering UCLA as a 4-year admit. Outcomes are shown, from top to bottom, graduating with a UCLA bachelor's degree, a bachelor's degree in the College of Letters and Science, a bachelor's degree in a Life Science Major, and a bachelor's degree in one of the three EEB undergraduate majors.

However, 4-year graduation rates in EEB departmental majors, for students who entered as EEB department majors, were substantially lower. Graduation rates of EEB majors increased over the review period, but only 37.4% of 4-year admits entered UCLA as declared EEB majors graduated in an EEB major in 4 years, compared to 24.4% in 2010. This figure increases to 43.8% for the 6-year graduation rate. Although this rate represents a relatively low percentage of declared EEB majors completing an EEB major, a much larger percentage of entering EEB majors complete a degree within the Life Sciences. In total, 73.8% of EEB majors entering in

2019 completed a Life Science degree vs 55.1% for the 2010 entering cohort (75/3% for 6-year graduation rate).

Examination of graduation rates, disaggregating the three EEB undergraduate majors shows that graduation rates (i.e. graduating in the major declared at entry) was lowest in the Biology major, with only 35.4% of Biology majors entering as first-year students in 2019 graduating within 4 years (Figure D-11). In comparison, 55% of Marine Biology majors entering in 2019 completed their degree in 4 years, and 65% of Ecology, Behavior and Evolution majors did the same. All of these numbers represent a substantial increase over the entering 2010 cohort.

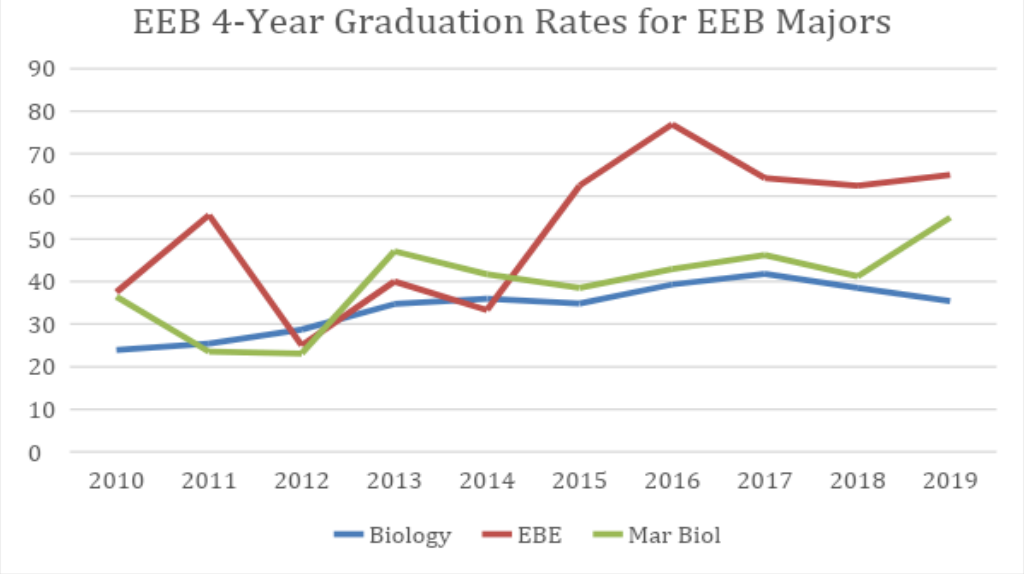


Figure D-11. Four-year graduation rates for students entering UCLA as a 4-year admit in each of the three EEB majors, Biology, EBE, and Marine Biology

Overall, EEB had the lowest 4-year departmental major graduation rates across 7 units of Life Sciences, with only 37.5% of students declared in EEB majors graduating in those specified majors within 4 years as of 2019 (Figure D-12). Although this represents an improvement over the 24.4% rate in 2010, it is substantially less than other LS departments like Psychology (69.4%), Institute for Society and Genetics (59.6%), MCDB (54.9%), Integrative Biology and Physiology (53.6%), Neuroscience interdepartmental major (48.6%) and Computational and Systems Biology (41.7%).

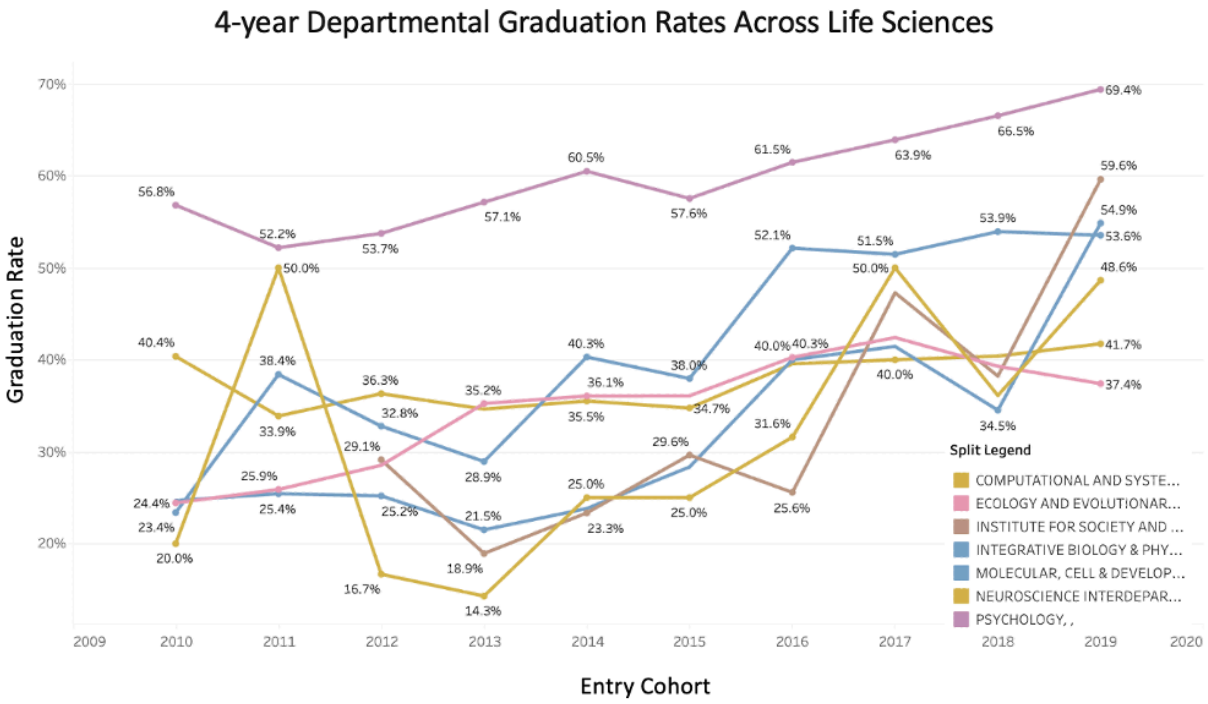


Figure D-12. Comparison of four-year graduation rates across 7 degree granting units of Life Sciences for students entering UCLA as a 4-year admit. Data reflect the percentage of students who enter UCLA declared as a specific Life Science major complete their degree in their original declared major

These numbers improve substantially for Divisional 4-year graduation rates, (Figure D-13). In total, 73.8% of 4-year students entering UCLA declared EEB majors in the 2019 cohort completed a Life Sciences degree within 4 years, a number that, while still lower than most Life Science degree granting units, is more comparable. The higher 4-year graduation rate for the more specialized EBE and Marine Biology majors, relative to the Biology major, suggests that these differences relative to other Life Science departments may result from the vast majority of our majors being Biology majors. Students entering a more generalized major may be more likely to switch to more specialized majors during the course of their studies as exposure to more specific fields of biology may inspire them to switch to a more specialized major. In contrast, other Life Science degree granting units offer majors that are far more specialized (including “competitive majors” that have highly selective entry requirements), potentially attracting students who are already more focused on a specialized field of study, reducing the likelihood that they leave these majors.

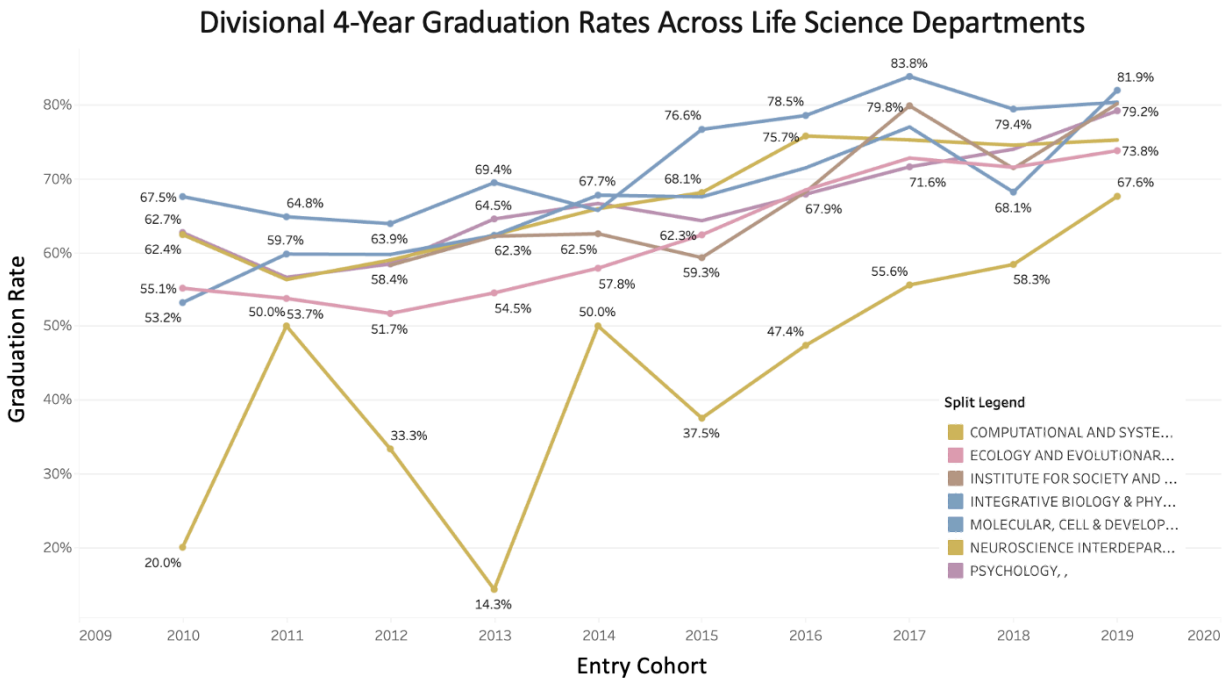


Figure D-13. Comparison of four-year graduation rates across 7 degree granting units of Life Sciences for students entering UCLA as a 4-year admit. Data reflect the percentage of students who enter UCLA declared as a specific Life Science major complete their degree in any undergraduate major in the Division of Life Sciences.

2-year graduation rates

Overall 2-year graduation rates for transfer students entering EEB majors between 2010 and 2021 also show an upward trajectory toward higher graduation rates, and were markedly higher than graduation time for 4-year students (Figure D-14). For transfer students that entered in 2021 as one of three declared EEB majors, a total of 89.8% earned a bachelor's degree in some UCLA undergraduate major within 2 years, compared to 64.0% in 2010. Similarly, 87.8% of transfer students entering as a declared EEB major completed an undergraduate degree in the College of Letters and Science within 2 years, relative to 57.0% in 2010. Even for transfer students graduating in the specific EEB major they declared at entry, 76.5% completed that intended degree in 2 years, compared to 28.4% in 2010, showing a strong improvement in serving the needs of transfer students at UCLA. There are two potential explanations for this pattern, which are not mutually exclusive. First, transfer students enter UCLA having had 2 more years to explore and refine their choice of major. As such, it is less likely that transfer students will change majors after entry, a pattern supported by the data. In addition, another key difference between transfer students and 4-year students at UCLA is that transfer students take their core curriculum in biology, calculus, chemistry and physics at community college, where classes are smaller and more personal relative to the large introductory lecture courses at UCLA. As such, the higher success in transfer students relative to 4-year admits may be the more approachable introductory STEM curriculum at community colleges.

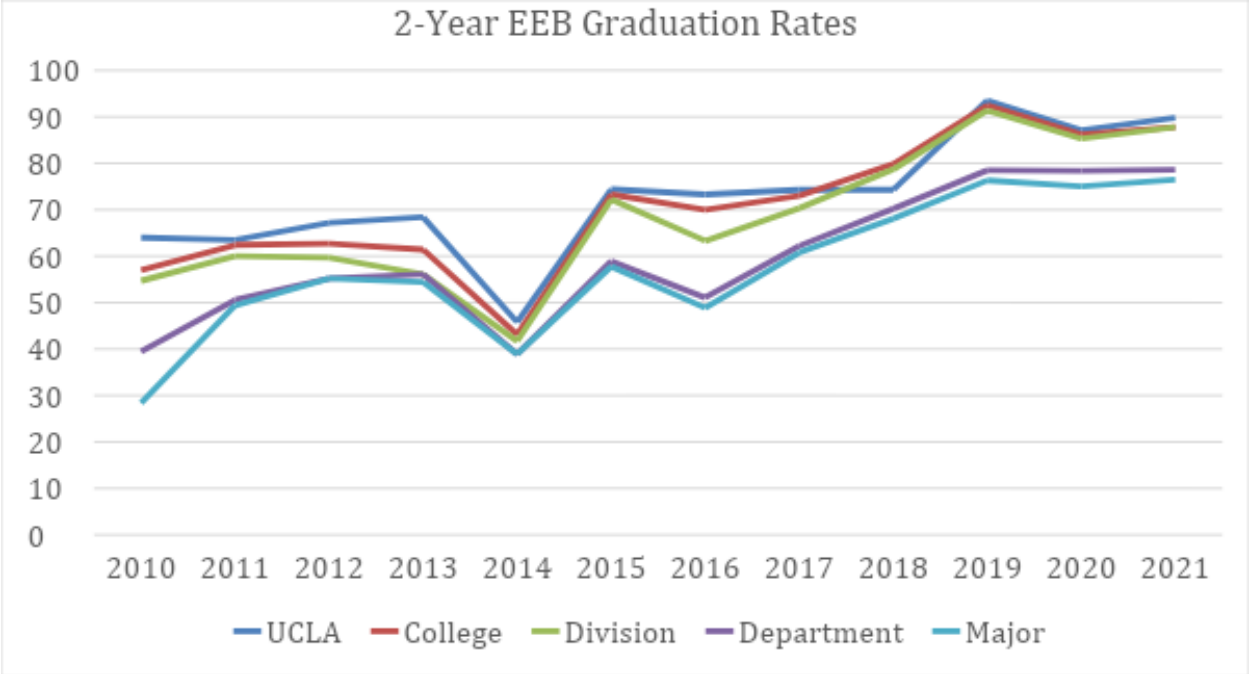


Figure D-14. Two-year graduation rates for students entering UCLA as a transfer student. Outcomes are shown, from top to bottom, graduating with a UCLA bachelor's degree, a bachelor's degree in the College of Letters and Science, a bachelor's degree in a Life Science Major, and a bachelor's degree in one of the three EEB undergraduate majors.

No data is presented differentiating Biology, EBE, and Marine Biology majors because relatively few transfer students select EBE or Marine Biology.

The 2-year graduation rates of transfer students completing EEB majors, having declared EEB majors at entry, has grown substantially from 39.5% in 2010 to 78.6% in 2021 (Figure D-15). Not only is the current 2-year graduation rate of transfer students in EEB much higher than that of 4-year admits, it is also higher than several other Life Science departments, and is broadly comparable to the best performing Life Science departments.

Similarly, EEB has a very high 2-year divisional graduation rate, with 87.8% of transfer students entering as declared EEB majors complete a Life Science degree in 2 years (Figure D-16). This represents a substantial increase over the 54.7% 2-year divisional graduation rate in 2010.

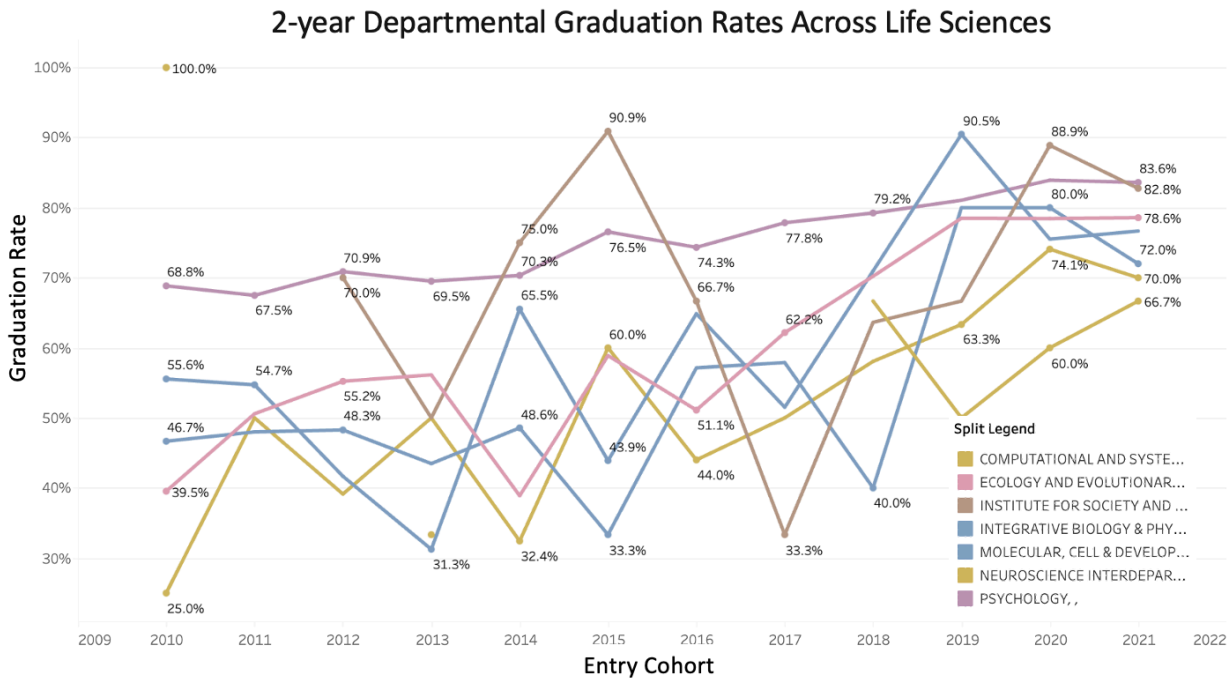


Figure D-15. Comparison of two-year graduation rates across 7 degree granting units of Life Sciences for students entering UCLA as transfer students. Data reflect the percentage of students who enter UCLA declared as a specific Life Science major complete their degree in their original declared major.

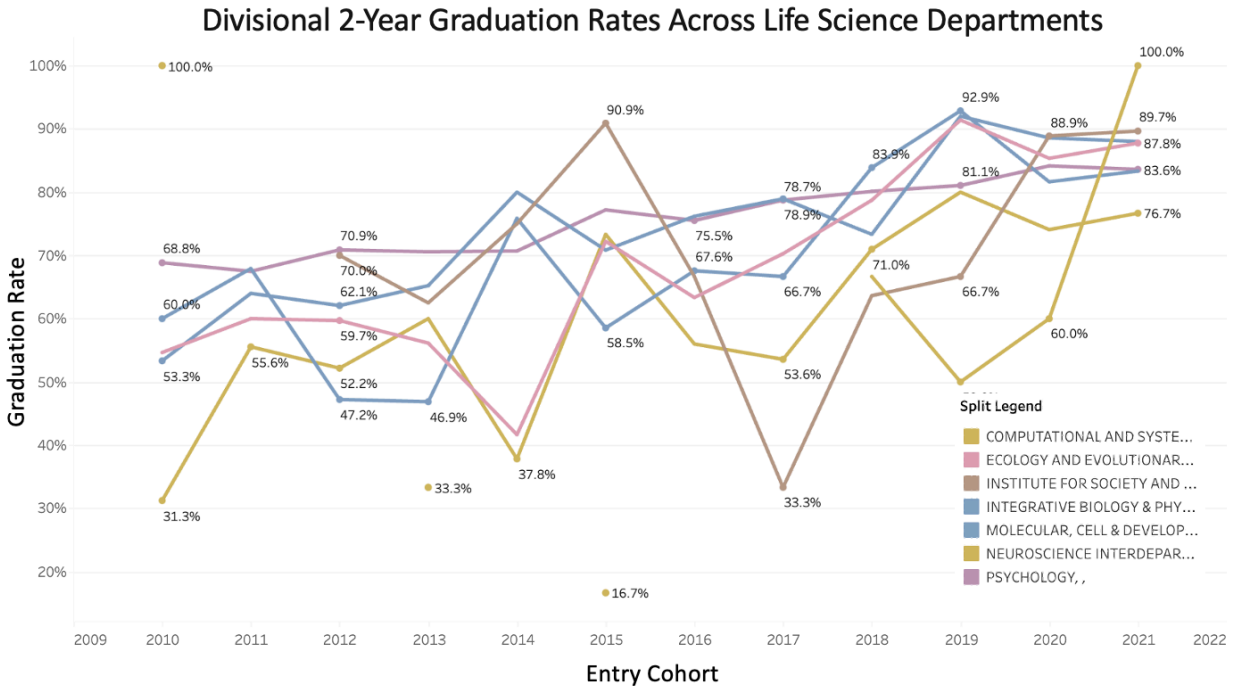


Figure D-16. Comparison of four-year graduation rates across 7 degree granting units of Life Sciences for students entering UCLA as a 4-year admit. Data reflect the percentage of students who enter UCLA declared as a specific Life Science major and complete their degree in any undergraduate major in the Division of Life Sciences.

Major Migration

Degree completion rates in EEB majors are partially explained by students changing majors. For example, of 4243 students who entered UCLA as declared Biology majors from 2014-2024, the majority (63.5%) earned a BS in biology, while a small percentage (2.0%) switched into, and graduated in, other EEB majors (either EBE or Marine Biology) (Figure D-17). However 24.8% of entering Biology majors switch to, and graduate in, other Life Science majors; 8% switch to, and graduate in, other Physical Science Majors, while 1.7% graduate in other majors across UCLA, ranging from Anthropology, Climate Science, and Engineering. Similar patterns are seen in the Ecology, Behavior and Evolution major, as well as Marine Biology, except a substantial portion of students that leave these majors (40%, and 39.5% respectively), complete their bachelor's degree in an alternate EEB major. Given that major specific graduation rates are calculated based on the percentage of students entering as a declared major who graduate with that same declared major, this substantial movement of entering EEB majors into other majors may partially explain the surprisingly low major degree completion rates combined with the much higher departmental, divisional, college, and UCLA graduation rates of entering EEB majors.

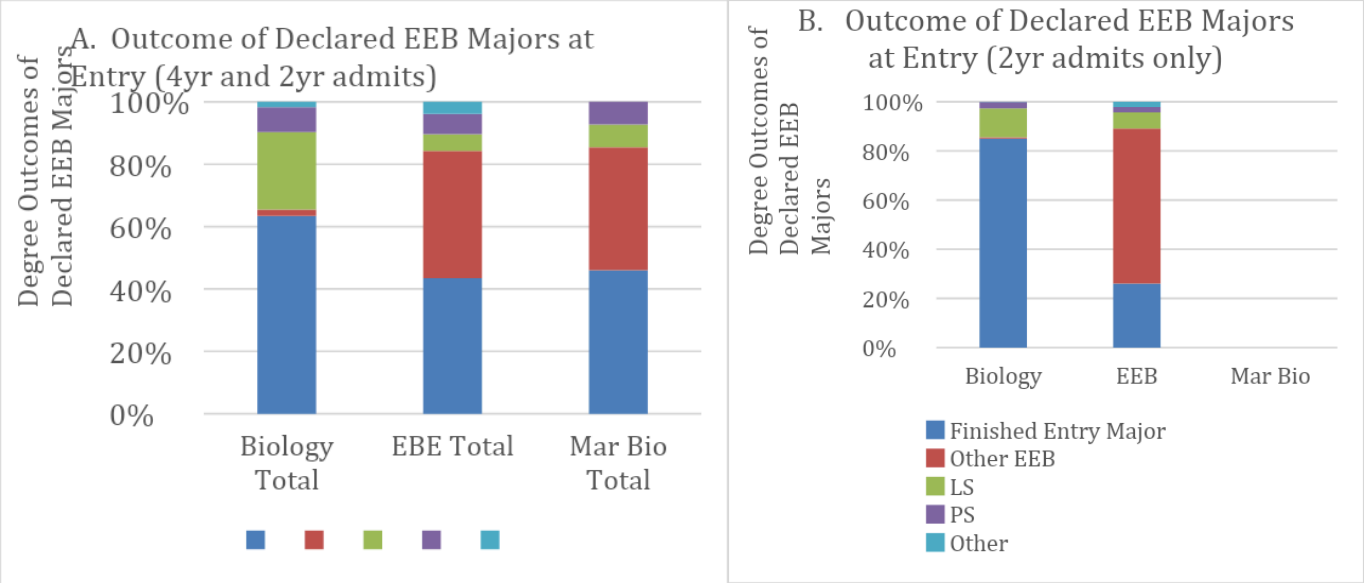


Figure D-17. A. Final degree outcomes of 4551 total students (4-year and 2-year admits) who entered UCLA and graduated between 2014-2024 as one of three declared EEB majors, Biology; Ecology, Behavior, and Evolution; and Marine Biology. B. Final degree outcomes of 872 transfer students entered UCLA and graduated between 2014-2024 as one of three declared EEB majors, Biology; Ecology, Behavior, and Evolution; and Marine Biology.

Although a substantial number of students who enter UCLA declared in one of three EEB majors will graduate in a major outside of the EEB department, there is a similarly large influx of students from other departments into EEB majors. For example, of 9131 students who graduated in the Biology major from 2014-2024, 78.1% of them started in the Biology major, while 6.8% started in another EEB major, 14.0% from other Life Science majors, 4.5% from Physical Science Majors, and 2.8% from other UCLA undergraduate majors as diverse as

Anthropology, Education Electrical Engineering, Music, and World Arts and Dance (Figure D-18). Similar patterns are seen in the EBE and Marine Biology majors, although a substantial portion (14.9%, and 8.8%, respectively) come into these majors from other EEB majors. In total, 1935 students who graduated in the Biology major, transferred into this major from outside of EEB. This large number of students transferring into the Biology major is likely a function of the broad nature of the major, and the flexibility built into fulfilling the major requirements. As such, particularly for other Life Science and Physical Science students, it is likely that most STEM classes they have completed will fulfill the Biology major requirements, making it simple to change to this major. As expected for more specialized majors, the number of students who finished with EBE (194) and Marine Biology (75) that began their studies at UCLA is much smaller than in Biology.

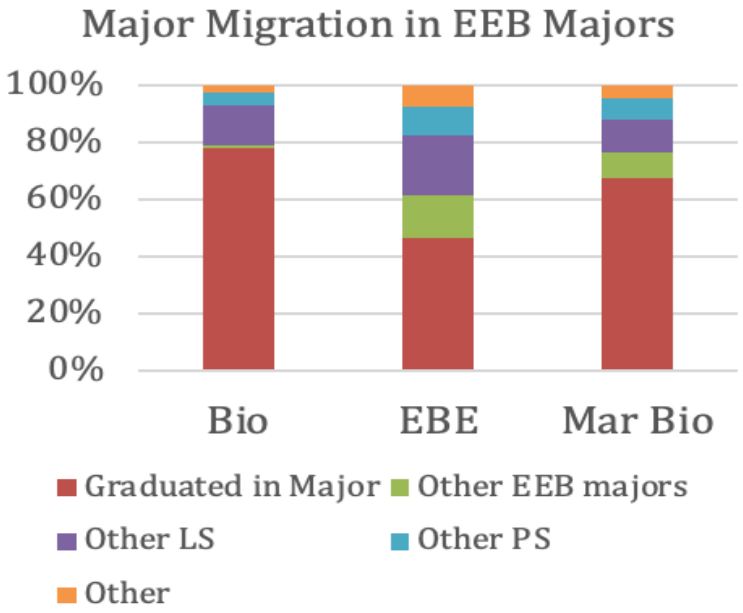


Figure D-18. Entry majors or major Division for 9951 students who completed a degree in one of three EEB majors (Biology; Ecology Behavior and Evolution; Marine Biology) between 2014-2024.

5. Student satisfaction with the majors and instruction

Since 2014, all UCLA graduating seniors have been systematically surveyed about their experiences at the university and their satisfaction with their degree programs. The results for Ecology and Evolutionary Biology contain both positive and negative elements (Appendix D10 – data from 2014 forward was available), with the lowest level of satisfaction given to the availability of courses in the major (e.g., 59% from Biology major respondents reported being satisfied compared to 67% from Life Sciences majors overall). On the positive side, the vast majority of UCLA students were “satisfied” or “very satisfied” with major aspects of their experience in the Department of Ecology and Evolutionary Biology, including curriculum,

intellectual challenge, and quality of faculty instruction (in all cases 83%-88% of the respondents reported that they were pleased).

In subsequent years, Ecology and Evolutionary Biology has improved in these metrics relative to other Life and Physical Science departments. There are smaller differences in student satisfaction among the departmental majors in comparison with the life sciences overall as well as with the students who participated in the survey. For instance, in the area of “intellectual challenge by faculty,” the percentage of Biology majors who were “satisfied” or “very satisfied” was 88% in comparison with 91% of all life sciences majors and 91% of all students in the College of Letters and Science.

The recent revisions of the department majors should improve satisfaction levels in these areas in many ways. The increased flexibility of the Biology major, improved advising, and our efforts to increase course availability, provide students a wider range of options for completing their degree. Moreover, the Division of Life Sciences, as a whole, is planning a major overhaul of the lower division major curriculum in response to the need for life scientists to acquire more quantitative skills.

The creation of a “two-tier” system of upper-division courses has allowed large classes such as EEB 100 (Introduction to Ecology and Behavior) to introduce topics and provide plenty of course capacity. However, having a few large classes also allows the department to continue (and increase) offerings of smaller, more advanced classes. This diversity allows the majority of our department’s majors to experience a small class setting and get to know some of the professors (and vice versa!), with positive carryover into the quality of instruction and curriculum assessments.

While offering a larger number of smaller classes has a positive impact for student satisfaction, it will have negative impacts on departmental finances. Under the new Bruin Budget model, departmental financial support is directly tied to the numbers of student credit hours taught by departmental faculty. As such, high-impact capstone courses like the MBQ, FBQ and FMBQ where 2 faculty teach 15-16 students yield a total of 256 credit hours (16 students times 16 units), or 128 student credit hours per faculty. In contrast, a 5-unit 200 student lecture course taught by a single faculty member, common in other Life Science departments, yields 1000 student credit hours. As such, one of the hallmarks of the EBE and Marine Biology majors will become increasingly challenging to offer, as these courses are expensive to offer, but also bring in the least revenue under the new Bruin Budget model.

6. Learning Objectives

A. Description

During the last WSCUC re-accreditation process, EEB developed 4 learning outcomes for each of our 3 Majors.

a) Biology Major Learning outcomes

1. Acquire a broad foundation in basic biological concepts and principles across different levels of biological organization from molecules to ecosystems.
2. Effectively communicate scientific information both orally and in writing.
3. Demonstrate an understanding of the processes involved in generation of new knowledge, including the scientific method, data collection and analysis.
4. Be able to critically evaluate scientific concepts that are presented in a diversity of venues, from scientific articles to the popular press.

b) Capstone Major (Marine Biology and Ecology, Behavior, and Evolution) Learning Outcomes

1. Demonstrate broad-based knowledge of the fundamentals of Ecology, Behavior and Evolution or Marine Biology acquired through coursework, including general knowledge as well as developing skills in library research, interpreting data, synthesis, and scientific writing.
2. Utilize the current primary scientific literature, including searching databases, identifying appropriate sources, and reading and understanding papers. This includes understanding key questions and hypotheses, interpreting results and conclusions, and discriminating quality through critique.
3. Use knowledge gained in the classroom and during discussions to conceive and execute their own project, including developing their own questions and hypotheses, designing an appropriate theoretical or empirical/experimental approach, executing that approach, and analysis and interpretation of data.
4. Exhibit strong teamwork and problem-solving skills during the entire capstone experience. As both of these are key to the scientific process, especially in field-oriented research, we prioritize training in both of these life-skills.

B. Curricular Support for Learning Outcomes

For our two capstone majors, Marine Biology and Ecology, Behavior, and Evolution majors, Learning outcomes are focused around the capstone experience, and, to a lesser extent, around the coursework that prepares students for this capstone experience. The first LO is for students to “demonstrate broad-based knowledge of the fundamentals of Ecology, Behavior and Evolution or Marine Biology acquired through coursework, including general knowledge as well as developing skills in library research, interpreting data, synthesis, and scientific writing”. This knowledge and these skills are part of every required fundamental course as well as the electives that can be used to fulfill the requirements of these capstone majors.

For the Marine Biology Major, the fundamentals are covered in EEB 109, Introduction to Marine Science. The laboratory complement to this course, EEB 109L, requires that students develop and carry out a major project that requires library research, interpreting data, synthesis, and scientific writing. Subsequent courses are chosen from five categories of courses: 1) EEB lab courses where students learn hands-on skills and data interpretation and synthesis, 2) EEB organismic biology or physiology where students gain general knowledge, 3) EEB ecology or behavior where students gain general knowledge and writing skills, 4) Evolution where students gain general knowledge and set this knowledge in context of evolution, and 5) an addition choice from physical, chemical, or geological oceanography where students learn to set their knowledge of biology into the larger context of ocean science.

For the Ecology, Behavior and Evolution major, the fundamentals are taught in EEB 100, Introduction to Ecology and Animal Behavior and EEB 120, Evolution. The laboratory component of this course, EEB 100L, is also centered around student projects and writing. Subsequent courses are chosen from four categories of courses: 1) courses in morphology and systematics where students get the background needed to understand the diversity of a clade, 2) courses in physiology where students gain general knowledge, 3) courses in ecology, behavior, and evolution where students gain background in specific disciplines, and 4) an additional choice from a long list of upper division electives, where student can expand their discipline-based investigations.

For both of these capstone majors, successful mastery of Learning Objective 1 is met through the foundation course (EEB 100 and 109 for EBE and MB, respectively) and Learning Objectives 2-4 are accomplished in the capstone experience, participation in either a Field Biology Quarter or Marine Biology Quarter. These capstone experiences are comprised of one entire quarter of courses devoted to field research (or a recognized equivalent from outside UCLA). Students carry 16 units of coursework that is spread over two to four courses, each with a significant research component. The site of the Marine Biology Quarter (MBQ) currently alternates between the Richard B. Gump South Pacific Research Station, Moorea, French Polynesia, which is administered by the University of California, Berkeley, and the Wrigley Marine Lab on Catalina Island, CA administered by the University of Southern California. For the Field Biology Quarter (FBQ), students travel either internationally to field sites in Australia, Ecuador, Nicaragua, or Thailand, or locally to Californian or Mexican desserts. Please refer to Section IV.A.3.g for more information about these programs.

C. Assessment of Learning Outcomes

An initial assessment Plan for the Learning Outcomes (Appendix D11) was approved by the faculty by vote (18 Yes, 0 No, 1 Abstain, 6 did not vote) on April 29th and 30th 2014. This plan was developed with guidance from assessment experts that included:
Jennifer A. Lindholm, Ph.D., Assistant Vice Provost, Division of Undergraduate Education
Erin R. Sanders, Ph.D., Director of the Center for Education Innovation in Life Sciences (CEILS), MIMG Adjunct Assistant Professor, Instructional Labs Academic Coordinator
Joanne Valli-Meredith, Ph.D., Director, Evaluation and Educational Assessment, UCLA, Office of Instructional Development.

Although we approved this assessment plan, it has yet to be operationalized. Central to this issue is that the assessment plan was viewed by the above experts as overly ambitious. This plan largely focuses on graduating (or nearly) seniors, but it was unclear how to make it practical, and not overwhelming both for students and for the department to administer. Similarly, it was unclear how to embed these assessments into our curriculum in a way that would provide meaningful results without having to incentivize students to participate in the assessment. The final issue was trying to develop this in a way that would provide useful information to faculty, so that curricular needs could be modified to assess any potential shortcomings in the outcomes of the learning objectives.

The plan, following the results of our last 8 year review, was to choose one LO for each of our three majors and develop a plan to assess it in the next four years. The idea being that we can use this preliminary data and what we learn from the process to refine our preliminary plans for the rest of the LOs (Appendix D11). However, progress towards this plan was slowed by substantial leadership turnover, of both chairs and vice chairs, and then the COVID-19 pandemic, where all efforts were invested in transitioning between in-person and remote learning, and rising to the challenges of the day. Although we have returned largely to normal operations of in-person instruction, educational activities have been substantially disrupted by graduate student strikes and protests. We are in need of a period of departmental stability in order to continue to progress with this important goal.

7. Summer Enrollment

Overall summer instruction enrollment has declined from a high of 987 in 2016 to 766 in 2023, marking a 29% decrease. This pattern runs counter to an overall increase in summer session enrollment in other non-Psychology Life Science departments. As of summer 2024, EEB offered a total of 9 lecture and 2 laboratory courses, largely taught asynchronously to accommodate more varied student summer schedules, and most of these offerings are near enrollment capacity. Decline in overall summer enrollment may be a result of overall decline in time to degree; with more students completing degrees under 12 quarters (4 year admits) and 6 quarters (transfer students), there may be less need to take summer school to maintain normative degree progress. However, because summer instructional revenue provides substantial financial resources to the department, declines in summer instructional revenue could have a major impact on departmental finances.

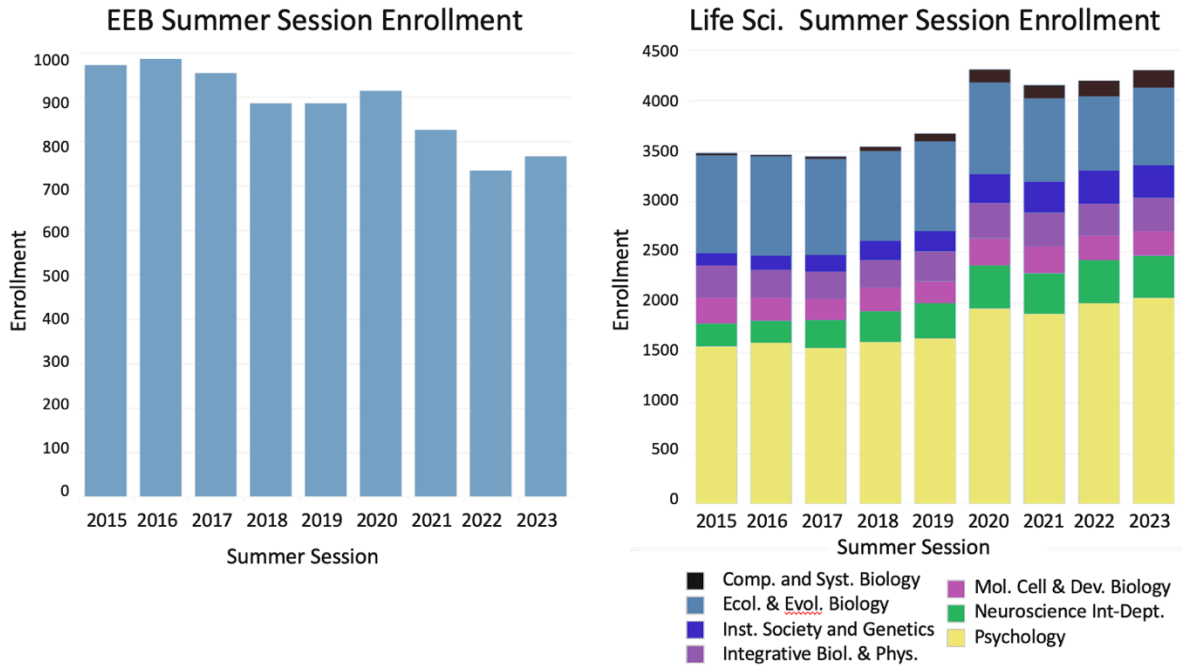


Figure D-19 Summer session enrollment for EEB and Life Science Departments for academic years 2014-2024

8. Other Contributions to Undergraduate Education

A. Contributions to the Life Sciences Core

Faculty and lecturers in Ecology and Evolutionary Biology contribute to the teaching mission of the Life Sciences Core through teaching the following courses.
LS 7B: Genetics, Evolution and Ecology
LS 40: Statistics of Biological Systems
LS 110 Career Exploration in Life Sciences

LS 7B is a required core course for all life science majors, so most students that take it are not our majors. In Winter Quarter, 2024, 1474 students enrolled in LS 7B across 5 sections. This course is taught by both ladder faculty, but more frequently by EEB lecturers. There is a push from the LS Core to have more ladder faculty participation in teaching of the LS series.

LS 40 is a Life Science-focused introduction to statistics. Unlike traditional formula-based statistics courses, LS 40 emphasizes computer simulation to allow for deeper understanding of statistical concepts. To do this, students learn a simple programming language, increasing familiarity and confidence with computational biology. This class is taught by EEB faculty once per year, typically serving 100 students per offering.

LS 110 is focused on preparing Life Science majors for a diversity of careers. Through career panels and discussions, students learn about a diversity of life science career options. Students also work on practical skills like CV development and writing cover letters. This class is taught multiple times a year by EEB faculty, serving approximately 100 students per offering.

In addition to the Life Science Core, EEB faculty members also participate in a General Education cluster, which are year-long courses (three quarters) taught by teams of faculty on a single, integrated theme. GE 70, the Cosmos Cluster, has included an instructor from our department since its inception (Drs. Van Valkenburgh, and more recently Jacobs). The College has made these cluster courses one of the centerpieces of the first-year experience at UCLA, and Ecology and Evolutionary Biology is proud to have been a part of this successful innovation.

EEB also plays a significant role in teaching Research Practice 97Xa and 97Xb, with EEB faculty member, Paul Barber, serving as the director of the Program for Excellence in Education and Research in the Sciences (PEERS). PEERS is] a 2-year academic support program for underrepresented and underserved STEM majors in Life Sciences and Physical Sciences, and 97Xa helps students transition effectively from high school to college, while 97Xb focuses on career exploration and preparation for engaging in undergraduate research. Combined, these courses support approximately 350 students per year.

B. EEB Minors and Individual Courses Routinely Taken by Other Majors

We offer a minor, Conservation Biology, which is frequently taken by students outside of our department. The Conservation Biology Minor is popular with students in the Environmental Science Major (47% of total majors), and is offered as one of the main tracks for this major. Other non-EEB students constitute 24% of the major, including students from Geography/Environmental Science, Physics, Economics, and Sociology. The Evolutionary Medicine Minor is very new (last year), and, at present, is composed of all Biology Majors in our department. However, we predict this will change quickly and include other life science majors.

Several EEB courses draw substantial enrollments from non-EEB majors, reflecting the department's interdisciplinary reach and contribution to broader undergraduate education. For instance, **EEB 100: Introduction to Ecology and Animal Behavior** and **EEB 129: Animal Behavior** typically host Psychobiology and Psychology students, who occupy 10–25% of the available seats. Similarly, **EEB 119A: Mathematical Models in Ecology and Evolution** and **EEB 119B: Advanced Modeling in Ecology and Evolution**, taught by Jamie Lloyd-Smith, have seen a majority of students from non-EEB majors in recent years. These include Computational and Systems Biology (CASB), pre-CASB, Environmental Science, Mathematics/Applied Science, Applied Mathematics, and other Life Sciences majors.

C. Honors and Fiat Lux Courses

Honors courses are taught generally as special sections affiliated with regular Ecology and Evolutionary Biology courses, and are taken by undergraduates in the College Honors Program. Several departmental faculty are regular volunteer participants in Fiat Lux courses, one-unit

small group seminar courses offered primarily to first-year students. The UCLA Office of Instructional Development evaluates all of these courses.

D. Center for Education Innovation and Learning in the Sciences (CEILS)

The Center for Education Innovation and Learning in the Sciences began in 2014, to create a community of instructors that advance teaching excellence, assessment, diversity, and scholarship. CEILS serves as a resource to benefit the campus goals, to close degree attainment gaps, and to support student success. CEILS works with individual departments on curricular revisions across the sciences to reduce inequities through implementation of evidence-based and inclusive teaching strategies, through multiple levels of engagement: faculty development, undergraduate curricular initiatives, faculty learning communities, and classroom observation projects.

Inclusive and anti-racist teaching has shaped the evolution of CEILS becoming the umbrella under which the educational initiatives and pedagogical principles of faculty development efforts now fall and include workshops, consultations, resources, and other pedagogical training opportunities. The Department of Ecology and Evolutionary Biology has consistently engaged with CEILS for tailored workshops to suit their specific needs. Examples include a CEILS staff workshop to improve interpersonal workplace relationships, and three workshops for faculty on Interrupting Microaggressions, How to Effectively Apologize, Effective Mentoring, and Creating a Compact. CEILS also supports inclusive curriculum development and worked with EEB faculty to revise and instruct the Graduate Student Introduction to EEB course (200B) using a design justice approach. EEB faculty have also participated in CEILS professional development workshops such as the annual Faculty Workshop for Inclusive Teaching, the Becoming and Antiracist Educator Conversation Series, and the Summer Course Design Institute.

E. Ecology and Evolutionary Biology Departmental Honors Program

Students in the four majors in Ecology and Evolutionary Biology who wish to graduate with departmental honors must have a final grade point average of at least 3.4 in their upper-division major courses and their overall UCLA coursework. Those wishing to graduate with departmental highest honors must have a final GPA of at least 3.6 in both these categories, and must complete EEB 198A and 198B (Honors Research), writing a research report/thesis at the end of the 198B quarter. Approximately 70 students graduate with departmental honors each year, and about eleven with departmental highest honors. See Table 2 below.

Table 2: Departmental Honors and Highest Honors Awarded 2006-2014

Year	Honors	Highest Honors
2014-15	55	2

2015-16	152	24
2016-17	296	24
2017-18	388	37
2018-19	426	32
2019-20	432	14
2021-21	88	9
2022-23	103	8
2023-24	505	35
Average	**272**	**21**

F. Annual Biology Research Symposium

The EEB Undergraduate Office sponsors and organizes the department-wide annual Biology Research Symposium each spring. This event brings together the department's faculty, graduate students, postdoctoral fellows, and undergraduates to discuss their research. Financial awards are given to undergraduate and graduate researchers for outstanding posters and can be used to support attendance at scientific conferences.

G. Course Development

Courses are developed by individual faculty or teams of faculty and are evaluated by students every quarter they are offered. The UCLA Center for Educational Assessment evaluates all of these courses.

H. Promotion of Teaching Excellence.

Since 2014 four EEB faculty have won the UCLA Academic Senate Distinguished Teaching Award (Barber, Buth, Grether, Lloyd-Smith). Including two previous winners (Blumstein and Fong), 5 active EEB faculty are winners of this significant campus-wide award.
To further promote teaching excellence, the EEB department is among a select group of UCLA departments working to pilot a novel approach to teaching evaluation, Holistic Evaluation of Teaching. Through this program, in addition to traditional student teaching evaluations, faculty are evaluated on their implementation of research-based pedagogical best practices.

I. Funding for Undergraduate Research.

The department has utilized the following funding sources to award scholarships for undergraduate research:

- Whitcome Fellowship (Summers, funded by the Dean's Office)
    - Funds research in research of undergraduate students in biology and biochemistry research programs.
    - 2 undergraduate student(s) from Biology with a $4000 stipend each to conduct biological or biochemical research with a member of the UCLA faculty during the summer of 2024.
- Park Nobel (Development fund; to students in FBQ programs & competitive-based)
    - $500 awarded to student participants in the Marine Biology Quarter in the 18-19 academic year.
    - $225 awarded to a student participant in the Field Biology Quarter in the 18-19 academic year.
    - $4125, total, awarded to the Field/Marine Biology Quarter in the 15-16, 17-18, and 21-22 academic years.
- AR Wallace (Development fund; to students in MBQ programs)
    - $1500 awarded to students participating in the Marine Biology Quarter in each of 15-16, 16-17, 17-18, and 18-19 academic years.
    - $1500 awarded to students participating in the Field Biology Quarter in the 18-19 academic year.
    - $4700, total, awarded to the Field/Marine Biology Quarter in the 17-18, 19-20, and 21-22 academic years.
- Joseph Epperson (Development fund; to students in FBQ programs)
    - $4975, total, awarded to students participating in the Field Biology Quarter in 14-15, 15-16, 16-17, and 17-18 academic years.
    - $4450, total, awarded to students participating in the Field/Marine Biology Quarter in the 15-16, 16-17, 19-20, and 21-22 academic years.
- Marine Biology Fund (Development fund; to students in MBQ programs)
    - $5500 awarded to students participating in the Marine Biology Quarter in the 21-22 academic year.
    - $1400, total, awarded to the Field/Marine Biology Quarter in the 19-20 academic year.
- Vance-Fahey Gift Fund:
    - $10,500 awarded to students participating in the Marine Biology Quarter in the 22-23 academic year.

9. Hybrid and Online Learning

Following innovations in the Life Sciences Core, other classes in EEB have adopted a flipped classroom approach. In particular, EEB100 has been developed into a flipped course, where students engage with course materials online prior to coming to class where they discuss that material.

As with most educational institutions, the COVID-19 pandemic forced a rapid transition to complete online learning. While EEB courses have largely returned to full in-person instruction, the implementation of asynchronous online course has been adopted for many summer session offerings. By offering asynchronous instruction during the summer, we are facilitating the participation of more students whose summer work schedule might otherwise interfere with taking summer courses.

A. Equity Gaps

Although overall graduation rates are up and time to degree is down for all EEB students, these gains are not equally distributed. Examining time to degree by gender indicates a very minor difference with females graduating slightly faster than males for both 4-year and transfer admits (Figure D-20)

However, while data for both underrepresented minority (URM) and non-URM students show a clear downward trend in time to degree, URM students have a longer average time to degree than non-URM students, with the exception of 2019-22, where URM transfer students had a slightly shorter time to degree than non-URM transfer students (Figure D-21).

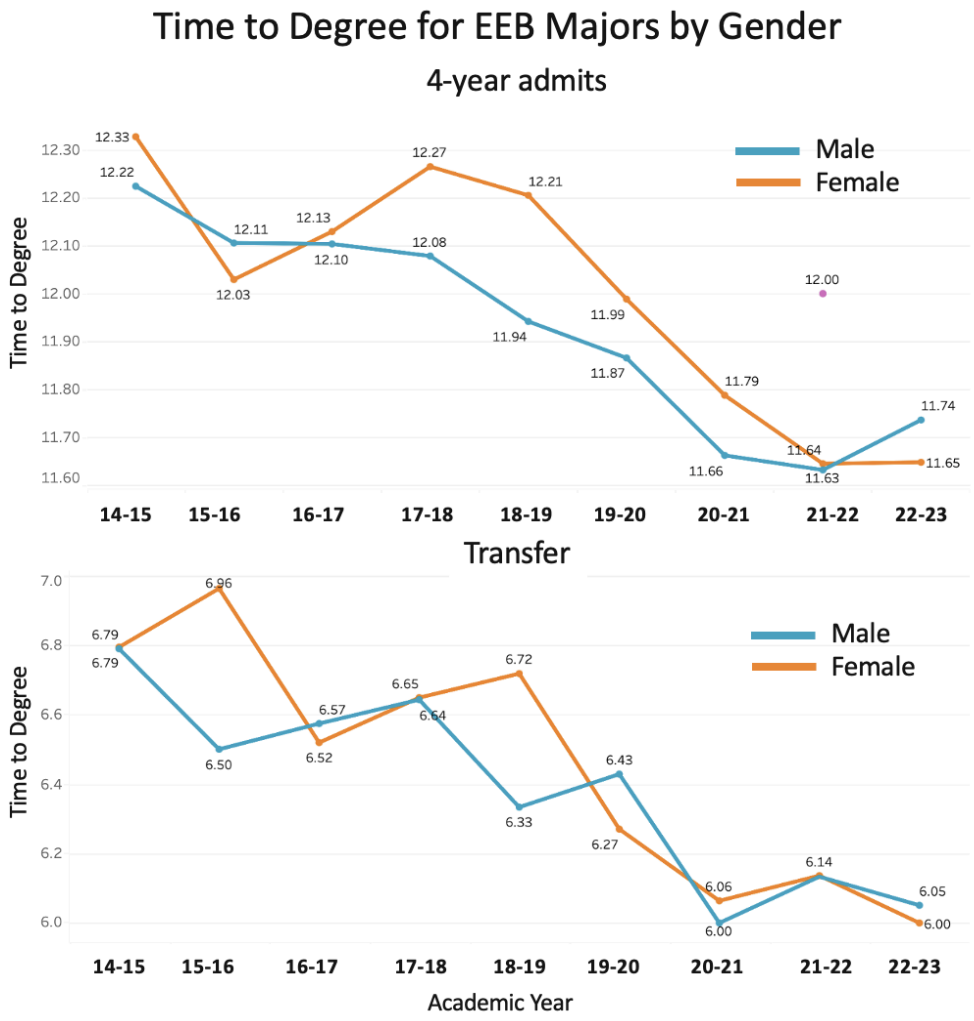


Figure D-20. Time to degree (in quarters) for all three Ecology and Evolutionary Biology majors (Biology, EBE, Marine Biology) combined across 2014–2023 for 4-year admits (top) and 2-year transfer students (bottom), disaggregated by gender.

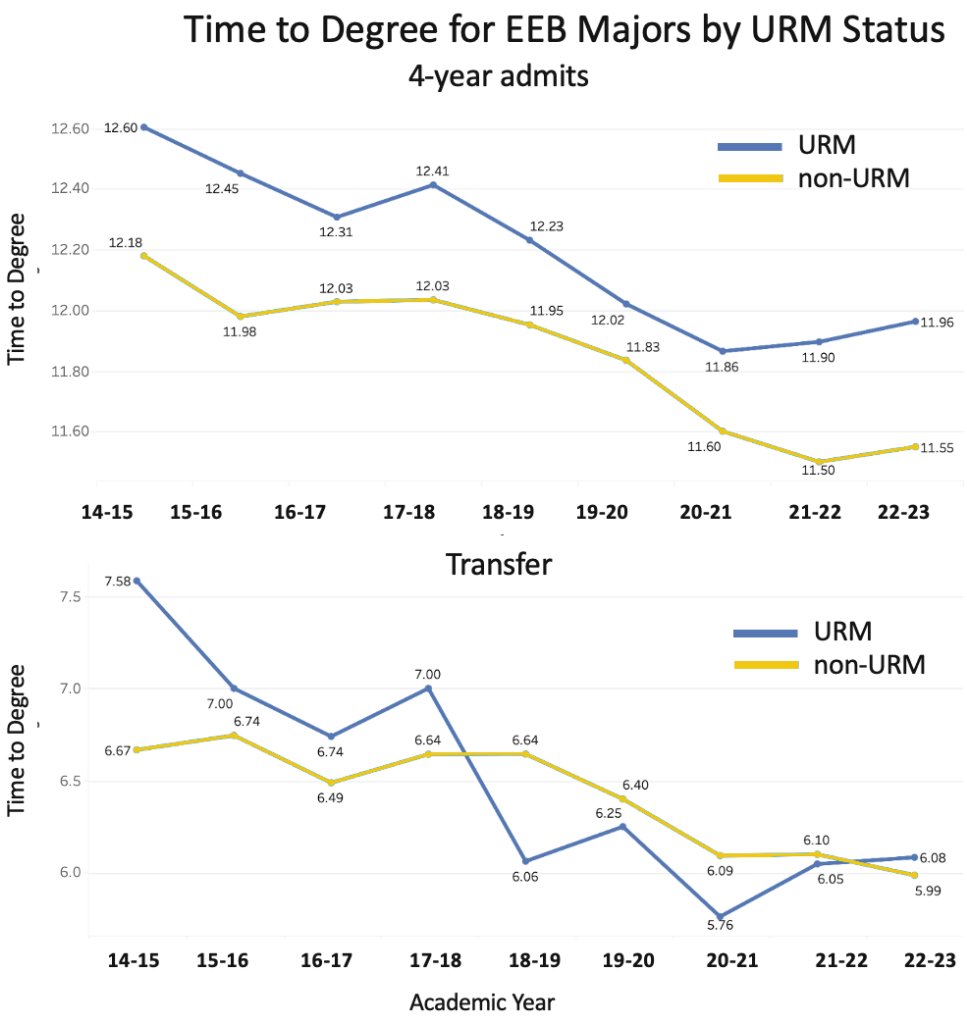
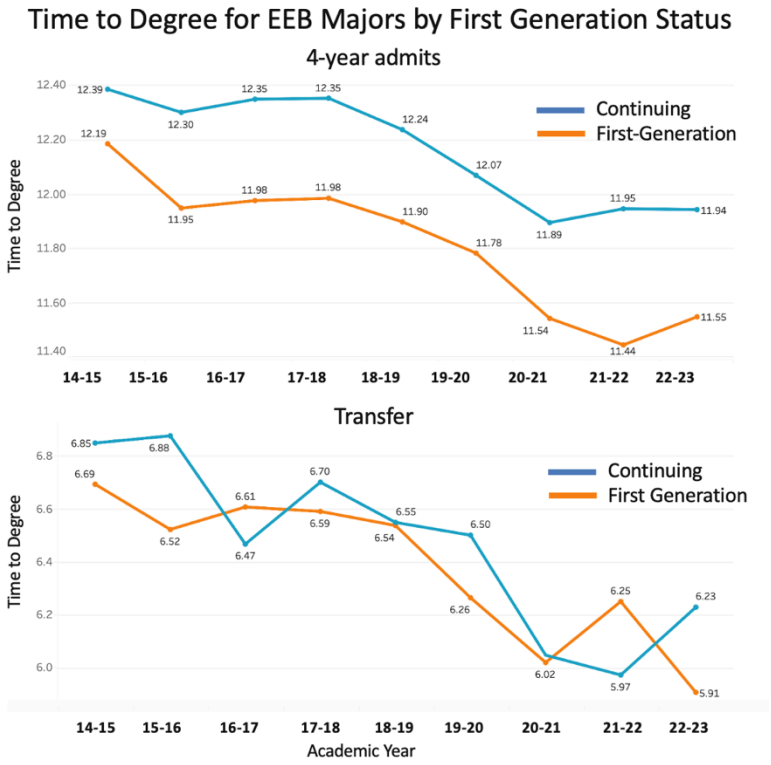


Fig D-21. Time to degree (in quarters) for all three Ecology and Evolutionary Biology majors (Biology, EBE, Marine Biology) combined across 2014–2023 for 4-year admits (top) and 2-year transfer students (bottom), disaggregated by status as an underrepresented minority (URM).

Similar patterns are seen for comparisons of first-generation college students vs. continuing students (Figure D-22) as well as for Pell grant and non-Pell grant students (Figure D-23). Because of strong correlations between URM, first-generation, and Pell grant status among UCLA students, subsequent analyses of equity gaps of these three groups focus on URM status.

Fig D-22. Graduation rates for all three Ecology and Evolutionary Biology majors (Biology, EBE, Marine Biology) combined from 2014-2023 for 4-year admits (top) and 2-year transfer students (bottom), disaggregated by status a first-generation college student or continuing student.



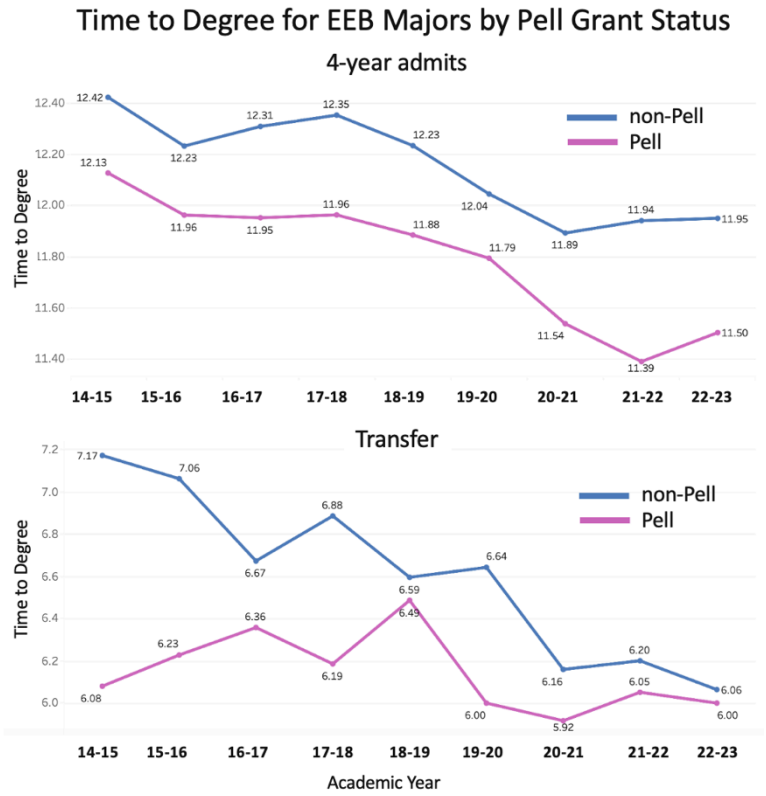
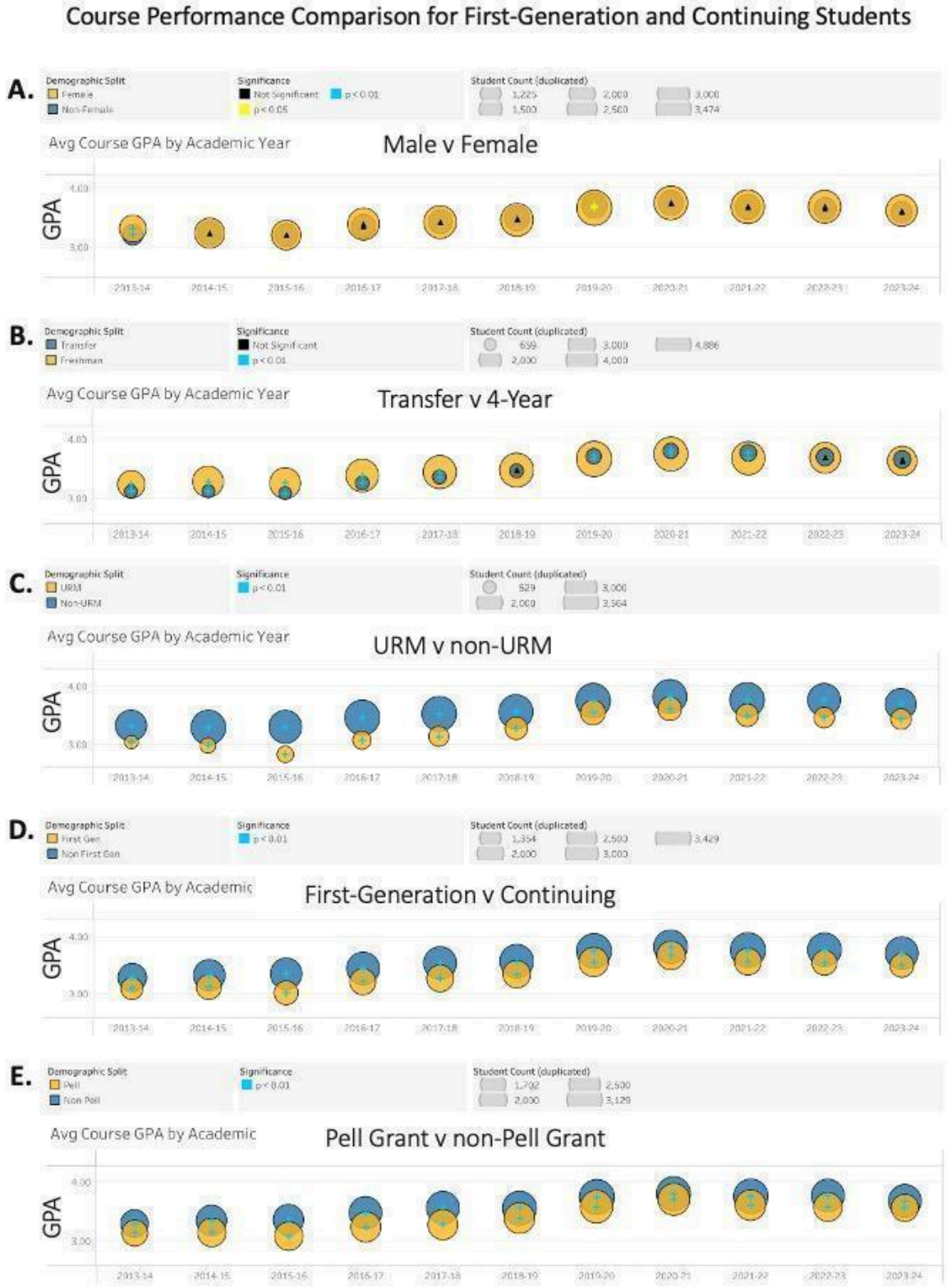


Fig D-23. Graduation rates for all three Ecology and Evolutionary Biology majors (Biology, EBE, Marine Biology) combined from 2014-2023 for 4-year admits (top) and 2-year transfer students (bottom), disaggregated by status a Pell grant recipient and non-Pell grant recipients

Longer time to graduation for URM students is likely a function of performance gaps between URM and non-URM students in their science classes. Although there are no significant differences in average GPA performance between male and female students, and minimal differences between transfer students and 4 year admits, there are substantial and significant ($p < 0.01$) difference in average course GPA for 1) URM vs. non URM students, 2) first-generation vs continuing, and 3) Pell grant vs. non-Pell grant recipients (Figure D-24, a, b, c, d, e). Even when comparing URM vs. non URM, including on students who are 4 year admits, non-first-generation, and non-Pell grant receiving students to control for other known risk factors in STEM student success, URM students have significantly lower average grades in EEB courses than non-URM students (Figure D-24, f). Although these grade performance gaps are lower, suggesting that being a transfer student, first-generation college student, or low-income student do influence academic performance, even when controlling for these variables, URM students still have significantly lower grades, suggesting that racial and ethnic identity alone is associated with lower academic success in EEB courses.


Course Performance Comparison for First-Generation and Continuing Students
A.
Demographic Split
Female
Non-Female
Significance
Not Significant
p < 0.01
p < 0.05
Student Count (duplicated)
1,225
2,000
3,000
1,500
2,500
3,474
Avg Course GPA by Academic Year
Male v Female
GPA
4.00
3.00
2013-14
2014-15
2015-16
2016-17
2017-18
2018-19
2019-20
2020-21
2021-22
2022-23
2023-24
B.
Demographic Split
Transfer
Freshman
Significance
Not Significant
p < 0.01
Student Count (duplicated)
659
3,000
4,886
2,000
4,000
Avg Course GPA by Academic Year
Transfer v 4-Year
GPA
4.00
3.00
2013-14
2014-15
2015-16
2016-17
2017-18
2018-19
2019-20
2020-21
2021-22
2022-23
2023-24
C.
Demographic Split
URM
Non-URM
Significance
p < 0.01
Student Count (duplicated)
529
3,000
2,000
3,564
Avg Course GPA by Academic Year
URM v non-URM
GPA
4.00
3.00
2013-14
2014-15
2015-16
2016-17
2017-18
2018-19
2019-20
2020-21
2021-22
2022-23
2023-24
D.
Demographic Split
First Gen
Non First Gen
Significance
p < 0.01
Student Count (duplicated)
1,354
2,500
3,429
2,000
3,000
Avg Course GPA by Academic
First-Generation v Continuing
GPA
4.00
3.00
2013-14
2014-15
2015-16
2016-17
2017-18
2018-19
2019-20
2020-21
2021-22
2022-23
2023-24
E.
Demographic Split
Pell
Non Pell
Significance
p < 0.01
Student Count (duplicated)
1,792
2,500
2,000
3,129
Avg Course GPA by Academic
Pell Grant v non-Pell Grant
GPA
4.00
3.00
2013-14
2014-15
2015-16
2016-17
2017-18
2018-19
2019-20
2020-21
2021-22
2022-23
2023-24

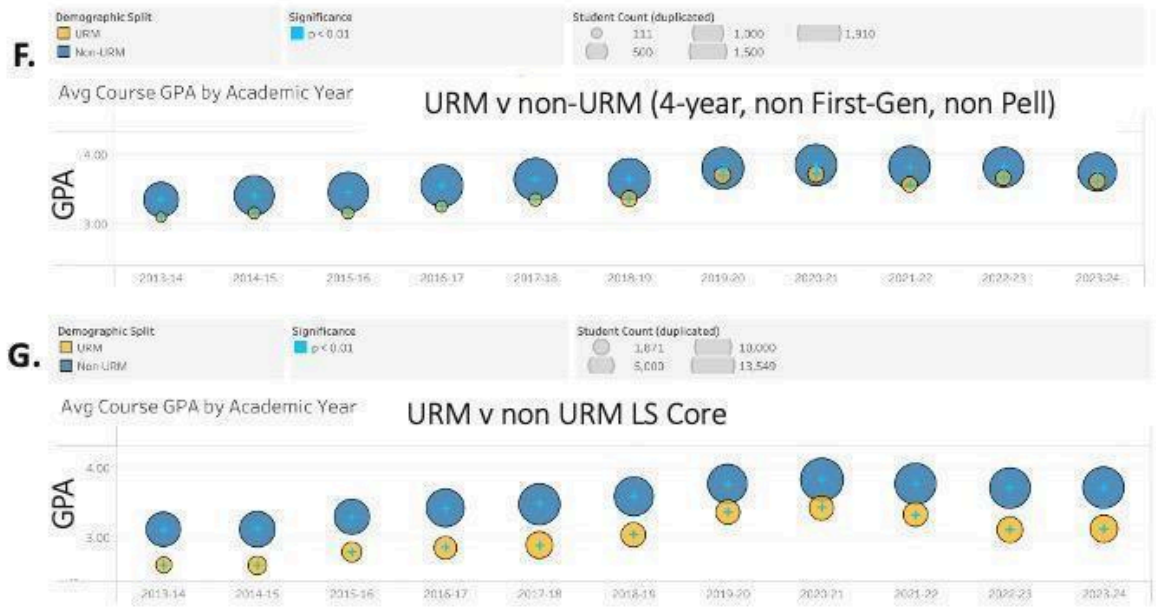
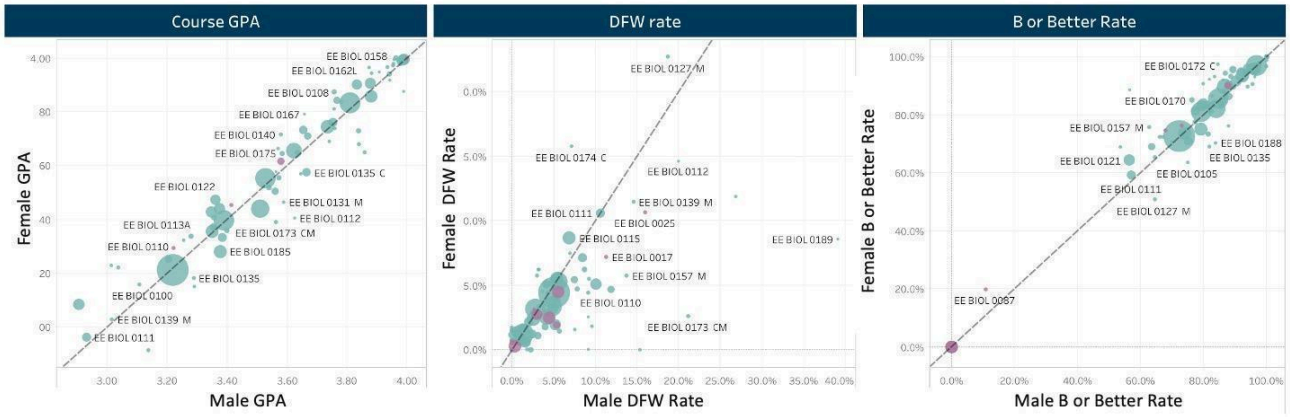


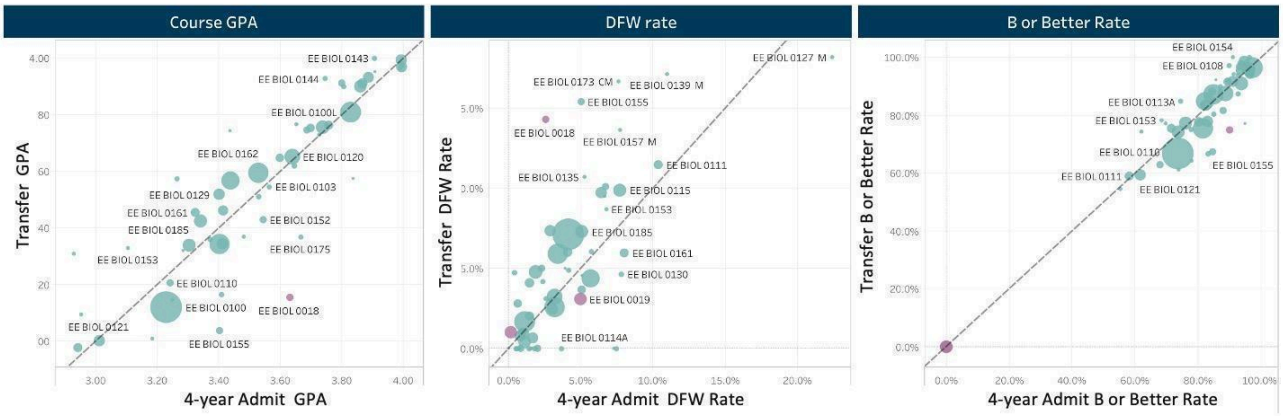
Figure D-24. Grade gaps in EEB courses based on a) gender, b) admit type, c) URM status, d) first-generation status, and e) socio-economic status as inferred by Pell grant status. F) shows grade gap between URM and non-URM students, including only 4-year admits, non-first-generation, non-Pell grant receiving students. G) shows grad gap data for the Life Science Core. Significant differences are indicated by bright blue cross.

Examining EEB course performance disparities in more detail show that there are very few disparities in individual courses based on gender and minimal differences between 4-year admits and transfer students. Rather, the average course performance gaps seen in Figure D-24 result from students who are URM, first-generation college students, and/or low socioeconomic status students doing worse in individual EEB courses than students who are not URM students, are not first generation, and are not Pell grant recipients. Examining course performance based on individual courses shows disparities in most EEB courses, particularly in some of the larger course offerings (Figure D-25). Given the disproportionate number of URM, first-generation college students, and students who are Pell grant recipients that receive a D, F or W, it is unsurprising that these demographics have a longer time to degree (Figures D-21, D-22, D-23). These student demographics are also less likely to earn a B or better in EEB courses, relative to non-URM, non-first-generation, and non-Pell grant receiving students.

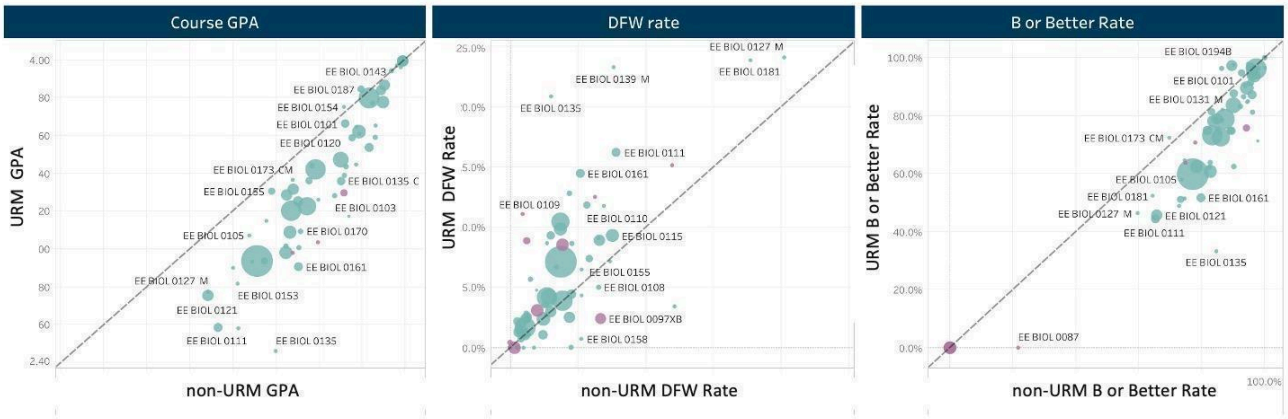
A. Course Performance Comparison for Male and Female Students



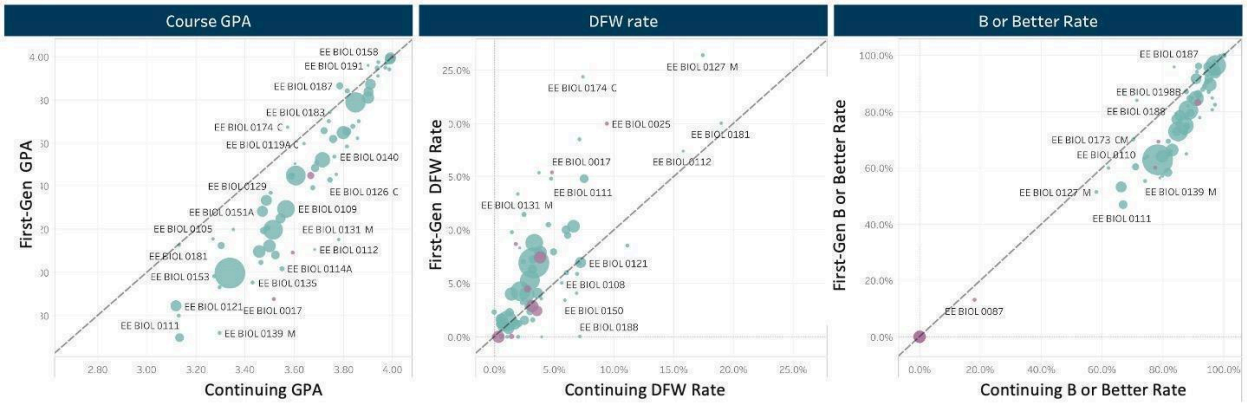
B. Course Performance Comparison for Transfer and 4-Year Students



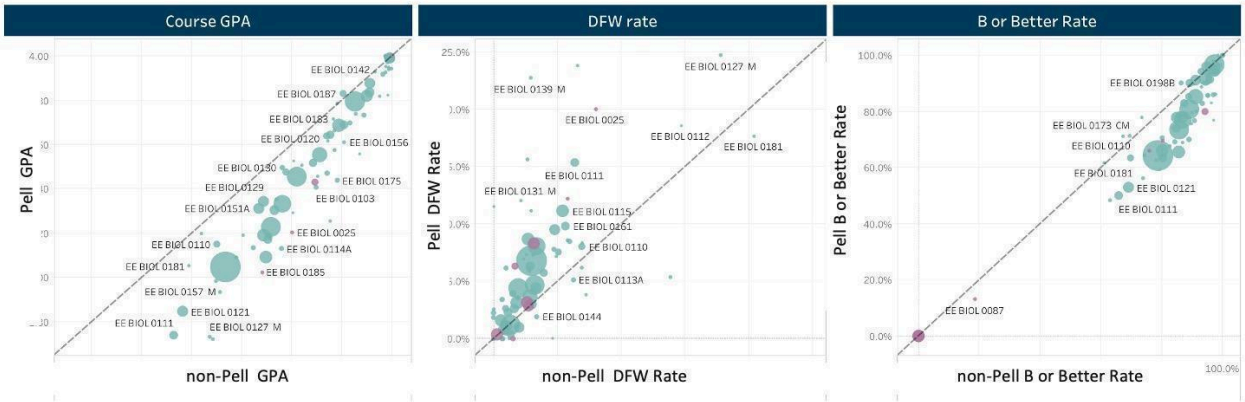
C. Course Performance Comparison for URM and non-URM Students



D. Course Performance Comparison for First-Generation and Continuing Students



E. Course Performance Comparison for Pell Grant and non Pell Grant Students



F. Course Performance Comparison for URM and non-URM Students Who Are 4 Year Admits, Not First-Generation, and Not Pell Grant Recipients

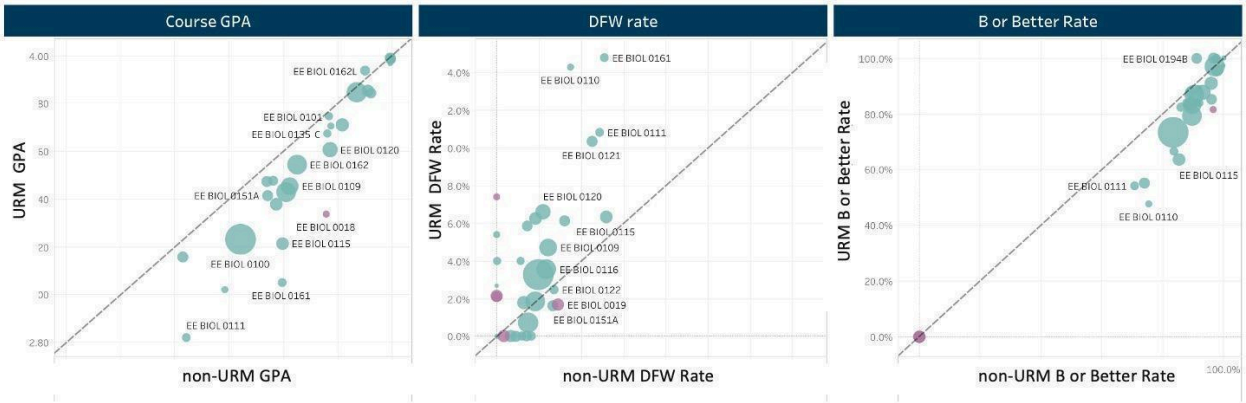


Figure D-25. Course performance gaps in EEB courses based on a) gender, b) admit type, c) URM status, d) first-generation status, and e) socio-economic status as inferred by Pell grant status. F) and URM status, including only 4-year admits, non-first-generation, non-Pell grant receiving students. Course metrics include average grade in a course (left panel), frequency of D, F, or Withdraw from a course (middle panel), and percentage of students earning a B or better (right panel).

Performance gaps are not unique to EEB. In fact, comparison of URM and non-URM average GPA across all Life Science departments and educational units shows that every department/unit has significant ($p<0.01$) disparities in academic success, with the exception of Computational and Systems Biology where differences are not significant, but only for low sample size (Figure D-26). The largest gaps in mean GPA between URM and non-URM students is seen in the LS Core and the Neurobiology interdepartmental program.

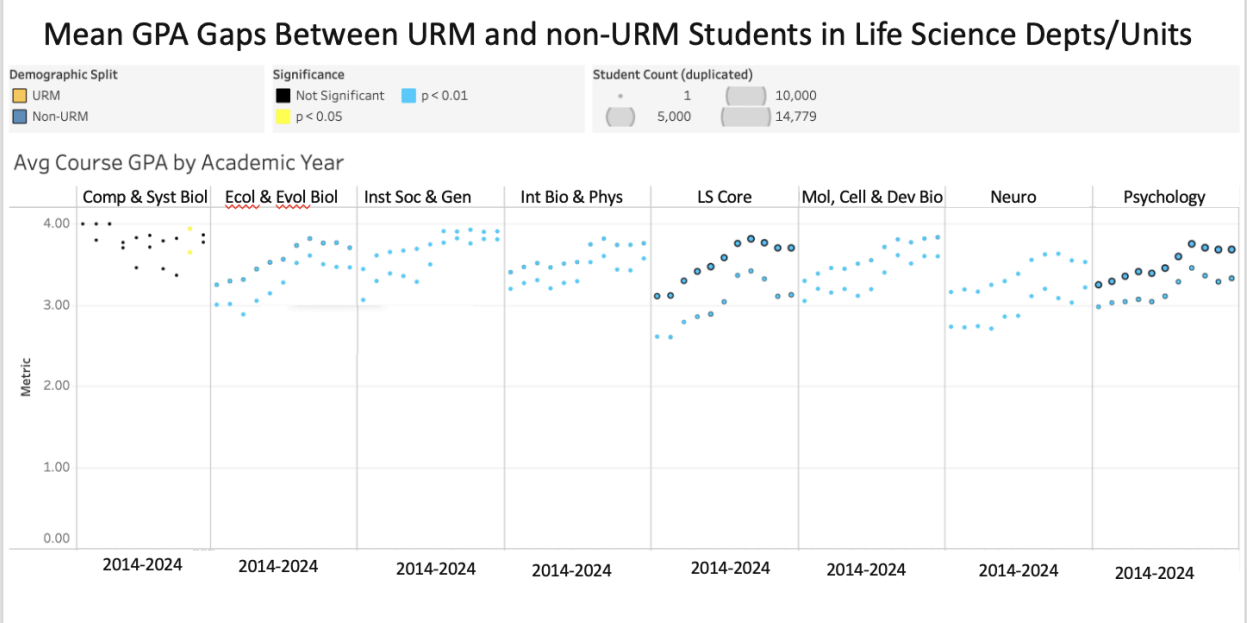


Figure D-26. Comparison of average course GPA for URM and non-URM students across 8 LS departments and educational units. All dots appear blue because all are significantly different at $p<0.01$, but URM in all cases are lower.

Although there are significant grade performance gaps in EEB classes based on race/ethnicity, first-generation status, and socio-economic status, students in EEB majors are also heavily impacted by course performance gaps in their foundational curriculum in the Life Science Core, as well as in the departments of Chemistry and Physics. Comparing URM vs. non-URM students shows a prominent grade performance disparity, with URM students earning significantly lower grades in introductory Life Science courses as well as in their core Chemistry and Physics Curriculum (Figure D-27). These average GPA disparities between URM and non-URM students result from substantially lower grades in the introductory curriculum taught by the LS core, as well as the Chemistry for Life Sciences series (Chem 14) and the Physics for Life Sciences series (Physics 5); URM students also earn substantially more Ds and Fs, and withdraw more from these courses due to poor course performance (Figure D-28). Although performance disparities in EEB courses indicate the need for improvements in how we deliver instruction to students from underrepresented and underserved groups, performance disparities in EEB are lower than the Life Science Core, Chemistry, and Physics, indicating that the ability of EEB to fully meet the needs of our diverse (racially, ethnically, socio-economically, etc) is

severely limited by pronounced equity gaps in other educational units that are responsible for the introductory curriculum for our majors.

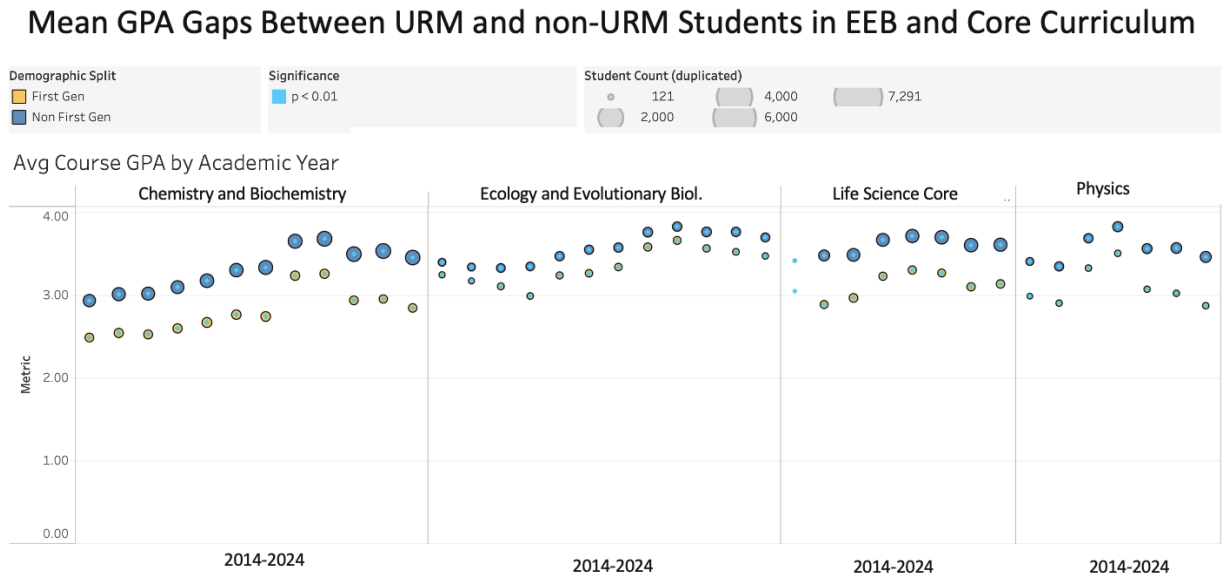


Figure D-27. Course performance gaps in introductory Chemistry and Physics courses based URM status. Course metrics include average grade in a course (left panel), frequency of D, F, or Withdraw from a course (middle panel), and percentage of students earning a B or better (right panel).

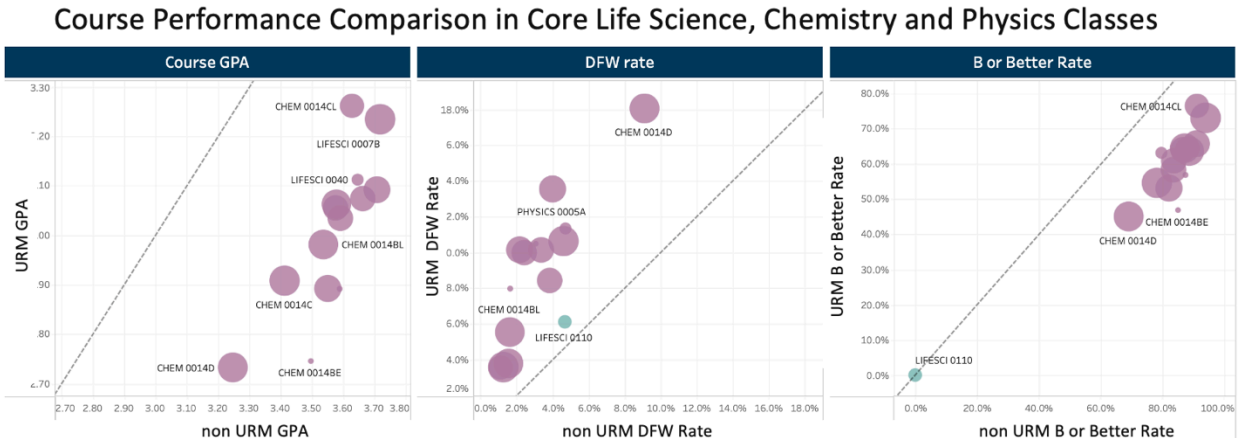


Figure D-28. Course performance gaps in introductory Life Science, Chemistry and Physics courses based URM status. Course metrics include average grade in a course (left panel), frequency of D, F, or Withdraw from a course (middle panel), and percentage of students earning a B or better (right panel).

10. Summary – Strengths, Opportunities, Challenges and Future Plans

A. Strengths

The department provides a broad curriculum in each of its majors, enabling students to benefit from the range of disciplines and academic experiences available at UCLA, without undue restrictions in these often-overlapping areas. The flexibility of these majors allows students who wish to specialize the ability to do so, while simultaneously allowing students from more specialized majors the ability to succeed in our general Biology major. These three majors, and the flexibility that they provide, prepare students who graduate with a degree from EEB for a broad range of jobs and graduate/professional schools.

A second strength is our two field-oriented capstone majors—Ecology, Behavior, and Evolution, and Marine Biology. These majors are very well-received by the students because (a) the upper division courses have small enrollment that enable students to personally interact with the professor; (b) the Field Biology Quarter and Marine Biology Quarters are cited almost without exception by our graduating seniors who have participated in them as peak experiences in their education which tied together everything that they had learned; (c) students develop their own research projects from concept to manuscript preparation. The large number of papers published by faculty and UCLA undergraduate students enrolled in FBQ or MBQ courses highlights the impact of these training activities.

A third strength is in the large number of undergraduate research opportunities for students. UCLA has a reputation for encouraging undergraduate research, and our department, unlike many, credits one quarter of undergraduate research toward our students' majors whether or not it is taken within our department. Many students work in the laboratories or field sites of our department. Many take advantage of the plethora of opportunities available in the biomedical areas where the clinical and basic science professors are delighted to have undergraduates in their laboratories. The large number of papers published by faculty and UCLA undergraduate students highlights the impact of these training activities

B. Opportunities

The proximity of the UCLA biomedical sciences provides a unique opportunity for our pre-professional students. They volunteer in the laboratories and clinics and obtain valuable experience and personal connections that prepare them for their medical careers. This has also aided in the development of our Evolutionary Medicine minor, and courses such as Health Equity and the Environment, and will doubtless contribute to their continued success.

The Southern California location and its proximity to the ocean and diverse terrestrial habitats provide a host of teaching and research opportunities for our students. Many of our students are from a very urban background, and exposure to local natural marine and terrestrial ecosystems is often a career- defining experience. This unique environment has supported the development

of a local field biology quarter, supporting research training opportunities for students that prepare them to address challenges of the Anthropocene.

Our capstone majors have evolved over time and we would like to expand the number of students within these majors. However, at present, the numbers have stabilized, most likely as a result of the limited number of students we can accommodate in our field courses. We need to devise solutions to this financial and logistic limitation and expand our numbers of capstone majors (see Challenges below).

C. Challenges

A major challenge for the EEB department to meet the needs of its undergraduate students is a substantial loss of faculty without sufficient replacements. Loss of 14 faculty since 2014 (Buth, Campbell-Staton, Gowaty, Gordon, Hubble, Kremer, Narins, Rundel, Smith, Taylor, Wayne, van Valkenberg, Zimmer, Zimmer-Faust) off-set by the recruitment of new faculty (Aguillon, Garud, Kraft, Ordway, Pinter-Wollman, Sears, Song, Tingley, Zapata) has left a major hole in our ability to staff our courses with ladder-rank faculty, leaving us to rely heavily on lecturers (permanent and temporary). This issue is compounded by the fact that many EEB faculty (Barber, Kennison, Shaffer, Sork) have leadership appointments outside of the department that limit their contributions to departmental teaching needs. Although two new faculty will start in EEB in fall 2024 (Aguillon, Song), more faculty growth is needed to maintain our undergraduate teaching mission.

Although younger faculty have engaged in teaching the field biology quarter, there has been no equivalent renewal in the marine biology quarter, largely because of the loss of the majority of marine focused faculty over the review period. With two of the remaining 4 marine focused faculty (and only two faculty who teach MBQ) of retirement age, EEB is close to losing its ability to deliver a key course needed by one of our two capstone majors. Lack of recruitment in marine sciences represents an existential threat to this major.

The department no longer has the word “organismic” in its title, but almost every ecological and evolutionary biologist will agree that understanding organisms is key to our discipline. One of the strengths of our faculty and our undergraduate and graduate programs is that we examine conceptual questions across a range of organisms. Among our most populated courses and valuable educational experiences are the courses that teach about the biology of non-human organisms. Increasingly, these courses have been taught primarily by temporary lecturers. Other universities have tried to address this problem by taking into account the ability of job candidates to teach courses like vertebrate biology, ornithology, or plant systematics when they are hired. As we rebuild the faculty ranks of our department, we hope to address these important teaching needs. The challenge will be to offer these traditional organismic-based courses, while at the same time, offering courses that reflect the emerging sub-disciplines and necessary topics for a 21st century biology curriculum.

The analysis of course performance data highlights significant gaps in the success of students from underrepresented and underserved backgrounds. While data on EEB courses show that

our department has work to do to address these issues, data from departments providing instruction in our required introductory curriculum indicate even larger performance gaps. These results suggest that even if EEB reduces or eliminates performance gaps in courses that we offer, underrepresented and underserved students will still be heavily impacted by performance disparities in our core curriculum. This suggests that reforms to this curriculum are necessary.

Increased graduate student salaries resulting from negotiations with the UAW (the union that represents UC graduate students) has resulted in a smaller graduate program and fewer graduate students to work as teaching assistants in our undergraduate courses. This change, combined with policies that prevent us continuing a long-established practice of hiring recent graduate as non-student tutors to serve as teaching assistants leaves us with a critical shortage of teaching assistants to staff our courses, resulting in courses and sections being canceled, likely increasing time to degree, as students will not be able to obtain needed courses in a timely manner.

The new Bruin Budget model disincentivizes one of the strengths of our teaching offerings—small courses. Given that departmental funding will be directly tied to total student credit hours, there will be pressure to shift towards larger courses. This change is particularly problematic given that some of our most impactful courses are the FBQ and MBQ capstone courses that, in addition to being expensive to run, have small student credit hours, bringing in minimal departmental revenue.

Growth in undergraduate enrollment in Life Sciences has not been met with a concomitant growth in resources to support undergraduate instruction. As such, demand for courses frequently exceeds supply. At the same time, the number of classes that students are required to graduate has increased. This competition for seats combined with the need for students to obtain courses in a timely manner to make degree progress has turned every enrollment period for STEM majors at UCLA into an existential crisis, where inability to enroll in a particular course in a particular quarter can have cascading effects, particularly for those courses required as prerequisites for upper division courses. Although one potential solution to the enrollment issue is increasing the number of seats in class, UCLA currently lacks the teaching space to support such expansion. Since 2006, Life Science undergraduate enrollment has nearly doubled, from 5,372 to 10,144 students. Physical Sciences has seen a similar doubling of student enrollment., However, this enrollment growth has not been met by a concomitant increase in classroom space. As such, it is extremely difficult to schedule large rooms to accommodate more students, and even if we could find that space, there are insufficient resources to support the teaching assistants needed to provide instructional support. This issue is even worse when considering laboratory course offerings. Although EEB added some instructional lab space with the recent renovation of the Botany building, lab space in the Life Sciences building is both limited and not well-suited to modern instructional needs. The need is particularly great for marine science labs–although UCLA recently upgraded the seawater facilities in the Life Sciences Building to better support faculty research needs (see marine facilities, Appendix D12), a similar investment is needed for the teaching labs.

Although the above are existing challenges, there are coming challenges that will significantly impact the ability of EEB to conduct its undergraduate teaching mission. UCLA is in the process of shifting to a new “Bruin Budget” Model. This funding model ties funding of departments and divisions to student enrollment and is designed to help funding be redistributed among campus units in ways that accounts for shifting patterns in student enrollment, correcting for years of structural underfunding for divisions like Life Sciences that have experienced tremendous growth in student enrollment without concomitant increases in funding. While this new budget model should be a net positive for EEB, there is a key issue that will result in the structural underfunding of the department. Rather than being based on student enrollment, the Bruin Budget model is based on student credit hours, or SCH, the number of students enrolled in a course multiplied by the number of credit hours. However, although EEB faculty teach, on average, more courses and have similar enrollments to similar Life Science departments, EEB classes that consist of 3 50 minute lectures and 1 50 minute discussion section only earn 4 units of credit, while comparable courses in similar Life Science departments earn 5 units of credit, resulting in a substantial disparity (Table 3). Because EEB courses receive 20% fewer units for equivalent courses, EEB will receive 20% less funding than similar Life Science departments to support our departmental teaching mission. Similarly, because undergraduate students at UCLA must complete 60 upper division units to satisfy their graduation requirements, EEB majors must take 15 upper division courses valued at 4 units each to graduate, compared to 12 courses valued at 5 units each for similar Life Science departments. By taking more classes, our students are more likely to experience challenges with enrollment, will have to take more courses per quarter, and/or take more courses in summer school. We have filed a request with the UCLA Academic Senate to correct this inequity in course credit, but it is unclear whether our request will be granted.

Table 3. Average number of courses, course enrollment and student credit hours per ladder faculty FTE for academic years 2021-2024. Average Faculty FTE is from 2020-2022; more recent data was not available. Results show that despite teaching more courses per FTE and having similar student enrollment per FTE, EEB student credit hours are approximately 33% less than in IBP and MCDB. Data from UCLA Office of Academic Planning and Budget and the Office of Diversity, Equity and Inclusion.

	Average Faculty FTE 2020-22	Average Courses/Faculty FTE, 2021-24	Average Enrollment/Faculty FTE, 2021-24	Average Student Credit Hours/FTE, 2021-24
EEB	23.8	1.9	94.5	315
IBP	20.2	1.7	86.1	401
MCDB	19.7	1.7	97.9	415

11. Future Plans

Lecturers. Given the new Bruin Budget model and reduced faculty size, it is critical to invest in hiring lecturers who can take on a larger amount of teaching. However, we need to do this strategically, hiring permanent lecturers with experience in developing and implementing evidence-based pedagogy, and who—in addition to serving larger numbers of students—can use their knowledge and experience to serve as a resource to help other EEB elevate their teaching, particularly with respect to inclusive teaching practices.

Curricular reform. Biological knowledge has advanced exponentially over the past 30 years, yet our approach, broadly, to the biology curriculum is based on how the discipline has been taught for 100s of years. As the amount of information that a modern biology student needs to know has grown, resulting in increases in the amount of information and courses students are required to know for graduation, little attention has focused on reducing other areas of the curriculum to offset new areas of growth. It is essential to rethink and re-envision what we are asking our students to learn and master. In particular, our students may no longer be served by a traditional STEM curriculum where they take a year or more each of math, chemistry and physics. The Life Sciences took the bold step to replace the standard year-long calculus for Life Sciences series with a novel, problem-based 2-quarter Mathematics for Life Sciences series that teaches students to use python as a computational tool to solve biological problems. Assessment of this course showed significantly better outcomes in subsequent Physics classes compared to students in the traditional calculus series, highlighting that when done thoughtfully, reducing and refocusing our preparatory courses can serve the educational needs of our students better. Given that the largest course performance gaps are seen in the introductory Life Science, Chemistry, and Physics curriculum, it is time to revisit these requirements to best serve the needs of our students.

Undergraduate Program Appendices Not Referenced
Appendix D13: Biology Major Requirements from UCLA General Catalog
Appendix D14: Ecology, Behavior, and Evolution Major Requirements from UCLA General Catalog
Appendix D15: Marine Biology Major Requirements from UCLA General Catalog

Section E. Graduate Programs

1. Overview

The graduate program in ecology and evolutionary biology aims "*to provide new knowledge of the ecological and evolutionary processes that produce and sustain life on Earth and to educate the next generation of scholars, professionals, and citizens for the biological, environmental and biotechnological challenges of the future*." To this end, our graduate program offers broad training in ecology and evolutionary biology including in related fields of animal behavior, marine biology, and systematics. We train graduate students to be leaders in their chosen fields by immersing them in a dynamic and interdisciplinary research atmosphere, teaching them required skills, and providing ongoing seminars and colloquia in which graduate students can discuss cutting edge ideas and interact with leaders in the field. Students in our program work throughout the world, and on a wide variety of taxa, from bacteria to redwoods and from ants to whales. Maintaining this intellectual diversity is, in itself, an important educational goal that requires admitting an excellent cohort of students each year, supporting them, and mentoring them to completion. Graduates from our program are highly competitive in a wide variety of careers – with many going on to postdoctoral fellowships then tenure-track positions – but our graduates are also successfully employed in government, NGOs, industry, and many other sectors (Appendix E1: DGE Report Biology PhD). While most faculty and students pursue basic science academic research, many projects also have an applied focus, and this mix of academic and applied research makes some of our students particularly marketable.

The EEB department offers two MS degrees (Biology MS and Evolutionary Medicine MS) and one PhD degree (Biology). We emphasize the doctoral program in terms of admissions and financial support. Over the last decade, approximately 65% of our matriculated students have been in the PhD program (n = 191) while the remaining 35% (n = 102) have been in one of the two MS programs. The intellectual and disciplinary aims of the MS and PhD programs in Biology are similar, with an ultimate goal of preparing students for research- or teaching-oriented careers in ecology or evolutionary biology. In contrast, the MS specializing in the ecology and evolution of medicine ("EvMed"), which was created since our last review, is partly pre-professional, preparing graduates for future medical and veterinary careers as well as academic graduate programs in the biomedical sciences. To this end, the EvMed MS draws from faculty expertise across multiple departments, including the David Geffen School of Medicine. The EvMed MS involves both coursework and experiences within the Ronald Reagan UCLA Medical Center, including Grand Round lectures at the David Geffen School of Medicine and other medical experiences.

2. Admissions

Graduate Admissions is overseen by a faculty-led committee that is appointed each year by the Department Chair. Our exact policies, guidelines, and practices are reevaluated and potentially revised each year to maintain rigorous standards (e.g., standards of fairness and equity) and to adhere to current best practices. Over the last 5 years, revisions have specifically sought to engage best practices in holistic evaluation of applicants, as such practices have been developed. Due to the changing nature of standards and policies, descriptions below are necessarily limited to describing the current admissions policies (as of fall 2023). Where notable changes to policies or procedures have been made, the year of this change is noted below. A complete current SOP for the Graduate Admissions and Support committee can be found in Appendix E2.

A. Application Process

Recruitment of our graduate applicant pool is done in a decentralized manner through a variety of formats. Collective efforts of recruitment are done primarily through distribution of physical and/or electronic posters and brochures at conferences (e.g., Ecological Society of America, Society for Advancement of Chicanos/Hispanics and Native Americans in Science [SACNAS]) and through targeted under-represented in graduate school (URG) recruitment events (e.g., CSU Summer Symposium and Graduate Fair, organized by UCLA Graduate Program in Bioscience). As our admissions program follows a PI-admit model (i.e., students are admitted to specific labs, rather than to a general pool with rotations), individual recruiting faculty frequently advertise for potential graduate students via a variety of venues and mediums.

Applicants apply to our graduate programs via online submission of the UCLA Application for Graduate Admission. Our departmental website also includes detailed instructions for PhD, Biology MS, and EvMed MS applicants. Students are requested to submit the names of two potential research advisors, CV and transcripts, three letters of recommendation, and two essays: a summary of research interests and professional experience (UCLA's "Statement of Purpose"), and a 'personal history' (see also §G.6).

The specific requirements for applicants to the department have changed over time, specifically with regard to how the department treats and/or evaluates contributions to diversity. Briefly, the department began evaluating potential contributions to diversity in 2016–2017 and stopped formally evaluating students by their GRE scores beginning in 2017–2018. A more detailed history of these changes can be found in §G.6.

Applications to the PhD and Biology MS programs are due annually on December 1st, while the EvMed MS program has a later application deadline of May 1st in order to better reach students on pre-health tracks.

B. Applicant Pool

The number of applicants to our graduate programs varies by year (**Figure E1**), with an 11-year average (± sd) of 100 (± 19) PhD applicants and 48 (± 18) MS applicants (combined across Biology and EvMed programs). The number of applicants to the MS programs increased between 2014–2018 and has remained nearly constant since 2019. The number of PhD applicants has varied more from year to year, with a minimum of 74 (for fall 2023 entry) and a maximum of 139 (for fall 2015). An important caveat to interpreting these data is that the PI-admit model of our department means that students reach out to faculty prior to submitting applications and faculty pre-screen, discouraging applications from students with poor fit.

The composition of the applicant pool has remained relatively stable over the last 11 years. Our applicant pool is consistently female-biased, with a long-term average of 58% of applicants identifying as female (40% identify at male; 2% as non-binary, although the university only allowed students to identify as non-binary beginning in 2020), with the percentage female varying between 54% (for 2021) and 63% (2019). Relatedly, the long-term average proportion of applicants who are underrepresented in graduate school (URG; a classification designated by UCLA as domestic students including American Indian and Alaskan Native, Black and Afro-American, South or Central American, Chicano and Mexican American, Latino and other Spanish-American, and Hawaiian, Native Hawaiian and other Pacific Islander) has remained undeviating, with a long-term average of 20% of applicants holding URG identities, varying across the decade between 16% (for 2016) and 24% (for 2019 and 2023). The proportion of URG applicants differs by degree program, where on average 17% of MS applicants are URG whereas 22% of PhD applicants are URG.

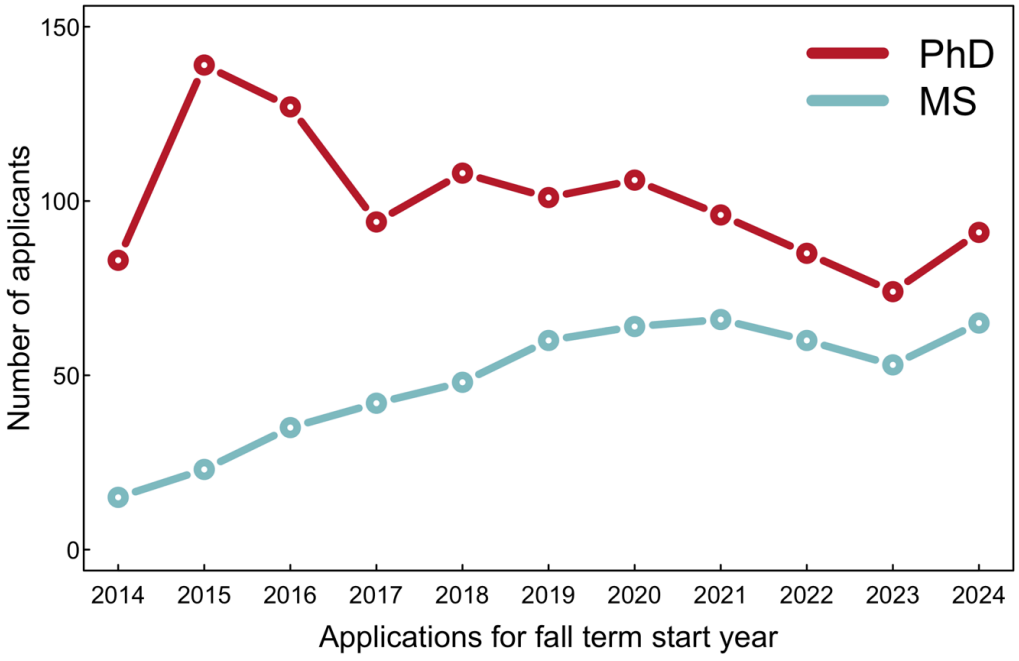


Figure E1. Number of graduate school applicants to EEB by year. Count for MS combines applicants across two programs, Biology MS and EvMed MS.

C. Application Review

As we follow a PI-admit model – whereby students contact, identify, and ultimately apply to 1–2 graduate-admitting labs run by faculty in EEB – initial filtering and selection of applicants for review is conducted by those individual faculty. All applications that list an individual PI as a potential mentor are forwarded to that PI who is given 1–2 weeks in December to review applications, contact and/or interview applicants (as appropriate), and nominate up to 2 PhD and 2 MS applicants for committee review and potential admission. In special cases (e.g., new faculty or faculty with new grants), >2 applications can be nominated. Faculty reviewing applications and nominating individuals are given the criteria used by the Graduate Admissions and Support committee (GAS) for ranking applicants (see below) and are advised to review holistically and consider the strong potential for implicit bias in initial selection. Additionally, as of fall 2022, the GAS Chair reviews all PhD applicants and identifies all applicants who are competitive for Cota-Robles fellowships, using UCLA's rubric to score each individual. If the top-scoring individual (or individuals, in case of a tie) is not nominated by faculty, then this applicant will additionally be added to the ranking pool by the GAS Chair. Nominations from faculty for applicant review are due in mid-December, annually.

Review and ranking of all nominated applicants is conducted by the Graduate Admissions and Support committee (GAS). The voting members of the GAS consist of the Vice Chair for Graduate Studies (who chairs the committee), the Faculty Graduate Advisor, two or three more faculty members (selected to reflect EEB research diversity), and a graduate student member (who is recommended not to be a student of any faculty on the committee, but ultimately this position is decided by the graduate student body). Graduate advising staff are non-voting committee members.

Each nominated application is rated on a five-point scale (1–5; 5 being high) in each of five categories by all voting members of the GAS. The categories include: (1) personal statements; (2) letters of reference; (3) academic scores; (4) research experience and preparation; and (5) potential contributions to diversity. For personal statements and letters, reviewers are advised to look for evidence of the following: knowledge and creativity, communication skills, resilience, planning and organization, ethics and integrity. For academic scores, EEB voted to eliminate consideration of the GRE (although students could still optionally report scores) beginning in fall 2017, so current evaluation of this category is based on undergraduate (and graduate, as available) transcripts. EEB began formally requiring diversity statements of applicants also in fall 2017. For categories 1–4 above, committee members are asked to benchmark scores off of the following scale:

5. Outstanding; truly exceptional relative to the expectations of the program
4. Above average relative to our expectations for incoming students
3. Meets the expectations for incoming students in EEB
2. Somewhat deficient in area; could succeed with additional mentoring and guidance in program
1. Deficient in this area; serious concerns

For rating potential contributions to diversity, reviewers are asked to use the following scoring benchmarks, with scores of 2 and 4 being intermediate:

> 5. Students have unique experiences that are underrepresented in higher education/ecology or have a history of working with underrepresented communities on issues that pertain to them. Demonstrates mature self-awareness of both the opportunities and barriers in their journey to graduate school and is able to communicate effectively across cultural, socioeconomic, and gender differences.
>
> 3. The student is self-aware and empathetic but has no or limited lived or working experience with underrepresented communities. They do not demonstrate effective communication across cultural, socioeconomic, and gender differences.
>
> 1. Students do not demonstrate an awareness of diversity and/or are antagonistic or paternalistic when speaking of underrepresented communities. The student does not have demonstrable experience communicating effectively across cultural, socioeconomic, or gender differences.

Finally, reviewers are asked to complete two additional tasks for each applicant. First, reviewers are asked whether the applicant is – regardless of priority – qualified for admission to the applied graduate program. Applicants must receive a "Yes" answer to this question from all reviewers in order to be admitted. Second, reviewers are required to give an overall score to each candidate which need not be an average of the 5 component scores. Scores are benchmarked off the following scale:

> 5. Highest priority for admission and recruitment
> 4. Strong priority for admission and recruitment
> 3. Admissible to the program as is, but not high priority
> 2. Deficient but could be considered for conditional admission (e.g., academic probation in first year)
> 1. Not admissible (if answered "No" above on admissibility)

Scores for each applicant are submitted independently by reviewers, but the full GAS meets in late December to discuss every nominated applicant. During this meeting, reviewers are allowed to (anonymously) adjust their scores, as needed, based on the conversation, allowing overlooked strengths or weaknesses of applications to be reflected. At the outcome of discussion, the committee agrees on an overall rank and final determination of admissibility for each nominated applicant. To limit potential bias, reviewers make all decisions and agree on ranks blind to: (1) knowledge of which faculty member nominated each individual; (2) potential sources or availability of funding. Committee members are not allowed to review applicants that they have nominated (or, for students, applicants to their lab) and must recuse themselves from all deliberations on such individuals.

This review process is identical for both PhD and Biology MS applicants; however, the process is slightly different for EvMed MS applicants. EvMed applications have a final deadline of May

1st each year. Following that deadline, the full EvMed applicant pool is sent to a separate EvMed committee who coordinates with potential supervising faculty to seek nominations. The EvMed committee reviews, selects, and ranks (as needed) the nominated applications, and sends on to the GAS no more than the number of applicants that the department has remaining TA-ship funding to support. The GAS then does a final review of these selected applicants, deciding solely whether they meet the criteria of admissibility.

D. Recruitment and Admissions

EEB holds an annual recruitment event for top-ranked applicants, generally over two days in early February. EEB reserves funding (including travel, room and board) for PhD applicants, although individual PIs can use grants to support the attendance for any ranked applicant (e.g., Biology MS applicants). The schedule of events for recruitment varies from year to year, reflecting experimentation to maximize yield, but generally prioritizes meeting time with faculty and current students, open social time to network and talk, and tours of campus and its resources. In recent years, events have been planned to ensure that URG applicants have opportunities to meet with and talk to current graduate students with whom the applicants share identities. In recent years, 10–25 individuals have been invited to and/or attended our recruitment event (including 2 years of remote recruitment during the COVID-19 pandemic). EEB's philosophy is that the recruitment is not an interview (i.e., all applicants have already been ranked) and that the shared intention is that the department should fund the attendance only of students it hopes and intends to be able to admit.

Offers of admissions are also sent to applicants during winter quarter, potentially beginning in January and lasting through April (or June, for EvMed MS students). Through the 2021-2022 academic year, some highly-recruited applicants were admitted prior to their attendance at the recruitment event (especially URG students who were being nominated for Cota-Robles fellowships). Beginning in the 2022-2023 cycle, admissions offers have been held off for all applicants until immediately after the recruitment event.

Decisions on admissions are coordinated by the GAS Chair (i.e., Grad VC) with support of EEB Grad staff and final oversight belonging to the Department Chair. Order and pacing of offers depends on the financial details of each offer, which generally falls into one of three categories:

1. *Department-supported offers*: The department pledges to provide the applicant with financial support in the form of fellowships and TA-ships (see §E.2.E).
2. *PI-funded offers*: The PI pledges to cover all financial support not covered by TA-ships. If a PI can cover some but not all of a minimal PhD offer (see §E.2.E), then it is classified as a departmental-supported offer.
3. *TA-only offer* (MS only): MS applicants can be admitted with financial support backed only through TA-ships.

Applicants (strictly PhD) receiving departmental funding support (i.e., department-supported offers) are admitted in an order based on three principles. First, all things being equal, applicants receive department support in order following the final consensus ranking of the

GAS. Second, as policy, no faculty member will be given the chance to make a second offer that would result in two department-supported offers until all faculty who requested a student that was favorably reviewed by the GAS have been given a chance to make an offer. Third, the department will invest in the success of junior faculty and new hires by giving them additional prioritization for student offers when their applicants are favorably reviewed. Since PI-funded and TA-only offers do not require departmental financial resources, such offers can generally be made at any time – regardless of ranking – given sufficient TA slots (with PhD offers prioritized over MS offers).

The number of admissions offers to our graduate programs varies by year (**Figure E2**), with an 11-year average (± sd) of 20 (± 4) PhD offers and 9 (± 5) MS offers (combined across Biology and EvMed programs). The number of admissions offers has varied over time, largely due to funding availability and admitting faculty size and interest. The number of admissions offers to the MS programs has varied between 4 (for a fall 2014 start) and 18 (for a fall 2016 start), while the number of PhD offers has varied between 16 (for fall 2018 and 2022) and 28 (for fall 2015).

The composition of the pool of students receiving offers has remained relatively stable over the last 10 years. The pool of students receiving offers is strongly female-biased, with a long-term average of 65% of offers for going to individuals identifying as female (34% identify at male; 1% as non-binary; although in recent years that has risen as high as 7%), with the percentage of females receiving offers varying between 58% (for fall 2016) and 79% (2023). Relatedly, the long-term average proportion of offers going to URG individuals has remained steady, with a long-term average of 26% of offers to those holding URG identities, varying across the decade between 11% (2016) and 38% (2024). The proportion of offers to URG applicants differs by degree program, where on average 24% of MS offers are to URG students whereas 27% of PhD offers are to URG students. Broadly speaking, on average, 49% of offers have been to white students, 19% to Asian students (including Asian-American and South Asian), 21% to Latin American students (including Chicano), and 8% to Black students.

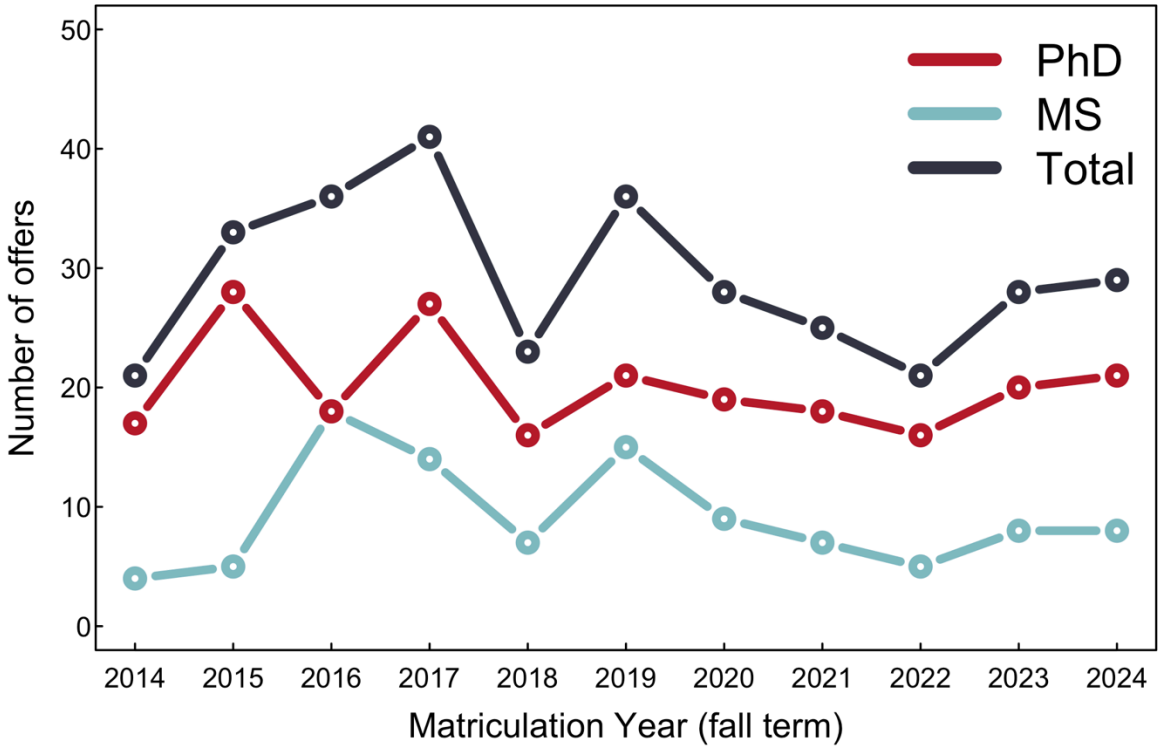


Figure E2. Number of graduate school admissions offers made by EEB each year. Count for MS combines applicants across two programs, Biology MS and EvMed MS.

Selectivity in a program is the ratio of admissions offers to applications for admissions. Lower selectivity indicates that fewer students are admitted relative to demand. Overall, our average selectivity is 20% and 19% for the PhD and MS programs, respectively. The selectivity is slightly higher (i.e., less selective) for URG applicants (25% for PhD; 27% for MS) than for domestic non-URG applicants (22% for PhD; 24% for MS). This standard metric of selectivity underestimates the true selectivity of our program, as faculty frequently pre-screen inquiries to discourage applications from weak prospective students. Our program is very selective against international applicants (6% for PhD; 3% for MS), reflecting both the fact that our program gets many international applicants who are not a strong fit for EEB and that UCLA charges additional supplemental tuition (NRT) for international students which makes them prohibitively expensive for most PIs to support.

E. Support

The department maintains a minimum funding package for admitted PhD and MS students. The exact dollar amount of annual support included in this minimum offer has changed over time (see §E.3), but for PhD students was approximately $26,000 in 2014, $30,000 in 2020, and $32,000 in 2023. Minimum funding offers for PhD students include salary and tuition for five years, beginning with two academic quarters of fellowship and four summers of fellowship ($7500 each summer). All other academic quarters, in a minimal offer, include guaranteed

funding via TA-ships (which currently pay substantially better than other forms of support). Minimal MS offers are supported exclusively via TA-ships, without guaranteed summer support (although MS students can apply for summer TA positions and other funding opportunities). Continued funding is guaranteed based on satisfactory progress in the program. If students receive external support from elsewhere on campus, or from non-UCLA sources (e.g., NSF fellowships, or foundation awards), this is used to replace departmental obligations. In addition to funds promised in admissions offers, students have annual opportunities to apply to the department (and university, as available) for additional sources of funding, including funds for conference travel, research support, summer and academic-quarter fellowships. Funding for graduate students is derived from UCLA Division of Graduate Education, the Life Science Dean's block grant, EEB endowed funds (for fellowships and summer funding), the instructional budget for teaching assistants, individual faculty grants, and departmental or campus training grants.

International students face additional funding challenges. The university charges supplemental NRT (currently $15,102 per year) to all non-resident students, which international students are classified as for the duration of their studies (unlike domestic out-of-state students who can become California residents beginning their 2nd year). An ongoing challenge is who pays for NRT? The cost of NRT is substantial enough that faculty are generally unwilling to foist the cost onto students (which would effectively cut their salary in half, below the Los Angeles poverty line). EEB’s current admissions support model is that it will pay NRT for the first year for all non-resident students, but then the PI (or the student, if via grants or fellowships) must cover NRT in subsequent years, as needed. Granters are generally unsupportive of paying tuition and/or have tuition caps, making NRT difficult in practice to find funding for. The result is that few international students are admitted to our graduate program (see stats in §E.2.D), depriving our program of top talent.

F. Yield

The number of matriculated students to our graduate programs varies by year (**Figure E3**), with an 11-year average (± sd) of 13 (± 5) PhD students and 8 (± 5) MS students (combined across Biology and EvMed programs). The number of matriculated students to the MS programs has varied between 2 (for a fall 2014 start) and 17 (for a fall 2016 start), while the number of PhD matriculated students has varied between 8 (for fall 2016) and 21 (for fall 2017). Since the number of matriculated students depends heavily on the number of admitting faculty – and not every faculty member wishes to admit a student each year – patterns of matriculation show more variability than, for example, the number of applications (**Figure E1**).

The composition of matriculated students has remained relatively stable over the last 10 years. The pool of matriculated students is female-biased, with a long-term average of 64% of graduate students identifying as female (35% as male; 1% as non-binary), with the percentage of matriculating females varying between 48% (for 2016) and 75% (2022). Relatedly, the long-term average proportion of matriculated students with URG identities has remained steady, with a long-term average of 27% of students having URG identities, varying across the decade between 12% (for 2016) and 47% (2020). The proportion of URG matriculated students differs

by degree program, where on average 23% of MS students are URG whereas 29% of PhD students are URG. Broadly speaking, on average, 47% of matriculated students are white, 23% are Latinx, 19% are Asian (including Asian-American and South Asian), and 6% are Black. For more context on the relative diversity of our graduate program and recent trends, see §G.6.

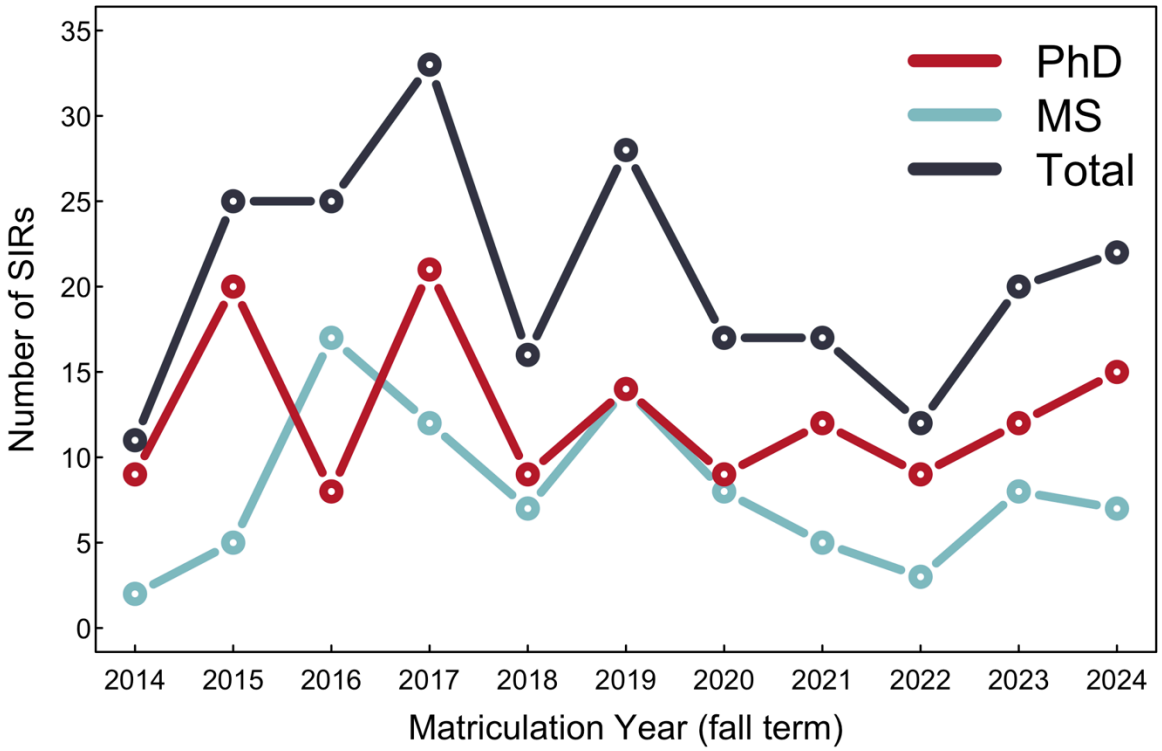


Figure E3. Number of EEB graduate students matriculating each year. Count for MS combines applicants across two programs, Biology MS and EvMed MS.

Yield in a program is the ratio of matriculated students to admissions offers. Lower yield indicates that fewer students choose to join the program relative to how many are offered. Overall, our average yield is 62% for the PhD program and 88% for the MS program, likely reflecting the fact that we are rare to have a funded MS program in EEB and many MS students are recruited directly from the UCLA undergraduate program. For the PhD program, our yield is substantially higher for URG students (67%) than domestic non-URG students (61%) or international students (62%). There are fewer differences in yield for our MS programs, where yield for URG students (83%) is slightly lower than for domestic non-URG students (90%) and higher than for international students (80%).

Using data provided by UCLA from the National Student Clearinghouse, we have some information on where PhD students who turn us down end up matriculating instead. Data from fall 2015 to fall 2020 indicate that students primarily went to other top programs in EEB, including UC Davis (13%), Stanford (9%), University of Chicago (7%), Yale (6%), UC Berkeley (6%), and Duke (6%).

G. Admissions Timelines

The general timeline for admissions is as follows:

Dec 1: Application deadline for PhD and Biology MS programs
Mid-December: Applications reviewed by potential faculty advisors and nominations submitted to GAS committee.
Late-December: GAS meets to discuss and finalize rankings of all nominated applicants.
Early January: Select applicants to receive invitations to recruitment event.
Early February: Recruitment event held over 2-3 days.
Late February through April: Admissions offers sent to successful applications.
April 15: National deadline for students to accept admission offers.
May 1: Application deadline for EvMed MS program.
May: Review of EvMed applications.
June: Admissions offers sent to successful EvMed applicants.

3. Time-to-degree and Student Progress

A. Sequence of Expected Milestones

Upon arriving at UCLA, all students immediately form a *Guidance Committee* consisting of their provisional major professor and two other (EEB) professors. The guidance committee reviews undergraduate coursework and experiences as well as desired research and career goals, and helps determine what courses each new student should take in their first year or two of study. The Guidance Committee serves in an advisory capacity until the student formalizes their *Dissertation Committee* when they Advance to Candidacy (ATC).

Students in the PhD program must enroll for a minimum of 12 units each quarter. To meet graduation requirements, students must: complete a minimum of 20 units of graduate-level courses (200-series; see §E.7 Curriculum); complete three quarters of teaching assistant experience; pass a departmental written qualifying exam (DWQE) at a level expected from a doctoral student; pass a university Oral Qualifying Exam (OQE), upon which they ATC and become Doctoral Candidates; and complete a doctoral dissertation.

The DWQE is required to be completed during the first year of study. Preparation for the DWQE is provided within required first-year graduate courses, including 200A (fall term) and 200B (winter term; see §E.7 Curriculum). The form of the DWQE is a written research proposal, starting with a literature review, of a total of 8–10 pages (plus references, figures, etc.). The proposal should be a serious attempt to develop a research project that might be carried out, but there is no requirement that it become the student's dissertation research. The first draft of the DWQE is due to the student's Guidance Committee by the end of week 1 of Spring quarter in their first academic year. Students then receive feedback on the DWQE which they

incorporate before submitting the final version of the DWQE by the end of week 8 of Spring quarter. The Guidance Committee grades the proposal as "not pass", "MS pass", or "PhD pass". PhD students who receive a "not pass" or "MS pass" are typically given additional time and opportunities to improve their DWQE in order to achieve "PhD pass" as judged by the faculty DWQE coordinator. Students must pass their DWQE before the beginning of their second academic year or risk academic disqualification.

The OQE (also referred to as the ATC exam) is the final milestone for PhD students prior to submitting their dissertations and represents the progression to PhD Candidate. Students are recommended to complete their OQE at the end of their second year, but no later than the end of winter quarter of their third academic year. Students who do not complete their ATC by this deadline risk losing fellowship support and access to departmental grants, and may eventually be recommended for academic disqualification by the Vice Chair for Graduate Studies. Prior to the OQE, students must officially form and register their Dissertation Committees (typically 4 faculty, with their major advisor as Chair), according to UCLA regulations. For the OQE, students prepare an original written dissertation proposal, which is then presented and defended during the OQE. OQE's typically last 2–3 hours in duration. After a brief presentation of the dissertation plans, committee members orally question the candidate on their dissertation plans as well as on the student’s broader understanding of subjects pertinent to the proposed work. The OQE is graded as "Pass" or "No Pass". After the OQE, the Dissertation Committee additionally decides whether the candidate will be required to sit for a PhD Defense upon completion of the written dissertation. In EEB, it is extremely rare for PhD Defenses to be required (i.e., no defenses have occurred in EEB in recent memory). Once progressing to PhD candidacy, students are required to meet annually with their Dissertation Committees which is documented via a brief form.

To graduate, PhD students must complete and file their dissertation and receive formal approval (i.e., signatures) from their Dissertation Committee. Finishing students are also strongly recommended to give a public seminar on their research (i.e., "exit talk"), such as during a weekly EEB seminar slot, although there is no formal requirement to do so. PhD students are expected to graduate within 5 years, which is the period of time in which the department guarantees funding (*n.b.* a 6th year of funding was guaranteed to all students enrolled in spring and fall 2020 due to disruptions from COVID-19 pandemic; see §E.4 Financial Support).

The Masters program (both Biology MS and EvMed) consists of at least nine courses completed in graduate standing, of which at least five must be graduate (200-series) courses (see §E.7 Curriculum). The remaining courses may be in the 100, 200, or 500 series. No more than two 596 courses (eight units) may be applied toward the nine courses required for the degree; only one 596 course (four units) may be applied toward the minimum five graduate courses required. Courses in the major that are taken for S/U grading may not be applied toward the minimum requirement. Courses outside of the major that are taken for S/U grading may be applied toward the minimum requirement if they are deemed applicable and provided that no more than one such course is taken per quarter. Specific course requirements are established individually for each student by their Guidance Committee. Masters students, like PhD students, are expected

to complete the DWQE in their first year, although the written expectations are reduced as students need only achieve the score of "MS Pass".

Masters students can elect to graduate through either the Capstone Plan or the Thesis Plan. Upon passing the DWQE, students automatically satisfy the conditions for graduating under the Capstone Plan. For the Thesis Plan, students must form a Thesis Committee (minimum of three faculty, chaired by Major Advisor, and following UCLA guidelines) and prepare and file a master's thesis in accordance with University guidelines. MS students are expected to graduate within 2 years, which is the period of time in which the department guarantees funding (see §E.4 Financial Support).

B. Normative Time-to-Degree

Using data for all EEB graduate students who completed their degree from fall 2014 to present, the median TTD is 6 years for PhD students and 2 years for MS students (combined across both MS programs). Ninety percent of PhD students finish in 7 or fewer academic years while 90% of MS students finish in 3 or fewer academic years (Figure E4).

Many factors can potentially influence TTD. For PhD students in particular, the COVID-19 pandemic beginning in spring 2020 delayed many students' research progress, leading to potential extensions of TTD for current students at that time (Figure E5). Other factors may have to do with preparation for graduate school, student background, and funding sources. Comparing URG and non-URG students, both MS and PhD, there is no difference in TTD as a function of this classification – both demographic pools graduate have nearly identical TTD distributions (Table E1).

Table E1. Time-to-degree (TTD) by degree program and URG status for students who completed their EEB program between fall 2014 and spring 2024.

Degree and Status	Mean TTD	10% Percentile TTD	Median TTD	90% Percentile TTD
MS: URG	2.2	2	2	3
MS: non-URG	2.4	2	2	3
PhD: URG	6.1	5	6	7
PhD: non-URG	6.2	5	6	7

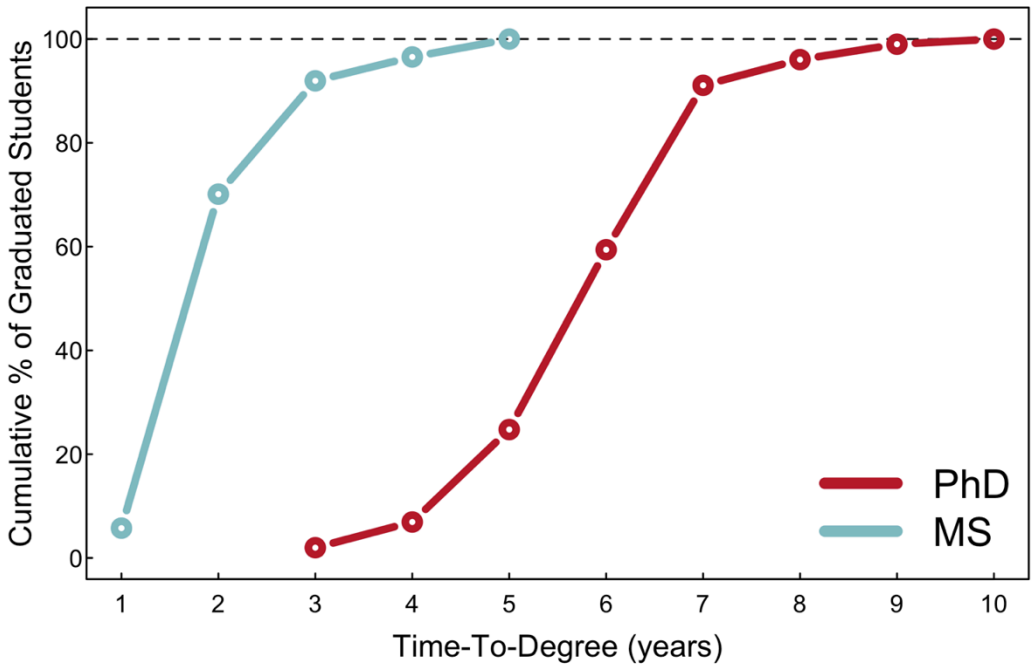


Figure E4. Cumulative % of EEB graduate students since 2014 that have graduated at or under a given TTD. Lines begin and end at the range of values (i.e., shortest to longest TTD).

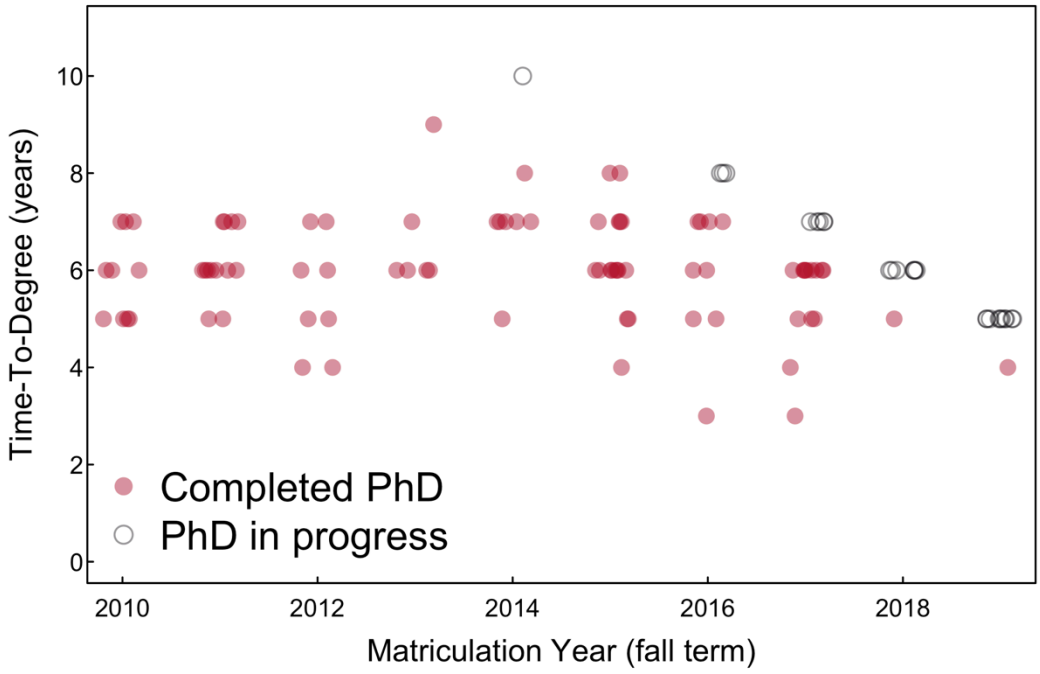


Figure E5. Time-To-Degree (TTD) as a function of matriculation year for PhD students in EEB. Open circles represent students still in the program who have not yet finished, demonstrating how TTD distributions for 2014 and 2016 onward are necessarily incomplete. Students beginning in the 2013 cohort potentially experienced delays due to the COVID-19 pandemic.

Another potential source of variation in TTD, for PhD students in particular, is their source of funding. Students are generally admitted with funding packages that are a mixture of fellowship and teaching assistantships (ASE; see §E.4 Financial Support), but individual students and labs have varying degrees of success in receiving internal or external fellowship (or GSR) support. Fellowships and GSRs generally provide more uninterrupted opportunities for students to progress their dissertation research, without the sometimes all-consuming distraction of teaching (although graduate student teaching remains an integral component of EEB's overall educational mission). In analyzing the data for students who have graduated beginning in 2014, we find a marginally significant effect (p = 0.055) of the proportion of time spent teaching as a TA on a student's TTD (Figure E6). As expected, teaching as a TA has the potential to extend one's TTD. The magnitude, however, is not massive: a student whose only source of funding is through TA has an expected TTD of approximately 11 months longer than a student who never TAs at all (Figure E6).

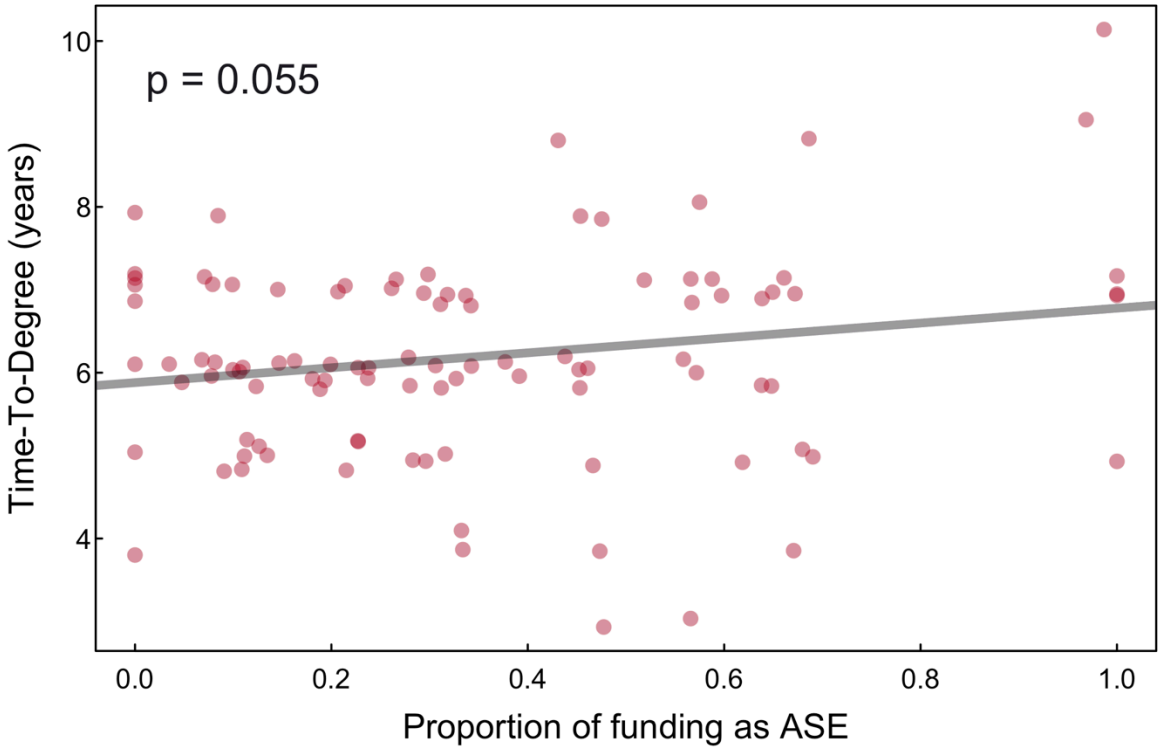


Figure E6. Time-To-Degree (TTD) as a function of the proportion of funding ($$) spent employed as a TA (ASE). Points show individual students who completed their PhDs between fall 2014 and spring 2024. A linear regression (solid line) is shown along with significance of the slope (p-value).

C. Program Attrition

Overall program attrition is low. Of 126 matriculated PhD students since fall 2014, 9 (7%) have left the university with no degree (i.e., withdraw prior to ATC) and an additional 9 (7%) have left the university with an MS degree (i.e., change of degree objective). Thus, in total over this period, 14% of matriculated PhD students did not complete their degree objectives. Of 91 matriculated MS students over the same time period, 2 (2%) have left without receiving a degree (i.e., withdrawn).

Reasons for students to fail to meet their degree objective are complex and not fully known, but include family, health, and financial concerns, changes in career objectives, lab transitions (e.g., when a major advisor leaves UCLA and the student transfers institutions), and failing to meet degree milestones (§E.3.A). In 2022, EEB graduate staff began conducting exit interviews with students who left the program prior to completing their degree objective, but current sample size of these interviews is insufficient to currently provide reliable inference on trends.

Demographically, combining across MS and PhD programs, students who left without completing their matriculated degree objective were 70% female and 30% male, compared to an overall EEB student body that is 62% female and 36% male. Similarly, students who left were 30% URG, compared to an overall student body that is 23% URG. Neither of these demographic summaries of withdrawing students are statistically different from what one would expect due to chance alone (χ^2-test: $p = 0.834$ for difference by gender; $p = 0.751$ for difference by URG status).

Another potential contributing factor to student attrition is the novel series of challenges posed by the COVID-19 pandemic and social upheaval that occurred from 2020 through 2021 (see §G.1 and §G.7). Substantially more students left the program in the aftermath (calendar years 2021 and 2022) than in the preceding 7 years (Figure E7).

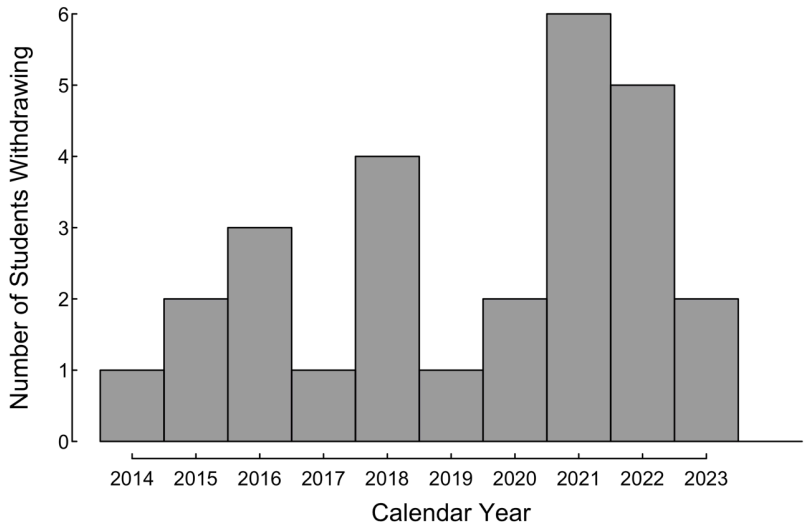


Figure E7. Number of students (combined MS and PhD) by calendar year who withdrew from their matriculated EEB degree program (including PhD students who left with a MS degree).

4. Financial Support

A. Typical Support Package

For the past decade, EEB has used a consistent minimum financial support package for all admitted graduate students. Although the dollar amounts have changed (see §E.4.C), the philosophy behind our support model has remained the same. For PhD students, our financial support has hinged on three underlying tenets: (1) PhD students have better outcomes when TA obligations are minimized (for evidence, see Figure E6); (2) guaranteed fellowships are more impactful earlier in the PhD when students are taking classes and developing their research ideas; (3) guaranteed summer funding alleviates the need for students to search and apply for funding, thus improving degree outcomes; and (4) guaranteed funding from any source is necessary for a minimum of 5 years. Given these tenets, our minimum support package for admitted PhD students has followed a model as follows:

Year 1: 2 quarters Fellowship, 1 quarter TA, summer Fellowship
Year 2: 3 quarters TA, summer Fellowship
Year 3: 3 quarters TA, summer Fellowship
Year 4: 3 quarters TA, summer Fellowship
Year 5: 3 quarters TA

The complete package includes not just stipends but also covers full tuition and (most) student fees (totaling $21,115 per year for in-state students, and $36,297 for out-of-state or international

students; students who are U.S. citizens are expected to establish residency within their first year). Many students receive additional funding, both intramural and extramural (see §E.4.B), which is primarily used to release students from TA obligations, replacing those quarters with funding via fellowships or Graduate Student Researcher (GSR) opportunities (e.g., via PI grants). Although our minimum support package only guarantees funding through Year 5 of the PhD, our department nearly always has enough surplus TA opportunities that students in Year 6 or beyond are able to continue supporting their academic progress through TA-ships (and/or PI grant support).

For MS students (including Biology MS and EvMed), we provide a guaranteed support package for two years following a model as follows:

Year 1: 3 quarters TA
Year 2: 3 quarters TA

This package also covers full tuition and fees, similar to PhD students. Although we do not provide summer fellowships for MS students, as we prioritize use of those monies for PhD students, many MS students are able to secure summer TAships, thus providing full-year income.

B. Intramural and Extramural Support

EEB students are eligible for and receive a wide variety of additional funding support, but from intramural and extramural sources. Many of our most generous intramural funds go to helping attract top talent and are thus awarded during the admissions period each spring. EEB has three primary endowed funds which provide sufficient income so as to support at least one admitted student each year. In each case, recipients receive 2 fellowship quarters in Year 1, 4 summer fellowships, $1000 research stipend available in Year 1, and tuition covered (thus, these awards provide the same funding as the standard departmental offer with the exception of an additional $1K research stipend in Year 1). The endowed funds and their disciplinary restrictions are as follows:

1. Vavra: organismal plant biology
2. Josephine Reich: environmental education/research
3. Lida Scott Brown: avian biology

In addition, each year, EEB receives funding from the Life Sciences Dean’s office to recruit top applicants. These additional awards and fellowships – which vary from year to year – are frequently used to (1) replace the TA quarter in Year 1 with a fellowship; (2) replace TA quarters in later years with fellowships; and/or (3) provide supplemental fellowship income beyond the departmental minimum offer.

Admitted students can compete for awards and fellowships offered by UCLA's Division of Graduate Education (DGE). These awards include the Graduate Dean’s Scholar Award, which is awarded annually to EEB's top applicant (based on Admissions Committee ranking) and

provides supplemental stipends in Year 1 and during two summers. DGE also awards competitive fellowships aimed at recruiting top PhD applicants from populations underrepresented in graduate education. One of these fellowships is strictly for graduates of HBCUs and provides 6 years of guaranteed funding, while the other is the Eugene V. Cota-Robles Fellowship, which provides full fellowships in Year 1 and Year 4. Between fall 2014 and fall 2023, EEB matriculated a total of 5 students with UC-HBCU fellowships and 25 students with Cota-Robles fellowships.

Both incoming and continuing students are eligible to apply for multi-year external fellowships, particularly through U.S. federal agencies. For EEB, the National Science Foundation (NSF) is the primary source of external student fellowships, but EEB students have also received fellowships from the National Institutes of Health (NIH), the National Aeronautics and Space Administration (NASA), and the Howard Hughes Medical Institute (HHMI). Between fall 2014 and fall 2023, a total of 38 EEB graduate students (13%) have been funded through NSF's Graduate Research Fellowship Program.

Finally, all ongoing students, PhD and MS, can apply for departmental support through a competitive awards call (annually, each spring quarter). Support from various gift funds, departmental monies, and endowed funds are all consolidated into a single application which is evaluated by the Graduate Admissions and Support Committee (composed primarily of faculty with 1 student representative). Awards are made on the basis of several factors, including eligibility, intellectual merit, progress to date, financial need, current support and past support. Awards can take multiple forms, including: quarter fellowships (stipend and fees), fee fellowships (fees only), summer fellowships (stipend only), non-resident student tuition fellowships (non-resident fees only), research grants (up to $2000), and conference travel grants (up to $1500).

C. Distribution of and Trends in Support

Graduate student funding can generally be divided into three categories: TAships, fellowships (combining intramural and extramural), and GSRs (primarily from faculty start-up or research grants). Between fall 2014 through spring 2024, EEB paid out over $24,000,000 in salary or stipend to graduate students. As a percentage of total $ awarded via stipends or salary, 41% of support came through TAs, 49% through fellowships, and 10% through GSRs. These percentages differ by degree program, however; for MS students, 88% of support came through TAs, 9% from fellowships, and 3% from GSR; whereas for PhD students, 31% of support came through TAs, 56% from fellowships, and 13% from GSR. As students from different backgrounds are eligible for different fellowships (e.g., Cota-Robles), the percentage of student support across these three sources also varies by URG status (Table E2), with URG students, on average, receiving a greater percentage of their income from fellowships.

Table E2. Type of funding support (as a % of total $ spent, fall 2014 to spring 2024) for EEB graduate students, separated by student type and URG status.

Degree and Status	% TA	% Fellowship	% GSR
All degrees: URG	42	52	7
All degrees: non-URG	52	37	11
MS: URG	85	14	1
MS: non-URG	89	8	4
PhD: URG	26	66	8
PhD: non-URG	33	53	14

These different types of support (TA, fellowship, and GSR) derive from different sources. As a percentage of total income awarded via stipends or salary from 2014–2024, 54% came from departmental sources (including Life Science allocations), 12% from faculty, 19% from Graduate Division, 12% from individual fellowships to students, and 3% from institutional training grants. Excluding dollars awarded via TAships (i.e., restricting to funds that generally derive from competitive fellowships or grants), 21% came from departmental sources, 21% from faculty, 32% from Graduate Division, 21% from individual fellowships, and 5% from training grants.

Income and support for EEB graduate students has generally increased over time (Figure E8). Comparing data from the 2014 EEB self-review to the 2023-2024 academic year, in 2014, EEB funded fellowships, GSRs, and TAs at a constant, equal rate of $6500 per quarter, including summer fellowships, for an expected minimum annual pre-tax income of $25,500. For students in the 2024-2025 academic year, EEB funds departmental fellowships at a rate of $8641 per quarter (matching UCLA's Step 1 rate for GSRs) and summer fellowships at $7500. The pay rate for TAs and GSRs has been increasing regularly over time due to UAW union contracts, particularly following new contracts ratified in early 2023. Thus, for fall 2024 incoming PhD students with the base funding package (see §E.4.A), they will receive an annual pre-tax salary of $36,948. This base salary represents an increase of 45% over the last decade, which exceeds Consumer Price Index inflation over the same time period (32%; U.S. Bureau of Labor Statistics).

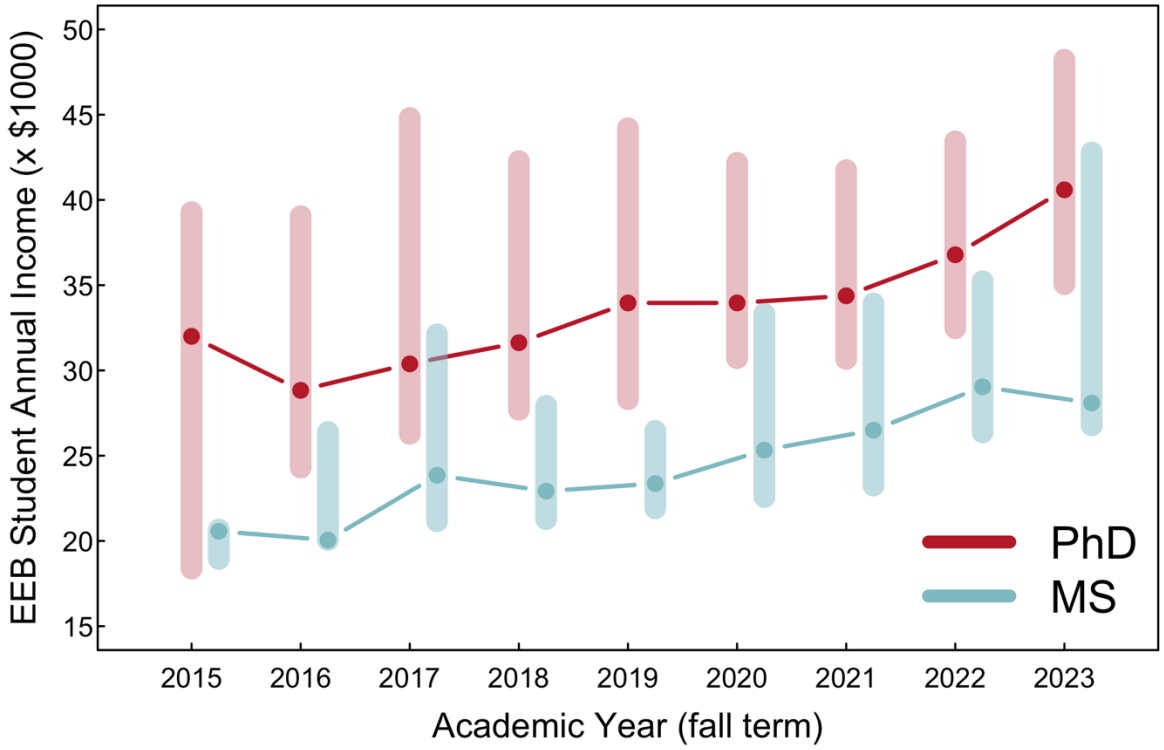


Figure E8. Trends in annual income from 2015-2016 academic year through 2023-2024 academic year. Points and line follow trends in the median annual income for MS and PhD students, while the bars represent the central 80% of student incomes.

The standard first-year student income is not necessarily typical of the EEB graduate student population as a whole. For PhD students, the median annual income from 2015 to 2017 was $29,730, while during the last academic year the median was $40,590. For MS students, who primarily receive funding via TAships but do not receive guaranteed summer funding awards, the median annual income for the same time period has risen from $20,050 to $28,090.

5. Student Mentorship and Success

A. Mentorship Model

EEB faculty are dedicated and talented mentors to graduate students. Mentorship in EEB is the primary responsibility of each student's research advisor and supervisor (who also serves as thesis or dissertation Chair), a relationship which is assigned during admissions (see §E.2.A). During the first two years, students also receive mentorship from their Guidance Committees, and after PhD students ATC, they receive mentorship from their full Dissertation Committees.

Faculty advisors are expected to meet regularly with their graduate students, with most EEB faculty meeting weekly to biweekly with their students during the academic year. Nearly all EEB faculty also hold regular lab meetings, where research updates are presented and/or new published research is discussed (i.e., journal club style). In addition, students are required to meet with their Guidance Committee at least once during their first year (at the beginning of fall term; documented via a form outlining recommended classes, see Appendix E3: 23-24 PhD First Year Course Plan), and then PhD students are required to meet annually with their Dissertation Committee once they are PhD Candidates. Beginning in the 2023–2024 academic year, Annual Committee Meetings require a summary and documentation to be submitted to the EEB Grad Office in order to track compliance (see Appendix E4: Annual Committee Meeting Summary).

All academic affairs for graduate students in the department are directed by the Vice Chair for Graduate Studies who is assisted by the Faculty Graduate Advisor and administrative staff. The Vice Chair for Graduate Studies generally assists students with issues of policy and conflict, whereas the Faculty Graduate Advisor serves primarily as an advocate for students who can help connect students to resources when problems arise. During the period of evaluation, the following faculty have held the position as Vice Chair for Graduate Studies:

2014–2017: Peter Nonacs
2017–2021: Nathan Kraft
2021–2022: *Vacant*
2022–present: Morgan Tingley

B. Professional Development Opportunities

Graduates from EEB with both MS and PhD degrees go on to a wide variety of careers (see §E.6), necessitating diverse and varied professional development opportunities. Individual graduate advisors are primarily responsible for the staged development of graduate students within their labs, where advisors are expected to instruct students in key skills such as: grant writing, experimental design, scientific programming, manuscript and report writing and preparation, field and lab techniques, scientific inference, and many more. Beyond this, EEB provides professional development via several other venues:

EEB 200 series: EEB provides two core courses for new graduate students, including EEB 200A (fall quarter) and 200B (winter quarter). While the exact curriculum of these courses has changed from year to year – and is subject to regular evaluation from the Graduate Curriculum Committee – current formulations of the course focus on the following key professional skills: (1) reading and discussing scientific literature from a wide variety of biology-related sub-fields; (2) synthesizing diverse literature into a coherent summary; (3) developing testable hypotheses and inferential methods; and (4) communicating and presenting one's ideas to diverse audiences.

EEB c237: A course on communication of science to informal audiences. This course was taught annually from 2015 to 2019, but not since.

EEB 250: A required course for PhD students and optional (but well-attended) for MS students, this course is titled "Professional Skills in the Biological Sciences." Through 2019, EEB 250 was primarily a grant-writing course whose aim was to guide and prepare students to submit NSF GRFP proposals in November of the fall term. However, in the late 2010s, NSF moved the GRFP deadline much earlier in October (close to the beginning of UCLA's fall term), making it difficult or impossible for 250 to serve as an effective grant-writing course. Coincidentally, changes in NSF eligibility meant that fewer current students could apply for GRFPs and more students were entering our program having already received them. Thus, in the 2017–2018 academic year, EEB 250 was moved to Winter or Spring quarter and pivoted to focus more on a diverse set of professional skills as well as preparing students for post-graduate careers. In its current form, EEB 250 includes 4 panel discussions with EEB alumni currently employed in diverse career sectors (e.g., academia, government, NGO, industry, creative), a postdoc panel, and activities and discussions aimed to demystify academia's 'hidden curriculum'.

EEB 297 Seminars: Each academic year, EEB faculty teach a wide variety of *ad hoc* graduate seminars under the listing of EEB 297. Many seminars are specific to certain scientific topics, but a wide variety are also aimed at building professional skills. Examples of seminars that develop professional skills are as follows:

W2021, *Theory and Practice of Peer Review*: Seminar devoted to exploring the role of peer review in science, including how to perform an external peer review, and how (as an author) to respond to peer review.

F2022, W2023, S2024, *The Academic Job Search*: Offered in many years coincident with ongoing EEB faculty searches, the seminar walks students (and postdocs) through the steps of an academic job search, providing insight on how to apply and be successful in faculty hiring.

There are also many opportunities for EEB graduate students to participate in departmental affairs, thus gaining professional development experience. The number and variety of committee positions available to graduate students has changed over the last 10 years; the current set of committee positions open to graduate students each year include:

Faculty-Student Liaison: Perhaps the most important role, the faculty-student liaison is a graduate student who, each year, serves as the primary conduit for communication and connection between faculty and the EEB graduate student body. The primary responsibilities of the liaison are to attend monthly faculty meetings, write and share notes of faculty meetings with graduate students, meet regularly with the Graduate VC and other faculty leaders as needed, and to recruit students for other committee positions.

Departmental Awards Committee: Decides departmental awards as part of annual (spring) call.

Graduate Admissions and Support Committee: Serves on the admissions committee to review and rank graduate applications; helps review and decide on allocation of financial awards, including both continuing student funding awards and fellowship awards made during the admissions process.

Committee for Anti-Racism and Equity (2 positions): Participate in regular CARE meetings and help write annual reports.

Graduate Curriculum Committee (2 positions): Help review, recommend, and change policy on graduate curriculum.

Seminar Committee: Organize EEB seminar series, including the selection, invitation, and hosting of internal and external speakers.

Finally, EEB has a standing student-run Professional Development Committee that aims to nimbly provide content and opportunities each year based on graduate student requests. The Professional Development Committee representatives work with the VC for Graduate Studies to put on events (e.g., career panels or discussions) and to develop resources. Over the past academic year, for example, the Professional Development Committee has worked with the Grad VC to develop a directory of EEB graduate student alumni who are open to hearing from current students in order to provide career advice and counseling.

C. Graduate Student Accomplishments

EEB graduate students are incredibly accomplished. Although we do not have data from all years covered under this report, we provide summaries of accomplishments in the following areas for the time periods available

(a) accomplishments of mentees

EEB graduate students are incredibly productive. In our field, peer-reviewed publications are the primary currency of academic accomplishment, and graduate advisors are encouraged to support students in the publications of the MS or PhD chapters as well as additional collaborative endeavors that may be created over the course of their time at UCLA. Many students publish chapters in scientific journals as they move through the program, although often chapters will not be accepted and published by journals for several years after the student has graduated.

To assemble a database of peer-reviewed publications by UCLA EEB graduate students during the review period, we (1) queried Google Scholar, attempting to match registered student names with Google Scholar profiles and (2) asked current faculty to input publications that they saw as missing from the query. Neither of these methods yields a complete sample, as Google Scholar only retrieves publications from individuals with registered Google Scholar profiles (only 45% of graduate students have registered profiles) and manually entering publications over a decadal period is onerous. The resulting list was de-duplicated and can be found in Appendix E5: EEB Grad Publications.

During the review period, UCLA EEB graduate students authored at least 667 peer-reviewed articles across 293 different scientific journals, including numerous papers in the best possible journals in the Life Sciences, including: 5 in *Science*, 3 in *Nature*, 4 in *Nature Communications*, 9 in *Proceedings of the National Academy of Sciences*, 7 in *Science of the Total Environment*, 1 in *Science Advances*, 5 in *Scientific Reports*, and 15 in *Proceedings of the Royal Society B*. Of the students for whom we have publication data available, the median number of publications during their time as graduate students is 4, with a range from 1 to 20. The number of papers published by graduate students per year ranged from 37–51 during the review period, with a peak of 96 papers in the year 2021.

(b) research awards or extramural fellowships

Over the period since the last reporting period, EEB graduate students have received at least 61 research awards or extramural fellowships (see Appendix E6: EEB Grad Research Awards and Extramural Fellowships). This includes 24 NSF Graduate Research Fellowships, 3 NASA FINESST Fellowships, and 1 Ford Foundation Fellowship.

(c) special distinctions

EEB does not officially track student distinctions, but an available selection is presented below:

Beichman, Annabel	2019	Fitch Award Finalist, Society for Molecular Biology and Evolution
Gostic, Katelyn	2016	Charles E. & Sue K. Young Graduate Student Fellowship Award
Helman, Sarah	2014	GRFP Honorable Mention, National Science Foundation
Hongjamrassilp, Watch.	2021	Honorable Mention--Allee Award for Best Graduate Student Research, Animal Behavior Society
Kandlikar, Gaurav	2019	Student Research Award, American Society of Naturalists (ASN)
Kandlikar, Gaurav	2019	Murray F. Buell Award for Excellence in Ecology, Ecological Society of America (ESA)

Kyriazis, Chris	2019	Hamilton Award Finalist, Society for the Study of Evolution
Kyriazis, Chris	2020	Young Investigator Award, Society for Molecular Biology and Evolution
Kyriazis, Chris	2022	Graduate Student Excellence Award, Society for Molecular Biology and Evolution, Society for Molecular Biology and Evolution
McFadden, Ian	2017	Student Poster Award, International Biogeography Society
Tonelli, Ben	2021	AOS Council Best Student Presentation Award, American Ornithological Society
Wolff, Richard	2021	GRFP Honorable Mention, National Science Foundation
Wolff, Richard	2023	Best talk award, Microbiome Virtual International Forum

(d) successful transitions into professional employment and postdoctoral training

Based on data compiled by DGE from responders to the Doctoral Exit Survey conducted between 2019–2023 (see Appendix E1: DGE Report Biology PhD) 91% of graduates from our program left going directly into employment. Of those, 76% accepted work at an academic or research institution, presumably as postdoctoral researchers, while the remaining 24% were approximately equally divided between employment in industry, NGO, and government.

D. Evidence for Effective Mentoring

EEB faculty care extensively about the quality of mentoring, and in recent years, faculty have repeatedly held training sessions and meetings about strategies to improve mentoring.

Quantitative data on mentoring effectiveness is limited but comes from surveys sent out to EEB graduate students in winter 2021 by the Graduate Council Survey (see Appendix E7: Graduate Division Data Packet). Among MS students (response rate = 24%), 75% indicated satisfaction with their advising from faculty and 75% indicated satisfaction with the overall quality of faculty mentoring in the program. Among PhD students (response rate = 41%), 62% indicated satisfaction with advising from faculty and 54% indicated satisfaction with the overall quality of faculty mentoring in the program. This medium level of satisfaction is mirrored in the 2021 EEB Graduate Student Climate Survey (see §G.7 and Appendix G7: EEB Grad Climate Survey Report).

There are many reasons why we might expect mentoring satisfaction to be abnormally low in 2021 and it is unfortunate that our only two sources of data come from the same year during the aftermath of the COVID-19 pandemic and related disruptions. For more context, see §G.7. Mentoring quality was less frequently mentioned than other issues in the 2024 EEB Graduate Student Survey (§G.7.D).

6. Career Placement

Alumni from our graduate program go on to a wide variety of careers. EEB does not formally track the careers of its alumni, largely due to the lack of personnel support necessary to accomplish this task. In 2024, faculty attempted to create a directory of all alumni who had completed our graduate programs from 2014 to the present. Of the 27 respondents who offered to share their career status and information with current graduates, 44% indicated current employment in R1/R2 academic jobs, 22% were employed in governmental positions, and equal numbers (7%) were employed in either teaching-intensive academic jobs (e.g., PUI or M1/M2), NGOs, or industry. Other careers include in visual arts (e.g., filmmaker), law, and medicine.

Communication about career placement to current students primarily through our first-year curriculum. In its current form, EEB 250 (§E.7) includes 4 panel discussions with EEB alumni employed across different sectors (academia, government, NGO, industry, creative). These panel discussions are recorded and made available to all current graduate students each year by the course instructor.

7. Curriculum

Graduate Curriculum includes a series of required courses, recommended courses, and optional courses. Specifics on required courses and how they have changed over the reporting period are found in Appendix E8: EEB Grad Course Requirements. A full list of all courses offered (and the years in which they have been offered) can be found in Appendix E9: EEB Grad Courses.

Ph.D. students must take the following courses during their first year: Ecology and Evolutionary Biology 200A, 200B, and 250. Students also must complete an advanced course (100 or 200 series) in statistics, biomathematics, or bioinformatics (generally, c202 or c219). Other specific course recommendations are established individually for each student by the guidance committee.

8. Summary: Strengths, Opportunities and Challenges

A. Strengths

Recruitment of a diverse student body: EEB's graduate student body is an outlier within our field with respect to diversity (see §E.2.F and §G.6.B). On average, our student body has been composed of 34% individuals underrepresented in graduate school, while the national average for EEB graduate programs is somewhere between 7–13% (see §G.6.B). Reasons for this strength are complex – and geography is in our favor – but recruiting a strongly diverse student body has been a stated departmental goal throughout the current review period, with many different ongoing efforts at the individual PI level (e.g., The Diversity Project, led by Profs. Barber and Fong), department level (e.g., an early abandonment of the GRE; the adoption of

holistic admissions rubrics), and university level (e.g., strong fellowship support via Cota-Robles).

Scaffolded development of student research: Throughout graduate student feedback (see §G.7), students indicate that in retrospect, they felt well-prepared by the sequence of expected milestone (e.g., DWQE and OQE), the format of those milestones, and the support from PIs and classes to prepare for those milestones (see §E.3). This likely contributes strongly to our time-to-degree statistics (see §E.3.B) which have a median time-to-degree consistent with all UCLA graduate programs (6 years) but which has an upper tail much shorter than UCLA as a whole[1]. Our time-to-degree statistics also do not show an equity gap (i.e., TTD is equal for students based on gender and URG-status) and evidence indicates that our preferred funding model (i.e., prioritizing student fellowships over teaching, when possible) results in slightly reduced degree progression.

Mentored accomplishments: Our graduate students are enormously successful. Our graduate students regularly receive top competitive extramural fellowships (see §E.5.C.b); for example, during the review period, 24 students received Graduate Research Fellowships from the National Science Foundation (9% of all registered MS or PhD students during the period). EEB graduate students are also publishing frequently in the peer-reviewed scientific literature (see §E.5.C.a) – the primary form of scientific achievement in our field. During the review period, EEB graduate students published at least 667 scientific articles, with a median of 4 papers per student, including 17 articles in the highest-profile science journals (*Science*, *Nature*, and *Proceedings of the National Academy of Sciences*). While it is impossible to quantify the contribution of faculty mentoring to these metrics, faculty support and guidance are generally necessary components of these successes during graduate work.

B. Opportunities

More ability to recruit top talent: Running an internationally recognized graduate program requires being able to successfully recruit top international talent among our applicants. Currently, university assessment of supplemental non-resident tuition (NRT; see §E.2.G) hampers our faculty's ability to provide funding offers to international students, resulting in an extremely low rate of admissions offers to international students (§E.2.F). The graduate student union (UAW) has also identified NRT costs to graduate students as an impediment and equity burden with the goal of asking the university to treat international graduate students more like out-of-state students. Multiple pathways forward exist, but success likely will depend on the university changing or shifting policies on international graduate students.

A culture of celebrating successes: EEB has no current requirement of an 'exit talk' or public 'defense' (§E.3.A), leading many students to finish their PhDs with an anticlimactic electronic dissertation submission. In the last two years, efforts have begun to encourage finishing students to formally present their theses to the department at celebratory events and the Seminar Committee is currently reserving space for students to do so. Such voluntary exit talks

[1] https://grad.ucla.edu/graduate-program-statistics/outcomes/?t=TTDDash

have been very popular, although creating a student culture that voluntarily gives such talks has been challenging. However, given the positive momentum in this regard, the opportunity exists for the department to change its graduation requirements to require an exit talk (in some form) that would formally provide a venue to celebrate each graduate's successes.

Role of Evolutionary Medicine MS program: The Evolutionary Medicine (EvMed) masters program provides many opportunities to flexibly support the health of EEB and its graduate student body while serving a broader audience, but the program has suffered from inconsistent applications and variability in faculty interest in accepting students into the program. The program would strongly benefit from 1–2 faculty members who would commit to teaching annually as part of the curriculum and who would lead the program forward.

A richer graduate curriculum: The graduate curriculum has been overhauled multiple times since the last review period in response to evolving faculty and student feedback, with a focus on 'getting right' the handful of uniformly required courses. The current curriculum provides a broad introduction to the research happening in the department and provides training and professional skills that are helpful in a wide variety of careers, but the current curriculum lacks disciplinary depth. Students wishing to formally learn the foundations or recent advances in their specific discipline (e.g., ecology or evolution or behavior) are mostly left to self-learning or the whims of faculty teaching one-time 297 courses. The need for consistently-taught, elective, advanced graduate courses in EEB disciplines is broadly recognized.

Agreement on minimum graduate mentoring standards: In recent years, the faculty have worked hard (motivated in part by feedback from graduate students) to participate in mentorship training and to deepen our evidence-based mentoring practices. Successes include widespread faculty engagement in the CIMER Project 'Entering Mentoring' curriculum, the codification of annual committee meetings for PhD students, and the development of an EEB-specific advisor-advisee mentoring compact template. However, our department would benefit from investment in the development of a faculty-wide agreement on minimum standards for graduate mentoring.

C. Challenges

Graduate funding in a shifting financial landscape: In some ways, EEB is strongly positioned for graduate funding due to the rich array of TA opportunities that the department controls, affording us the ability to guarantee funding for 15–20 admitted students every year. However, worrisome trends exist: (1) flexible block grant funding which supports our first-year fellowships and summer awards has declined every year for 5 years (10% cumulatively during that time); (2) costs in funding students have drastically increased due to union victories and associated tuition increases; (3) potential future costs are uncertain given short union contract durations, disincentivizing funding graduate students in grants; and (4) DGE and Labor Relations recently forbade EEB's use of non-student tutors (see §B.5 and §D.4.C) to fill in teaching gaps, forcing a more careful alignment of teaching need and supply. The easy solution is to accept students without guaranteed funding (e.g., a partially funded MS degree), but such a policy could create equity gaps, thus working counter to the department's diversity goals.

Graduate student climate: Recent standardized metrics of graduate student climate (§G.7) are skewed negative. While there are reasons (see discussion in §G.7 and §G.12) why relying on benchmarks solely from 2021 are not representative of student climate either presently or throughout the review period, the faculty acknowledge that student morale continues to feel low and some underlying tensions remain between students and faculty. Many faculty attribute the current climate to the historic grad student strike in winter 2022–2023. Since the strike's conclusion, the faculty have tried to increase the frequency of low-stakes social gatherings, but with more strikes potentially in the near future, this remains a challenge.

Consistent and dedicated teaching of graduate courses: The EEB teaching model (1) prioritizes undergraduate classroom teaching and (2) provides faculty the flexibility to advocate strongly for teaching the classes that they prefer to teach. Thus, graduate curriculum and teaching is largely an afterthought, especially when faculty wish to avoid teaching core graduate education courses because they provide the same 'credit' on paper as teaching a small, seminar course. This issue could be solved by having a transparent and standardized way to determine equity in teaching loads for both undergraduate and graduate education across faculty, but this solution has so far remained elusive within the department.

Graduate staff workload and the role of student support services: Our Graduate Staff are at the front lines of maintaining the structure and function of our graduate program. Until recently, EEB graduate staff had the time and job duties to answer every student question promptly and meet individually to allay fears and calm nerves. The importance of these roles is echoed repeatedly in student feedback, including the 2021 EEB Graduate Student Climate Survey. In recent years, University and Departmental changes – such as the adoption of UCPath, departmental re-allocation of hiring duties to Graduate Staff, and the massive changes to hiring administration since the 2022 strike – have made our graduate staff positions nearly full-time administrative jobs, leaving less and less time to care for student concerns and needs. Combined with staff turnover in these positions (e.g., we have had 7 individuals working in graduate student advisor positions since fall 2016) and a pandemic legacy of remote work by students and staff (see §G.12.B), our graduate students often feel unsupported administratively.

Section F. Postdoctoral Scholars

Over the review time period (2014–2024) the Department of Ecology and Evolutionary Biology has had 162 formally appointed postdoctoral fellows (Figure F-2; Appendix F1). The average length of appointment for this group was 24 months. Other researchers are those hosted under different job titles but nonetheless acted as postdoctoral researchers.

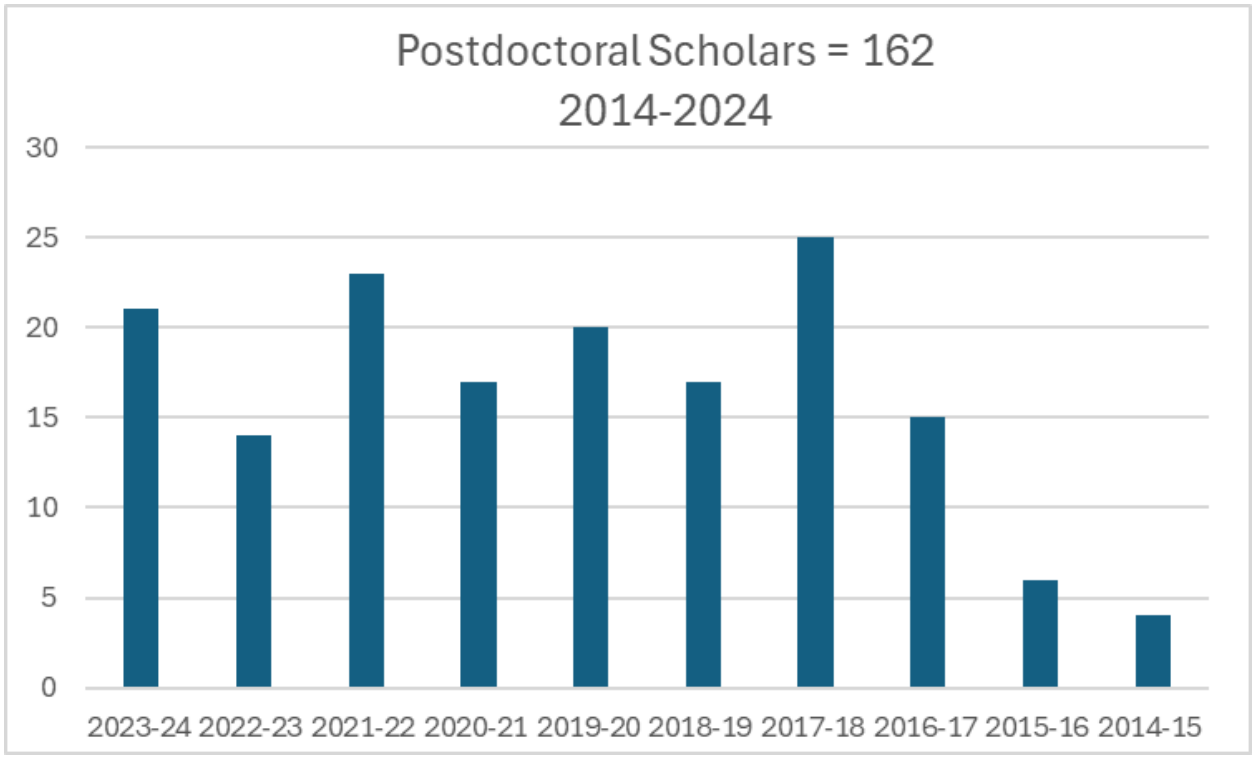


Figure F-2: Postdoctoral scholars in EEB by year.

The majority of Postdocs were supported by faculty funds, either on research grants or from start-up funding, which is often funded by Dean's Office Support or Departmental Indirect Cost Recovery. See Figure F-2 for a snap of the most recent funding sources.

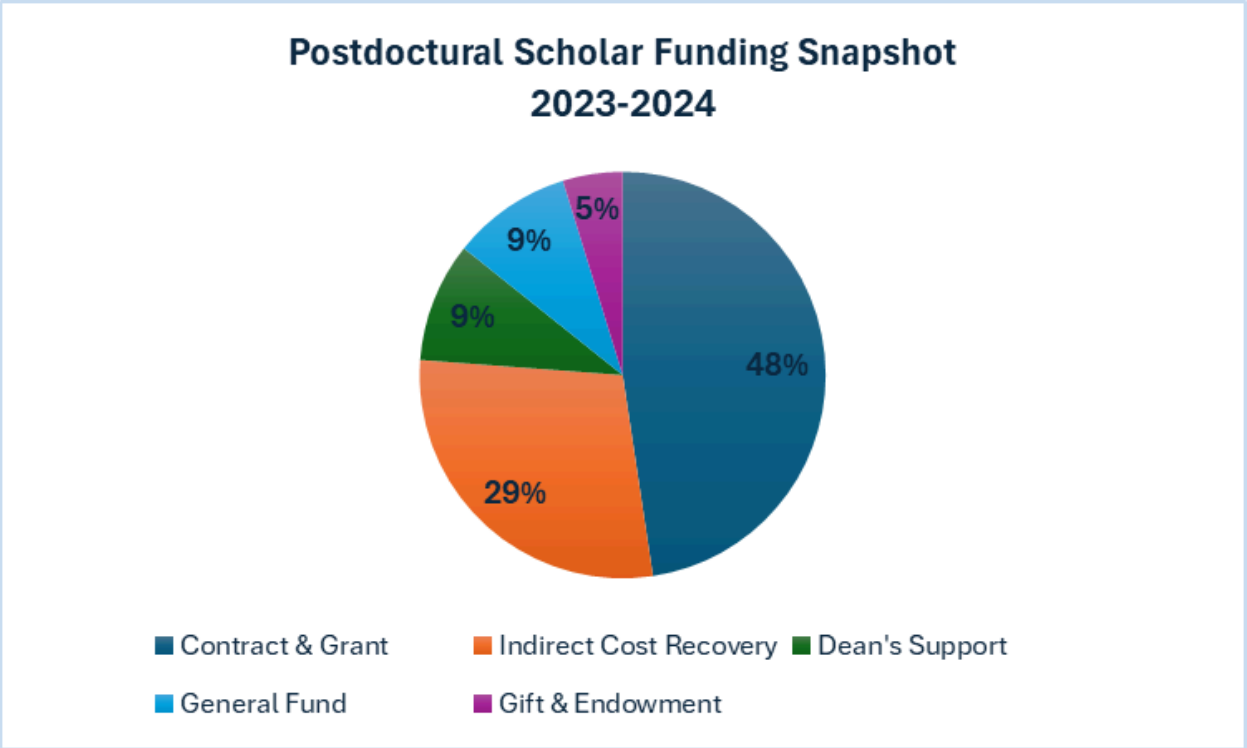


Figure F-2: Postdoctoral Scholar Funding Snapshot 2023-2024

Postdocs are fully integrated into PI's labs and many have become key players in inter-lab collaborations. Since many are supported on NSF or NIH grants, they have established mentoring plans. Postdocs are encouraged to participate in the Eco Evo Pub and some have become valuable departmental citizens through their interactions with our graduate and undergraduate students.

Apart from those who sought out lectureships, postdoctoral fellows had no formal teaching requirements. Nevertheless, a number of fellows actively participated in graduate courses and departmental seminars, and served on departmental committees. Many worked closely with our undergraduates and co-advised independent research and honors projects. Over the course of the review period, EEB postdocs published over 210 peer-reviewed publications (Appendix F2). Most postdoctoral fellows who have left UCLA did so for other positions in their field, such as other postdocs or jobs as professors, in government, at NGOs, or in tech (Appendix F1). Therefore, we conclude that their time at UCLA has been advantageous in furthering their career pursuits.

Section G. Diversity

1. Overview

The Department of Ecology and Evolutionary Biology has a long history of embracing efforts and initiatives to foster equity, diversity and inclusion among our faculty, staff, and students. Part of this culture comes from the strong tradition of leadership on these topics from within our own faculty ranks. For example, our current faculty includes the Executive Director of the Center for Education Innovation and Learning in the Sciences (CEILS), an essential source of trainings on equity-minded best practices on South Campus, a faculty member who served as Life Sciences EDI Equity Advisor for 5 years, and a faculty member who was co-chair of the Diversity Requirement Committee and then chair of the Diversity Requirement Implementation Committee. In addition, as Dean of Life Sciences, one of our faculty members created CEILS, the Life Sciences Diversity Advisory Committee, and the Mentor Professor Initiative, among many other accomplishments. During the review period, four of our faculty have received EDI awards from the UCLA Faculty Senate. This leadership from individuals within the department on EDI topics has helped to lead EEB to embrace department-wide EDI efforts.

Beyond these efforts from our individual faculty, our department has a longstanding history of embracing efforts to enhance equity, diversity and inclusion. Highlight from the last eight years include overhauling our graduate admissions criteria to include contributions to diversity and to drop the GRE as a requirement, graduate core course redesign following antiracist best practices, faculty and staff trainings and workshops centered on culturally aware best practices in mentoring and communication, and numerous faculty searches that have emphasized diversifying our faculty, including participation in both Division-wide and Departmental Mentor Professor searches. Details on these efforts as well as others are given in the sections below.

As is likely the case with many departments on campus and across the country, our departmental climate has suffered during the broader national and global events of the past eight years, but rebuilding positive departmental climate has been a major focus of departmental leadership in the last 3 years, and there are many promising signs that the efforts are paying off. Some of the major challenges that have impacted the climate of our department include the murder of George Floyd in 2020 and the broader reckoning with systemic racism that followed, the COVID pandemic and especially the extended shift to remote work, as well as the UC-wide graduate student and postdoctoral researcher strikes. In addition to these broader challenges, our departmental climate was negatively impacted by the restructuring of the Life Sciences Administration during the review period as well as by extensive public speculation and criticism of campus personnel actions related to one of our faculty members. Efforts to rebuild our climate in the last three years have included the creation of a new Vice Chair role focused on community, investment in community building events including seminars, social events, and networking opportunities, capacity building in communication and mentoring best practices, an

expansion of the opportunities for graduate students to participate in departmental committees, and investment in frequent meetings and communication with the graduate student body.

2. Departmental Leadership and Committees

Within EEB, all of the faculty in departmental leadership roles have responsibilities that include EDI efforts. The Vice Chair for Academic Personnel helps to ensure equity in the promotion and advancement faculty within EEB, and includes specific practices aimed at identifying and reducing inequities in the process, such as reviewing the achievements of faculty not seeking accelerations in a given year to see if they should be encouraged to pursue one. The Vice Chair for Graduate Studies plays a major role in EDI efforts within the graduate program, including chairing the graduate admissions committee, which helps to revise and implement the criteria for admissions, as well as by leading the graduate curriculum committee, which in EEB has helped to guide substantial EDI-focused changes in the graduate curriculum and in graduate mentoring practices in the last few years. The Vice Chair for Undergraduate Studies plays a major role in helping the department to understand how our curriculum structure impacts diversity in our undergraduate majors, as well as to recommend changes in policies and curriculum to promote diversity. Our newly created Vice Chair for Community has responsibilities to create activities that promote community, inclusiveness, and a sense of belonging among all members of our community, and also is charged with running the Committee for Anti-Racism and Equity (CARE), described in more detail below. Finally our Chair helps to guide all of the above activities, in addition to pursuing EDI opportunities arising across the University, such as Mentor Professor searches.

In addition to these departmental leadership roles, during the review period our department has been home to an impressive number of campus leaders who have had major impacts on campus efforts in equity, diversity and inclusion. In no particular order, this includes Prof. Paul Barber, who served as the inaugural Equity Advisor to the Division of Life Sciences for five years, as well as Prof. Victoria Sork, as Dean of Life Sciences, created many initiatives to improve EDI, including the Mentor Professor Initiative, the Center for Education Innovation and Learning in the Sciences (CEILS), and the Life Sciences Diversity Advisory Committee, among many other efforts. In addition, Prof. Michael Alfaro took major leadership roles in developing and implementing the campus Diversity Requirement, and Prof. Rachel Kennison serves as the director of CEILS and leads many efforts across the Division to improve equity outcomes in our teaching and mentoring.

Our department is advised by the Committee for Anti-Racism and Equity (CARE), which operates as a standing committee and includes representatives from our faculty, graduate students, postdocs and staff. CARE is charged with reviewing departmental policies and practices, recommending changes in policies and practices to reduce inequities or bias in outcomes, as well recommending new activities in the department to promote equity, diversity and inclusion (See Appendix G1). The committee meets monthly during the academic year, solicits input on its activities from the department through meetings and townhalls, and

generates a public annual report (CARE Annual Reports: Appendix G2; Appendix G3) to the EEB Chair that includes recommended changes for the coming year. The Chair then has the opportunity to assess and prioritize the recommendations for the coming year based on departmental resources and goals (Appendix G4), and CARE tracks progress on adopted recommendations and reports on this to the EEB community in the annual report. One of the features of CARE is that many of its recommendations, if selected for implementation by the EEB Chair, are carried out in practice by a range of other committees or leaders in EEB. This has the benefit of broadening the pool of department members involved in EDI efforts in EEB, and reduces the danger of burnout or overwork from those directly serving on CARE.

The origins of CARE in EEB date to the 2020 murder of George Floyd and the associated reckonings with structural racism in US institutions that played out nationally, within UCLA, and within EEB. In 2021-2022 EEB formed an ad-hoc Task Force Implementation Committee was to develop specific recommendations to implement in response to broader divisional and departmental brainstorming efforts in 2020-2021, and one of the major recommendations of this Task Force was the formation of CARE in fall of 2022.

3. Faculty hiring and recruitment

EEB works closely with the Life Science Equity Advisor in developing search plans. In recent searches in the review period, we have purposely implemented broad framing of our job description, sometimes as part of cluster hires, in order to encourage diverse applicant pools. In our most recent searches we have included mention of EDI-related teaching needs in our job postings. We have also participated in both Life Science and EEB-specific Mentor Professor searches. In our searches we typically appoint one committee member as Equity Advocate as a best hiring practice.

4. Current faculty climate

As described above, our department has experienced a number of challenges during the review period that have impacted the climate for all of EEB, including our faculty. In the last few years, the biggest challenges to a positive faculty climate have been the COVID pandemic and associated erosion of in-person work camaraderie and culture, the circumstances contributing to and fallout from the ongoing leave of one of our senior faculty members, and challenge to relationships and operations arising from recent staff turnover. While these difficulties created a challenging climate for faculty, especially in the first two years of the pandemic, there are signs that climate is improving. For example, going into the Fall 2022 Retreat (Appendix G5), many faculty were most concerned generally with the state of relationships in EEB (with each other, with graduate students and with staff) because of these recent challenges. Following a year of hard work focused on community building, improving the norms of relationships in EEB, and more time back together in person, many faculty expressed a positive sense of progress on climate and community in the Fall 2023 Retreat (Appendix G6). We expect that our work since

that time, as well as our successful hiring of a number of outstanding new faculty members, will continue to improve faculty climate in the coming year.

5. Approach to faculty retention

EEBs approach to retention starts with the recruitment of candidates who are great fits to the Department and whose research excellence encompasses the potential for collaboration with other colleagues within EEB and other south campus departments and/or interaction with other units such as the Institute of the Environment and Sustainability, Quantitative and Computational Biology, and the Institute for Society and Genetics. We also seek to recruit candidates whose research, broader impacts, and/or personal preferences are aligned with the opportunities and challenges of working at a top-tier public institution in an urban environment. Thus EEB seeks to minimize the need for retention by recruiting faculty who have a genuine desire to join our community. The Department is not characterized by lone wolf research groups whose lack of a strong identity within EEB and UCLA might predispose them to leave. We also endeavor to create a culture of support and positivity through a clearly articulated advancement process based upon merit, regular faculty meetings that give voice to all stakeholders, and a leadership team that is responsive to faculty concerns. We understand that outstanding faculty will draw attention from other institutions and in those instances where retention actions are needed EEBs general approach is to act as expeditiously to determine if the opportunities being offered at the new institution can be equaled or surpassed by opportunities at UCLA. To ascertain this the Chair meets as soon as possible with the faculty member that has been targeted for recruitment. The Chair will then typically work with the Dean to explore possible retention scenarios that would result in the faculty member staying. In spite of the steps outlined above, EEB lost two faculty to outside offers since the last review: Professor Shane Campbell-Staton (Princeton) and Professor Colin Kremer (University of Connecticut). These instances highlight some of the limits of EEB retention efforts. It is almost never possible for EEB to match the set-up resources and salaries that are available at the top private universities and this was a major contributing factor to Professor Campbell-Staton’s departure. Proximity to family members in Connecticut was a factor in Professor Kremer’s departure. Notably these faculty were hired shortly before the start of the Covid pandemic and the fragmentation of the EEB and campus community brought on by this event was also an important factor in their departure.

6. Efforts to diversify the graduate student applicant pool

Our department has a longstanding commitment to diversifying our graduate student body. This commitment has taken many different forms over the period of this review, which we detail below.

A. Changes to admissions requirements and application review

Beginning in the 2016-2017 admissions cycle, the Graduate Admissions Committee began to assess a new criterion described as “Potential Contributions to Diversity”, which continues to be

one of 5 criteria used to assess and rank applicants to the program (see §E.2). Originally this criterion was assessed using information from across the student's file, including letters of recommendation and the required student essay. Beginning in 2016-2017, our department opted to make the Graduate Division's optional second essay on the applicant's personal history required in order to facilitate a better assessment of this criterion. In 2017-2018 we further refined the language of the prompt for this essay to better articulate how it would be used to assess applicant's potential contributions to diversity. In 2022-2023 the graduate admissions committee, following encouragement from the EEB Committee on Anti-Racism and Equity (CARE) developed a detailed rubric for scoring applicant potential contributions to diversity (see §E.2.C).

In addition to changes to our admissions criteria, for the 2017-2018 admissions cycle, our graduate admissions committee and department moved to make the Graduate Record Examinations (GRE) optional instead of required for applicants. While applicants have been able to upload GRE scores and some individual faculty have viewed them as part of making recruitment decisions for their individual labs, the departmental Graduate Admissions Committee has not used GRE scores in its ranking and evaluation process since this change was made in 2017.

B. Trends in Underrepresented Students through the Admissions Cycle

The proportion of graduate applicants to our programs who hold identities underrepresented in graduate school (URG; a classification designated by UCLA including domestic students who are American Indian and Alaskan Native, Black and Afro-American, South or Central American, Chicano and Mexican American, Latino and other Spanish-American, and Hawaiian, Native Hawaiian and other Pacific Islander) has held relatively constant over time across the three main filters of the admissions cycle: applications (Figure G1), admissions offers (Figure G2), and matriculations (Figure G3). As the sample size gets smaller (e.g., from applications to matriculations, and comparing PhD to MS), year-to-year variation increases due to small sample stochasticity. Despite these inter-annual fluctuations, the proportion of our graduate program who hold URG identities has steadily increased over the last decade, from about 12.5% in fall 2014 to 25% in fall 2023 (Figure G4). This total increase is mirrored in our PhD program, but is even more dramatic in our MS program (Figures G4, G5). These increases keep pace with or exceed similar increases seen by other UCLA Life Sciences Ph.D. programs (Figure G6).

Overall, our graduate program is substantially more diverse than the national average for Ecology and Evolutionary Biology programs. Over the last 10 years, our graduate program on average has comprised 34% URG individuals, whereas the national average for EEB graduate programs is between 7%[2]and 13%[3]. As a related benchmark, white or Caucasian people has previously made up >90% of the membership of the Ecological Society of America[4].

[2]https://nap.nationalacademies.org/catalog/12994/a-data-based-assessment-of-research-doctorate-programs-in-the-united-states-with-cd
[3] https://ncses.nsf.gov/surveys/earned-doctorates/2023#data
[4] https://www.esa.org/wp-content/uploads/2012/12/profilesReport2006.pdf

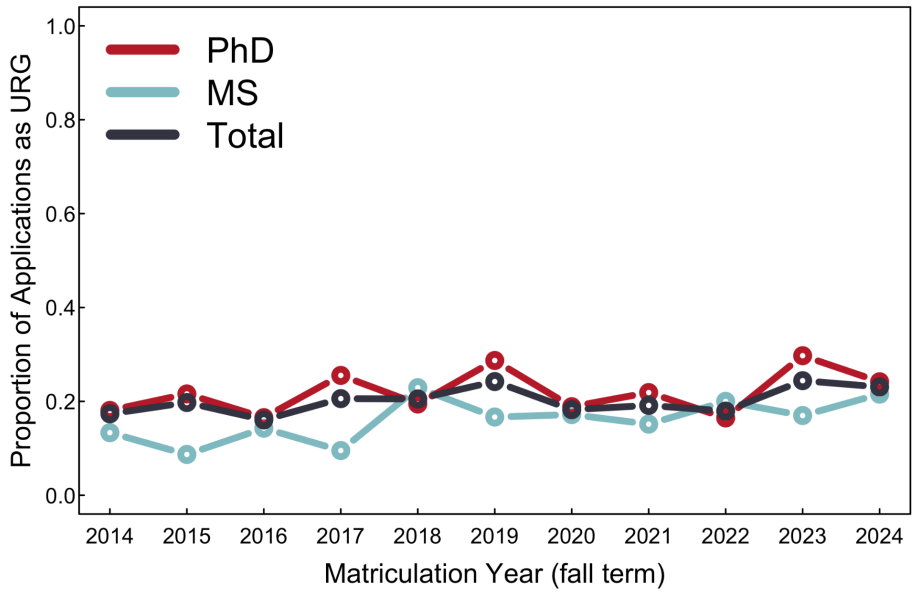


Figure G1. Trends in the proportion of EEB graduate applicants who are identified by UCLA as underrepresented in graduate education (URG). Count for MS combines applicants across two programs, Biology MS and EvMed MS.

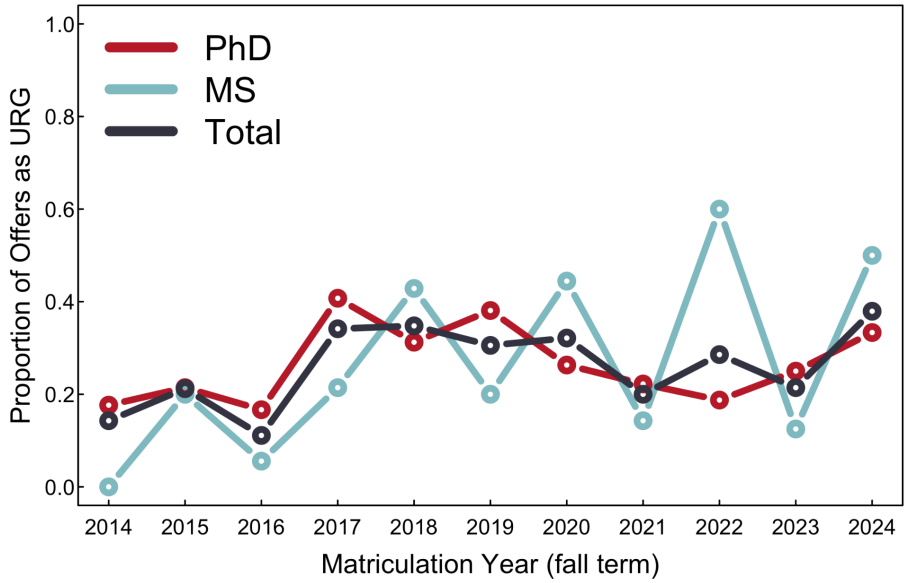


Figure G2. Trends in the proportion of EEB graduate admission offers to individuals who are identified by UCLA as underrepresented in graduate education (URG). Count for MS combines applicants across two programs, Biology MS and EvMed MS.

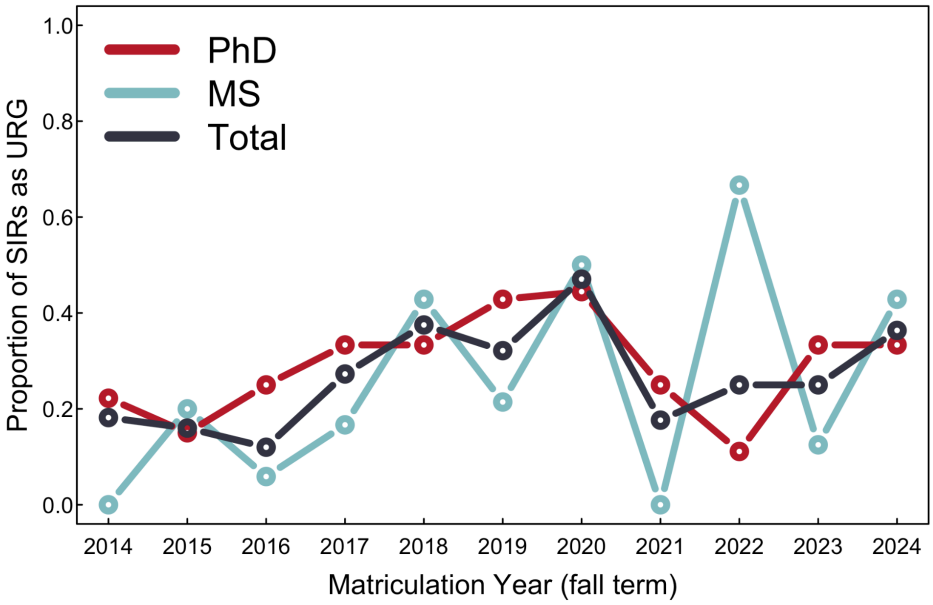


Figure G3. Trends in the proportion of matriculating EEB graduate students who are identified by UCLA as underrepresented in graduate education (URG). Count for MS combines applicants across two programs, Biology MS and EvMed MS.

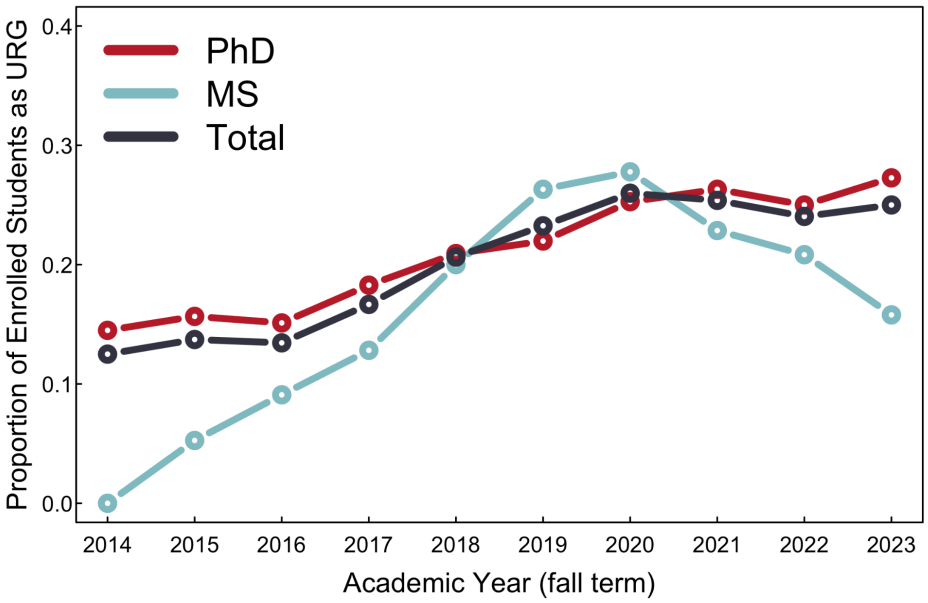


Figure G4. Trends in the proportion of enrolled EEB graduate students who are identified by UCLA as underrepresented in graduate education (URG). Count for MS combines applicants across two programs, Biology MS and EvMed MS.

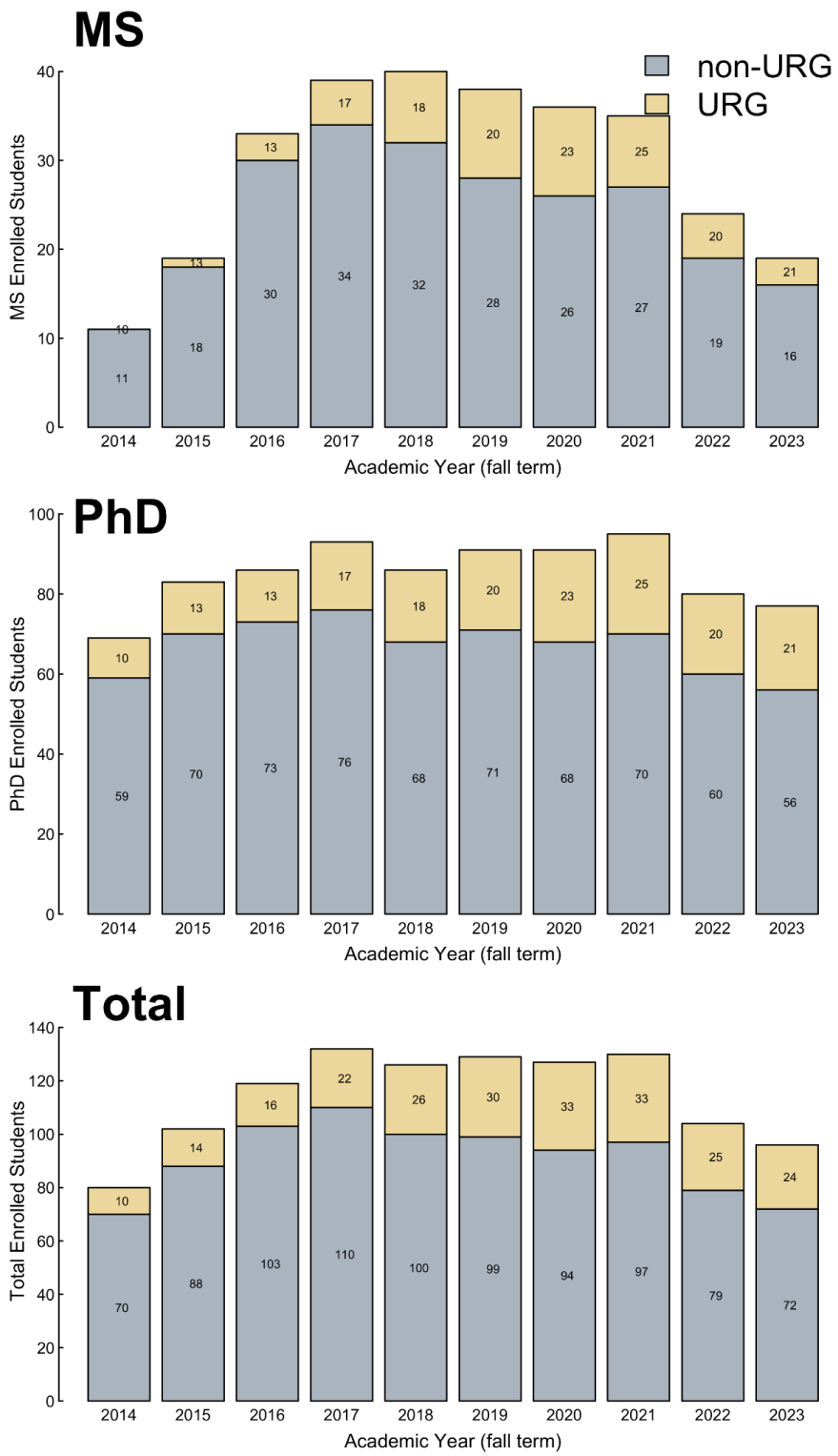


Figure G5. Total number of enrolled students in each academic year identified as URG or non-URG across MS (top), PhD (middle) and all (bottom) graduate programs.

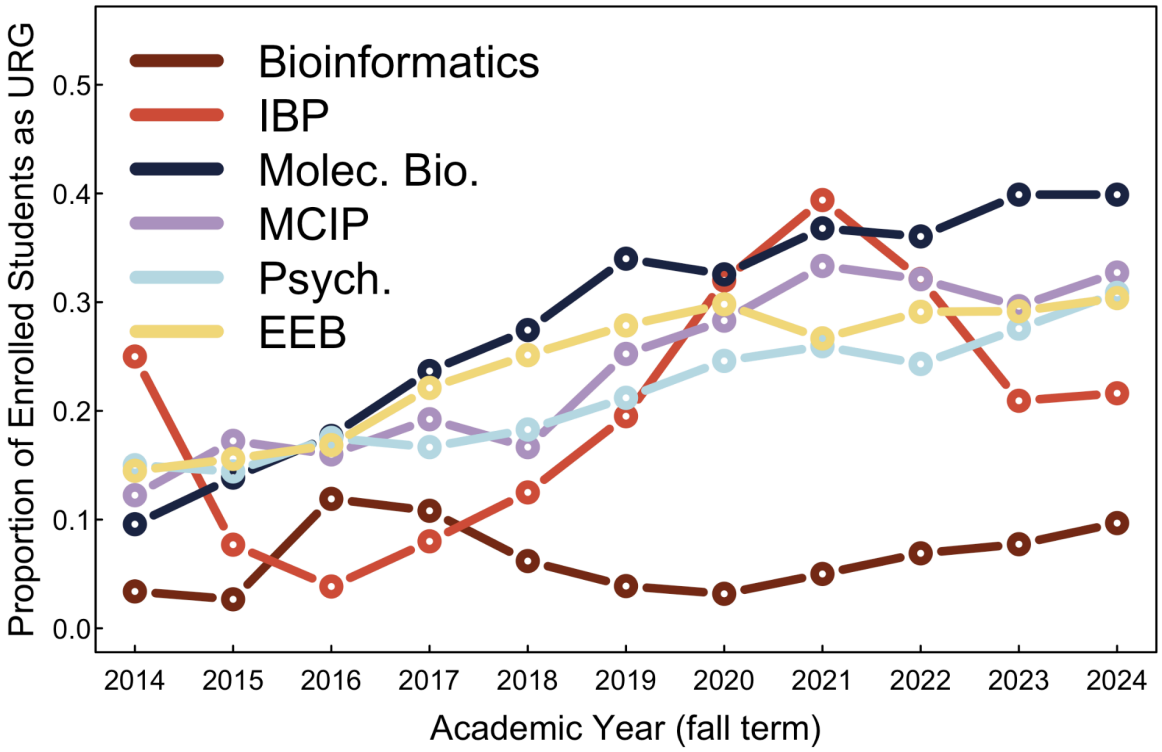


Figure G6. Percent of enrolled Ph.D. students in each academic year identified as URG across all UCLA Life Sciences graduate programs. EEB has shown increases in the proportion of URG students that keep pace with or exceed other Life Sciences departments.

7. Overview of Graduate Student Climate

In preparing this review, our department has three relatively recent surveys of our graduate students, in addition to ongoing input from our student body that has been relayed to the department leadership through the faculty-student liaison, at student townhalls and in regular meetings with our graduate student leadership. The three surveys are, chronologically, an EEB-graduate student run survey of graduate student climate, administered between February 17 - March 14, 2021 (hereafter “2021 EEB Grad Survey,” Appendix G7), the UCLA Graduate Council Survey, administered in Winter Quarter of 2021 (hereafter “2021 Grad Council Survey”), and, most recently, a brief survey of our graduate students conducted by the department in Winter of 2024 (hereafter “2024 EEB survey”, Appendix G8).

The surveys all differ wildly in the structure of their instruments. The 2021 EEB Grad Survey was a graduate student run and led effort, and it included 80 questions with a mix of multiple choice and free response questions, and it was completed by 55/104 (53%) of our graduate student body at the time. A consortium of EEB graduate students then led the analysis and interpretation of the data, which were shared with the EEB CARE Committee in Winter of 2023 in the form of an 81-page report (Appendix G7). The EEB CARE Committee discussed the report extensively and developed a set of recommendations to the EEB Chair in Spring of 2023

to address concerns highlighted by the authors of the report (link to CARE 2023 annual report). The 2021 Graduate Council Survey is a standard instrument administered to all departments prior to an 8-year review, and the survey yielded a 37% response rate, lower than the 53% for the 2021 EEB Grad Survey administered during roughly this same time. The data from the 2021 Graduate Council Survey were just made available to the department in Winter of 2024 as part of preparing this document. Finally, the 2024 EEB Survey was a brief two-question free response instrument administered to our current graduate students in Winter 2024 (Link to Winter 2024 survey), which garnered responses from 27/ 72 students (38%). These responses were analyzed by the department leadership with oversight from EEB graduate student leadership, who also had access to the raw survey data and reviewed the summary of those data presented below for accuracy.

A. 2021 EEB Graduate Student Climate Survey

Our department has had the most time to study and develop plans in response to the 2021 EEB Grad Survey, which was shared with the department on February 6th 2023. While best practices might favor a climate survey that was run by an external body rather than our graduate students themselves, the resources needed to fund such an effort have remained out of reach for EEB during the review period, and the graduate students who developed and analyzed the climate survey did so with tremendous attention to best practices, detailed in the report (link to Grad Climate Survey Report). Critically, the 9 “Ongoing Challenges” that the student authors highlight as key takeaways are entirely consistent with EEB leadership’s understanding of student concerns and stressors from the time period of the survey. Those nine ongoing challenges are, in the order presented in the student report, are:

1. Worsening mental health, especially as related to experiences of microaggressions, unsupportive interactions with faculty, work-life balance concerns, and a lack of departmental community
2. Barriers to accessing student mental health resources on campus (i.e. limited coverage through insurance)
3. Stress about career prospects and preparation
4. Class and program requirements (especially as they relate to workloads and our core courses)
5. Concerns about how to provide feedback to their mentors about mentoring relationships
6. Many students report experiencing harassment and discrimination while in the department
7. Many students experience stress around finances, particularly as it relates to the high cost of living in the area
8. Students noted that our Masters students feel less supported than our Doctoral students, both in terms of financial resources and in terms of access to certain opportunities such as doctoral student courses
9. Many students want better interactions with faculty outside of their committees, in addition to more support for community within the department

The formal recommendations from the EEB CARE committee in response to the 2021 EEB Grad Survey were made in spring of 2023 (link to CARE 2023 Annual Report), and the 2023-24 Academic year, CARE and EEB Leadership developed a formal system for tracking CARE recommendations and progress made towards addressing each recommendation (link to CARE recommendation tracking document). While we do not have space to summarize all of these documents here, we can present a brief overview of some of the highlights. First, given the two-year gap between when the survey was conducted and when the report was shared with EEB leadership, part of the CARE response (link to CARE 2023 annual report) highlights changes to departmental practices and structures in the two-year intervening period that we hope have started to move the needle on some of these concerns. These changes included the formation of a graduate curriculum committee to help coordinate and oversee the revision of both the structure and content of required graduate courses (beginning to address concerns 3,4 and 8), in-faculty meeting trainings from campus experts on supporting students in distress, on supporting LGBTQIA+ students, and on improving faculty and staff resources for managing and responding to microaggressions and discrimination in meetings and the classroom (concerns 1 and 6), as well as a renewed investment in fostering and supporting activities that create a sense of belonging in the department though an ongoing call for EEB funding for student-run community building events (concerns 1 and 9). Our hope is that these in-process changes, the new steps recommended by CARE in response to this report, as well as our system for tracking and keeping the community updated on progress towards CARE recommendations will all help to address the important issues raised in the 2021 EEB Grad Survey.

The most significant new investment in addressing graduate climate concerns in the 2023-24 year has been overhauling our mentoring and annual committee meeting expectations and documents (speaking to concerns 3,4 and 9). While this process is ongoing still this academic year, our goal is to 1. reinforce our expectation that students hold annual, low stakes committee meetings with their dissertation committee 2. Develop a reporting tool on those annual meetings that reinforces that these meetings include career development discussions as well as feedback on the mentor-mentee relationship 3. develop evidence-based best practice tools (such as a mentor-mentee compact builder) to better support the development of healthier mentor-mentee relationships (link to grad mentorship docs). To date faculty have explored tools at our fall 2023 retreat and beta-tested draft compact and committee meeting forms with some current mentees, with the goal of finalizing these documents for fall 2024.

B. 2021 Graduate Council Survey

The 2021 Graduate council survey overlaps in time with when the 2021 EEB Grad Student Survey was administered (Winter of 2021) and, while it asks different questions and had a slightly lower response rate (37% vs. 53%), we would argue it largely tells the same story as our student-run survey. In order to extract the top concerns from the Graduate Council Survey, we have simply extracted the 9 questions that had less than 40% positive responses (satisfied or very satisfied) from either the PhD or MS student responses. The table below shows those items ordered from least to most satisfied. The percentage shown is for the PhD student responses, and an asterisk denotes cases where the item was scored below 40% by both MS and PhD students. In order to facilitate comparison with the 2021 EEB Grad Survey, we have

added numbers that correspond to the 9 ongoing challenges our graduate students mapped out in their report on the survey.

Positive	Comment	Mapping to 2021 EEB Grad Survey Challenges
20%	the equity of distribution of financial support within your program	7,8
21%*	the extent to which faculty are sensitive to diversity issues	1,6
21%*	faculty explain the strategies for survival in graduate school	4
22%	the inclusion of graduate students in program governance	***new***
28%	The value of program requirements in facilitating your educational and professional goals	4
36%	The morale of students in your program	1,9
37%	Faculty insist we discuss my graduate program on a regular basis	***new***
38%	The quality of instruction in classes	4
39%	The extent to which faculty educate and assist students in finding employment	3

As we see it therefore, most of the areas of greatest concern from the 2021 Graduate Council Survey are largely congruent with the feedback the department has already been acting on from the 2021 EEB Grad Survey, with the exception of two points that we can comment on here. First, only 37% of our PhD students reported that faculty insist on regular discussions of student progress. Our plan to overhaul student mentoring and committee expectations, detailed above in the 2021 EEB Grad Survey section, includes a plan to redouble our efforts to enforce our program requirement of annual committee meetings, which was not formally monitored by our program previously. Second, only 22% of our PhD students were satisfied with the inclusion of graduate students in program governance. Since 2021, we have taken substantial steps to expand the opportunities for graduate students to participate in the running of the department. In the last two years we have added student positions on the new Graduate Curriculum Committee, the new Committee for Anti-Racism and Equity (CARE), and we have instituted a practice of regular meetings between EEB faculty leadership and EEB graduate student leadership to foster communication and collaborative problem solving. Perhaps most importantly, since winter 2021 our graduate students have formed a new EEB Graduate Student Association (GSA). The GSA leaders have served a critical role in supporting communication between faculty and students, organizing student events, and facilitating important conversations within the graduate student body that are then often summarized and relayed to faculty leadership. The landscape for opportunities for student involvement in key areas of

departmental life have grown substantially since this survey was administered, to the point we think that have more opportunities for student involvement than perhaps there is demand, at least as of fall 2023.

C. Perspective on EEB Graduate Climate in Winter 2021

Both of the major student survey instruments that are available to us for this review were conducted during the same short time period (Winter 2021) that was one of the most challenging time periods in our department's history. Our faculty, students, and staff were reeling at the time from the twin challenges of COVID (which had major impacts on student research progress, on the challenging of teaching and TAing, and our capacity to connect with one another in a remote setting) and the national response to the murder of George Floyd in 2020, which set off important but painful reckonings with systemic racism both nationally and within EEB. We raise these issues not to undermine the important themes and calls to improvement that we have highlighted from the surveys conducted at this time, which we think are clearly important areas for our department to work to address, but rather to question the value of some of the benchmark comparisons in the Graduate Council Report to other departments from entirely pre-COVID times. We believe that a comparison to other departments during COVID would perhaps be more informative.

D. 2024 EEB Graduate Student Survey

In winter 2024 we administered a short two question free response survey inviting current students to comment on strengths and weaknesses of the graduate program. Twenty seven (38%) of our students responded, and EEB faculty leadership coded each phrase of the response into a category and also assessed if the comment was positive, negative or neutral. The raw data were shared with EEB grad student leadership in order to check for any inaccuracies or other concerns.

The most discussed and commented theme in the responses was EEB Curriculum and Requirements, which constituted over one third of the comments received. Overall the comments on this theme were evenly split between positive and negative comments, with our quantitative offerings for our graduate students drawing the most negative commentary. The next-most discussed theme was our Departmental Culture, with well over half of the overall comments being positive. The most positive topic under culture was our new practice of offering funding for community events throughout the year, while the most negative was continued concern about student-faculty interactions, especially with faculty who were not the main advisor to the student. The third most discussed theme was Departmental Structure and Processes, with over two thirds of the comments being negative. TA hiring and interactions with our Staff were the most negative topics in this theme. The major themes raised by the students in this survey are consistent with the faculty's ongoing understanding of challenges facing our graduate students. Our Graduate Curriculum Committee continues to work with the department on refining and updating our course offerings in response to student feedback, and staff-graduate

student interactions, especially around TA hiring processes, are an ongoing target for improvement.

8. Disparities in undergraduate progress and outcomes

Analysis of course performance data shows no gender-based disparities in course performance. However, transfer students, URMs, first-generation college students, and Pell grant recipients have longer time to degrees than 4-year admit, non-URM, continuing students, and non-Pell grant recipients. Similarly, there are substantial disparities in course performance based on URM, first-generation, and Pell grant status as well, characteristics that are highly auto-correlated in the UCLA undergraduate population. These disparities are discussed in depth in the Undergraduate Programs, section N-1.

9. Postdoctoral Scholar recruitment and mentoring

Given the diversity of ways in which postdocs are supported in EEB, we do not yet have formal practices around ensuring a diverse postdoctoral applicant pool. However, our department has long included contributions to diversity in the assessment of faculty for promotion, and therefore faculty have an incentive in their own promotions to seek out and recruit diverse candidates. In addition, our faculty are frequent mentors to UC PPFP recipients (five currently, nine since 2020). We expect that our recent efforts to update and improve mentoring practices for graduate students will also result in similar improvements in postdoctoral scholar mentoring.

10. Staff Hiring Practices

EEB is required to adhere to College and Campus HR practices. Open positions are posted using the iCIMS platform and advertised across internal and external platforms. Job applicants are reviewed by the hiring manager. Top candidates are interviewed by a panel of colleagues. Top candidates are then interviewed by department leadership. Compensation is reviewed and approved by ColHR. Once references are checked and validated, an offer letter is auto generated by iCIMS. Both internal transfers and outside candidates must complete successful fingerprinting and UBS. Promotions and equity increases are submitted for review and must be approved by ColHR. Annual merit increases are centrally managed by CHR.

11. Staff Climate

The general consensus of the staff is they are stretched thin and the additional staff would alleviate some of the daily burdens. While the department is increasing the number of faculty and also experiencing growth in student enrollment, administratively we have not been able to add to the staffing roster. The addition of faculty equals additional administrative needs to the financial team, payroll/human resources, academic personnel and student services.

In order to support the long-term growth and addition of 6 planned recruitments between 2024 and 2026, we will need additional staff added to the administrative teams.

Currently the department has 1 academic coordinator to handle all academic merits, promotions and actions across the department. This means to a single point of failure; staff turnover frequently results in the loss of historical knowledge which is an ongoing issue within EEB.

The finance team (Contract and Grant administration) are crucial to the success of any academic unit. There are currently 3 financial analysts, who are working at capacity and managing high volumes of pre and post award management

12. Summary - Strengths, Challenges and Opportunities

A. Strengths

Innovative and Dedicated EDI Leadership: Our department has a “deep bench” of campus leaders with demonstrated commitment and expertise in innovating and advancing EDI objectives at the department, divisional and campus level. This leadership is complimented by stellar recent faculty hires who have strong track records in this area and tremendous potential for future leadership.

Diverse graduate student body: EEB has a long history of investing in forming a diverse graduate student population for our graduate program. This is supported by the dedication of many of our faculty to attract diverse applicants to their labs, programmatic decisions in our admissions process, and the support the campus provides in the form of fellowships and other resources. The average proportion of URG students in EEB over the review period is 34%, which is far above national benchmarks, and is similar to or exceeds the proportion of URG students in other UCLA Life Sciences Ph.D. programs.

Structural investment in EDI work: Our department has a long history of investing considerable faculty, staff and student resources into EDI initiatives and committees. This results in deep thinking about structural improvements that can be made within EEB, as well as a culture of respecting and embracing efforts to improve EDI outcomes within our department.

Engaged and thoughtful graduate student body: Over the course of the review period, our graduate student community has invested intensively in efforts to communicate climate challenges to departmental leadership in a thoughtful and constructive manner.

B. Challenges

Role of faculty nominations in shaping the diversity of the graduate applicant pool: Our graduate admissions model, which restricts committee-level review to a smaller number of applicants already nominated by faculty, means that individual faculty members have strong

control over the diversity of the graduate applicants in our graduate program. While many of our faculty members request review of URG applicants, the composition of the final applicant pool is primarily determined by faculty nominations, rather than Admissions Committee practices.

Ongoing impact of remote work on departmental culture: The COVID-related work from home period during this review had a substantial negative impact on climate, and while we have recovered in many ways from this time, some changes, such as increased work-from-home schedules for staff and hybrid departmental seminars, do continue to slow efforts to improve climate in EEB.

Impact of labor actions: The graduate student and postdoctoral scholar strikes have led to substantial improvements in working conditions and pay for members of our community, but those periods of time have also created challenges in interactions between different groups within our department (see also §G.8.C).

C. Opportunities

Implementation of Recommendations: Our department is still working on closing the gap between recommendations that EDI-focused committees make and what is actually implemented in the department. While we have made strides to create structures and processes that allow framing recommendations as a dialogue rather than a mandate and that create public accountability, we still do not have a strong, established culture of following through on recommendations. We need to continue to nurture a positive feedback loop between EDI committee work resulting in meaningful changes in order to combat burnout and withdrawal of community members who invest in EDI work.

Continued investment in the success of our diverse graduate student body: While we have long been leaders in creating a diverse graduate student body, input from our students during the review period shows that we need to continue to improve the ways in which we meteor and support our students in the program. Our recent efforts towards improved mentorship development in EEB is part of this, but more work needs to be done.

Section H: Articulated, Concurrent, And Self-supporting Programs

EEB does not have articulated, concurrent, or self-supporting programs.

Section I. Comparison to Prior Reviews

EEB has undergone significant changes since the last review. Below we highlight accomplishments as well as major changes in leadership, restructuring, strategic planning, recruitment, diversity in ranks, adjuncts, postdocs and development…Since the last review, Ecology and Evolutionary Biology has achieved remarkable success and impact in its research and training but also experienced existential challenges, particularly during and after the pandemic. Below we describe key highlights during this period.

1. Changes in Departmental Leadership.

The department has experienced significant changes in leadership since the last review, including the Chair of the department and the CAO. Notably the department had external chairs (Barney Schlinger followed by Greg Miller) in the period immediately preceding the Chair Alfaro's tenure. The CAO position has similarly experienced a high degree of turnover with five different CAOs since 2021. This turnover marks a phase of uncertainty within EEB following the pandemic. The appointment of Chair Alfaro and the current Vice-Chair Leadership team reflects an effort to rebuild culture, trust, stability, and a sense of direction with EEB. As part of this change EEB has also established a new Vice-Chair position focused on Community building with the department. This position, currently filled by Professor Nathan Kraft, has played a key role in establishing workflows to track department efforts to implement recommendations from EEB's Committee For Anti-Racism and Equity (CARE), to help generate and implement solutions to improve the culture of collegial interaction within the department (such as departmental retreats, events, and trainings), and to cultivate and support events initiated by members of the department that increase everyone's sense of community and belonging (such as social events, networking opportunities, and professional development efforts).

Chief Administrative Officer	
April 2024	Shannon McClenton (Interim, made Permanent September 2024)
April 2022	Kimberly Linsey
February 2022	Brian Rubke (Interim - Administration) & Sabrina Lux (Interim - EEB Spaces)
September 2021	Eric Wells (Interim - Administration) & Sabrina Lux (Interim - EEB Spaces)
August 2018	Jay True
May 2018	Rania Soliman (Interim - Financial) & Barbara Van Nostrand (Interim - Student Services)
March 2018	Asiroh Cham
Prior to March 2018	Annette Klufas

Department Chair	
July 2022 - present	Michael Alfaro
July 2021 - June 2022	Greg Miller (Interim)
April 2020 - June 2021	Barney Schlinger (Interim)
July 2017 - March 2020	Karen Sears
August 2016 - June 2017	Interim Chairs
December 2009 - July 2016	Daniel Blumstein

2. Administrative restructuring

Since the last review the centralized administrative unit for all Life Science South Administration (LSSA)has been reorganized. Previously, LSSA serviced the departments of Ecology and Evolutionary Biology; Molecular, Cell, and Developmental Biology; and Integrative Biology and Physiology, as well as a variety of affiliated interdepartmental and research units. The centralized services were headed by the Chief Administrative Officer who worked closely with the Chairs, Vice-Chairs and Academic Administrators of the various departments and programs. LSSA was composed of five functional areas, including financial services, payroll/personnel, student affairs, facilities and operations, and computing support, which reported to the CAO.

In the subsequent restructuring the departments and supporting services were decentralized and EEB hired a Chief Administrative Officer to oversee operations including, financial services, payroll/personnel and student affairs. Facilities and computing support remained a centralized resource.

3. Strategic Planning

In 2018 following the hire of an outside chair (Professor Karen Sears) EEB underwent a Strategic Planning process that included faculty discussion at a weekend retreat, follow up half day retreats with outside consultants, the construction of special committees to reappraise the core graduate training program and to identify search priorities, and discussions about the overall mission and identity of the department. This process generated consensus that the department identified as a unit that conducts fundamental research in ecology, evolution, and behavior and seeks to apply these concepts to better address the challenges of identifying and ameliorating the threat to biodiversity from anthropogenic change. Following this process the department instituted a series of changes to the graduate curriculum and pursued new faculty hires that were aligned with the strategic plan. However, since the pandemic the department has

undergone significant changes in staffing, departmental leadership, and has experienced a significant loss of key senior faculty. These events coupled with the recruitment of new faculty (Tingley, Ordway, Aguillon, Song, and Medina) and two years of EEB leadership under a new chair and set of vice-chairs highlight a new direction for EEB and the need for a new round of strategic planning.

3. New Faculty

EEB has hired eleven faculty since the last review and lost two faculty (Professors Colin Kremer and Shane Campbell-Staton) to other institutions. EEB's new hires have strengthened our expertise in community ecology, behavior, developmental biology, theoretical and empirical population biology and ecology, plant biology, and marine science. EEBs recruitments have generated and strengthened connections to other Life Science and Campus units. The quantitative expertise of many of the new recruits has created natural collaborations between EEB and the Institute for Quantitative and Computational Biology (QCBio); EEB's behavior faculty interacts broadly with units like anthropology and sociology (though the Center for Behavior, Evolution and Culture) and shared research on questions of conservation biology leads to connections with the Institute of the Environment and Sustainability, the Center for Tropical Research, and the Congo Basin institute.

Recruitments since last review			
Candidate	**Year of Search**	**Search Title**	**Current Position**
Monica Medina	2023-24	Opportunity Hire	Professor
Stepfanie Aguillon	2023-24	Mentor Professor	Assistant Professor
Chuliang Song	2023-24	Biology of Populations	Assistant Professor
Elsa Ordway	2019-20	Plant Ecology	Assistant Professor
Colin Kremer	2018-19	Ecology	at University of Connecticut
Morgan Tingley	2018-19	Quantitative Biology	Professor
Shane Campbell-Staton	2017-18	ISG and EEB joint hire	at Princeton University
Nandita Garud	2017-18	Population Genetics	Associate Professor
Karen Sears	2016-17	Chair, Ecology & Evolutionary Biology	Professor

Noa Pinter-Wollman	2016-17	Opportunity Hire	Professor
Felipe Zapata	2015-16	Plant Biology	Associate Professor
Nathan Kraft	2014-15	Computational Plant Biology	Professor

4. Improving Departmental Diversity.

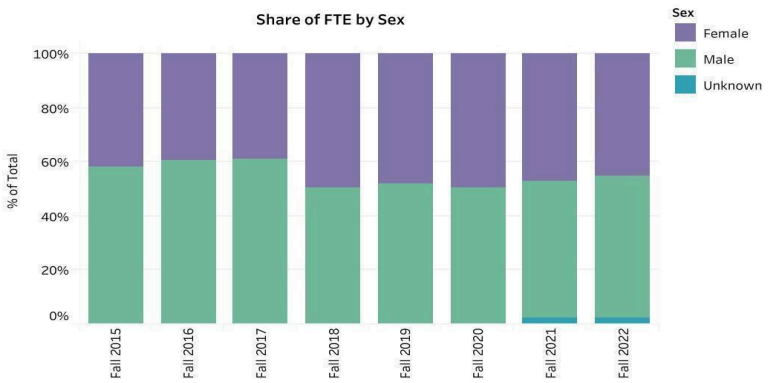
An important long term goal of the department has been to increase the diversity of the faculty, trainees, and undergraduates. Three recent hires (Aguillon and Song 2023, Medina 2024) have been recruited as Mentor Professors in EEB following the model developed by our former Dean Victoria Sork, meaning that these individuals have demonstrated potential for mentoring individuals from groups underrepresented in STEM and have received start-up funds from Life Science to support these efforts. Six of the nine EEB hires have been women which begins to address the historical gender imbalance within the department. Two recent hires are Latinx with a demonstrated commitment to promoting UCLA's strategic goal of becoming a Hispanic Serving Institution through programs like Latina Futures.

Ecology and Evolutionary Biology
Academic Workforce Diversity

Counting Method	Workforce Category	Academic Series	Faculty	Tenure	Senate	Academic Rank
FTE	All	All	All	All	All	All

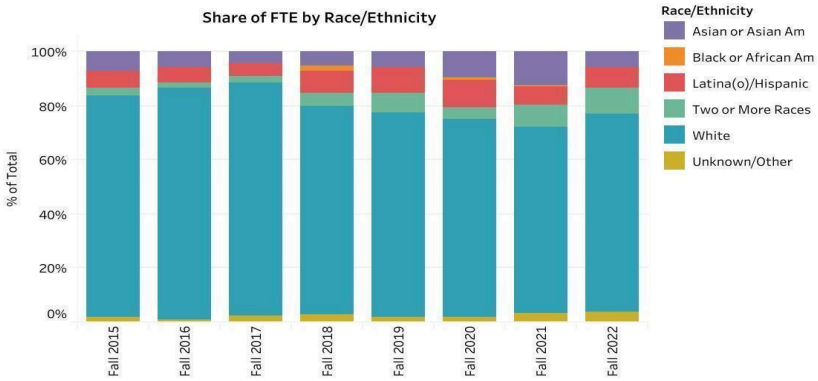
8-year change in percentage by Sex

	Fall 2015	Fall 2022	Difference
Female	41.7%	45.3%	3.6%
Male	58.3%	52.4%	-5.9%
Unknown	0.0%	2.3%	2.3%



8-year change in percentage by Race/Ethnicity

	Fall 2015	Fall 2022	Difference
Am Ind or Al Nat			
Asian or Asian Am	6.9%	5.6%	-1.3%
Black or African Am			
Latina(o)/Hispanic	6.2%	7.5%	1.2%
Nat Haw or Othr Pac Isl			
Two or More Races	3.3%	10.0%	6.7%
White	81.5%	73.2%	-8.3%
Unknown/Other	2.1%	3.7%	1.7%



5. Changes in adjunct faculty ranks.

Adjunct faculty play an important role in expanding the breadth of EEB's reach with local and global collaborators. Among new adjuncts since the last review are Dr. Anthony Baniaga and Dr. Jon Marcot. In 2021 Dr. Anthony Baniaga joined as EEB adjunct faculty to help support the shared mission of the EEB and the Botanical Garden to promote research and education in botany. Dr. Baniaga has been especially successful in promoting specimen-based research both through course development (including a museum science class where students develop projects from the herbarium or the Dickey collection) and through the mentorship of undergraduates for research credits using herbarium specimens. Dr. Jon Marcot is the collections manager for the Dickey Collection

	FACULTY NAME	FACULTY SPONSOR	RANK	RANK START DATE	CURRENT YRS AT RANK AS OF 7/2024	STEP START DATE	CURRENT YRS AT STEP AS OF 7/2024
BANIAGA	ANTHONY ERNEST FIORENTIONO	SORK	Asst. Adj. Prof.	07/01/21	3	07/01/21	3
KEELEY	JON	RUNDEL	Adj. Prof.	07/01/01	23	07/01/21	3
KNOWLES	BENJAMIN		Asst. Adj. Prof.	11/01/20	4	07/01/22	2
LARISON	BRENDA	WAYNE	Asst. Adj. Prof.	07/01/15	9	07/01/23	1
MARCOT	JONATHAN	VAN VALKENBURGH	Asst. Adj. Prof.	07/01/17	7	07/01/22	2
RILEY	SETH	WAYNE	Assoc. Adj. Prof.	07/01/15	9	07/01/21	3
SHIER GRETHER	DEBRA	NONACS	Asst. Adj. Prof.	07/01/10	14	07/01/18	6
WANG	XIAOMING	VAN VALKENBURGH	Assoc. Adj. Prof.	07/01/04	20	07/01/21	3

6. Formation of the Committee on Anti-Racism and Equity.

A key priority for the Department following the events surrounding murder of George Floyd has been to place EEB in all of its actions on a firmly anti-racist footing. In the summer of 2020 the department appointed a task force comprising graduate students, staff, and faculty to comprehensively review department practices. The report of the year-long work by this committee is available (Appendix I1. Task Force Recommendations). Following this report EEB appointed an Implementation Committee which made several key recommendations (Appendix I2. Implementation Committee Recommendations) based upon the task force report including the formation of a standing committee to promote antiracism and equity. Following this recommendation EEB appointed the CARE committee in the fall of 2022. Further details of CARE's operations and role in the department are explained Section G2: Diversity: Departmental Leadership and Committees

7. Updates on prior review recommendations and current relevance

The 2014 review provided six key recommendations for the department:

1. **Faculty Hiring and Strategic Planning**: Develop a long-term hiring plan and strengthen faculty diversity while addressing impending retirements.

2. **Merit and Promotion Processes**: Improve transparency and reduce bias in faculty advancement.
3. **Strengthening Collaborations**: Expand interdisciplinary research, particularly with quantitative biology and medicine.
4. **Graduate Student Funding**: Address the sustainability of student funding models amid increasing costs.
5. **Graduate Curriculum and Student-Faculty Interaction**: Refine the graduate curriculum and support systems to reduce workload stress and enhance training.
6. **TA Availability and Lecturer Planning**: Improve teaching resources through better planning and expanded lecturer support.

Here, we provide updates on these recommendations and their current relevance.

Faculty Hiring and Strategic Planning Update: Following the last review, the Department developed a five-year hiring plan, resulting in the recruitment of several new faculty members, including three women, five people of color, and an openly gay professor who actively supports the EEB LGBTQ+ community. With support from the Dean, EEB launched a Mentor Professor search which resulted in the recruitment of Professor Aguillon who brings a history of substantial commitment to DEI in her research, teaching, and professional activities to EEB. We have also strengthened interdisciplinary ties, actively collaborating on joint hires with QBC, the David Geffen School of Medicine, Biostatistics, and Human Genetics. Additionally, EEB's close partnership with the Institute of the Environment and Sustainability has facilitated the transition of leadership in a key tropical research program from Tom Smith to Felipe Zapata and Elsa Ordway, positioning the department for global research on questions of rainforest biodiversity across two continents.

Merit and Promotion Processes: The Senate recommended modifications to our highly-centralized advancement processes to reduce bias. In response, we have restructured our Academic Personnel (AP) Committee, instituting term limits and annual elections. Faculty input on promotions and advancements is now more inclusive, ensuring greater transparency and fairness in our review system, which was formalized with the adoption of revised by-laws in 2022.

Strengthening Collaborations: While our department had previously established connections with programs like QBC and the David Geffen School of Medicine, we have expanded our cross-disciplinary efforts by developing an evolutionary modeling group and pursuing joint hires that support a more integrated research landscape. Additionally, recent "Blue Sky" sessions have identified data science as a key area for growth, resulting in our participation in the DataX initiative, which will enhance our quantitative and computational capabilities.

Graduate Student Funding: More grant funding for student fellowships is a highly desired goal of our department. Unfortunately, the 2022 UC graduate student strike and subsequent contracts has significantly disrupted the existing student funding models at all levels of the university, and both the university and funding agencies are working to understand the implications of the new GSR and TA contracts (which are expected to be modified, again, in

spring 2025). EEB, like all other programs in life sciences, is grappling with ways to simply maintain the size of our graduate pool given these increased costs and our current model of providing two quarters of fellowship support plus summers. PIs with extramural funding are particularly being squeezed by the deficit created by existing budgets for student support and the new minimum rates for GSRs. The rapid and unanticipated rise in GSR stipends and graduate tuition (e.g., graduate tuition increased 19% between 2022 and 2024) has made budgeting for graduate students in grants nearly impossible as the university has been unable to match salary or tuition gaps in funded awards, leaving faculty unable to hire students from grants for which GSRs had been budgeted. Given the scope of this problem EEB will likely need to wait for institutional assessment and response before it is able to take meaningful action to address grad student support.

Graduate Curriculum and Student-Faculty Interaction: In ongoing discussions with recent graduate student cohorts it is clear to the current departmental leadership that the primary reason for the sense of a high workload on first-years is the Dissertation Written Qualifying exam (DWQE). There are no required classes in the spring and all students are supported on departmental fellowships for the first two quarters so the workload for students in the spring quarter is currently at the discretion of the student, their advisor, and the rest of their guidance committee. Anxiety around the DWQE is real and significant however (even though nearly every student in our program moves through this requirement without issue) and EEB has taken several steps to support students in developing their literature review and preliminary proposal (the elements of the DWQE). The core training courses (200a and 200b) now scaffold major elements of the DWQE through assignments that require students to develop a draft of the literature review and major research questions and that give them practice in presenting parts of the proposal to their peers for feedback. These changes reflect EEBs commitment to supporting first year students while finding the balance needed to develop the independence to develop and pursue their research proposals. In general, recent EEB student feedback has suggested that students are, in retrospect, grateful for the support with and requirement of the DWQE during the first academic year (see section G.7.D). Nevertheless, the goals and vision of graduate training within the department remain an area of active discussion. In particular, the newly-formed graduate curriculum committee (active since fall 2022) is thoroughly reviewing all aspects of training on a continuous basis to continue our movement toward a unified and progressive graduate training program.

TA Availability and Lecturer Planning: The department has made significant progress on this front in two ways. First, lecturers in the UC system are now unionized and the department hiring actions are consistent with the practices negotiated between the union and the UC system. Second, the department has hired a lecturer with security of employment so that the department has a nationally recognized expert on inclusive pedagogy in the sciences as part of its faculty. Third, the department has refined SOPs so that the teaching plan for the upcoming academic year is in the fall and winter quarters of the current year. A proactive teaching plan helps the department by identifying teaching needs while there is ample time to match our lecturers to these classes and to advertise for new instructors when necessary.

Section J. Resources

1. Operating Budget

Exclusive of academic salaries, Ecology and Evolutionary Biology receives annual operating funding of $3.3M for staff and instructional support. From this budget, the department directly covers staff and non ladder salaries, most course materials, seminars, commencement, graduate student recruitment, faculty travel costs for the Field and Marine Biology Quarters, classroom and research laboratory facilities expenses, office supplies and equipment. In addition to the annual operating budget , approximately $233,611 is received annually from contract and grants indirect cost recovery is used to support the research infrastructure of the department, including capital projects, common facilities and individual laboratories. This amount fluctuates based on C&G activity.

The department is also able to earn approximately $100,000 through its participation in Summer Sessions teaching. The department leans heavily into summer session teaching since these funds are used to subsidize budget shortfalls. EEB primarily uses these funds to further augment graduate student support and other operational expenses related to the department's teaching mission.

Faculty recruitment and retention are a financial concern of the department. The department's strategic plan is to recruit 5 new faculty over the next 2-3 years. With traditional support from the College Dean’s and Chancellor's offices becoming less robust than previous years, the department is required to support 35% to recruitment offers. This in addition to meeting outstanding recruitment and retention commitments has been a struggle. Unrestricted or donor funds are vital to EEB fulfilling these outstanding commitments.

Table J1 gives a general summary of these operational sources and their uses by the department

Table J1. Operational sources and uses.

Source	Operating Budget	Graduate Division	Summer Sessions	Indirect Cost Recovery	Annual Adjustments		TOTAL
Non Discreationary Resources	$3,389,683						$3,389,683
Discretionary Resources			$100,000	$233,611	$241,146		$333,611
							$3,723,294
Source	**Operating Budget**	**Graduate Division**		**Facilities**	**Annual Operating Expenditures**	**Recruitment & Retention**	**TOTAL**
Non Discretionary Expenses	$3,282,546						$3,282,546
Discretionary Expenses				$233,611	$208,840	$211,750	$654,201
							$3,936,747

EEB receives funding from the College of Life Sciences each year to support the hiring of full-time and temporary lecturers as well as teaching assistants (TAs). Appointments are considered 100% time or 1.0 FTE (full-time employed) when an individual teaches 2 courses per quarter for a full academic year (3 quarters). The typical TA appointment is 50% (20 hours/week), which is the most a graduate student may work while enrolled. Our department has utilized between 3.42 and 6.23 FTE to hire lecturers during the review period, with lecturers teaching an average of 45% of our courses each year (see **Table J2**). Our TA allocation has increased by an average of 83% since 2014 (**Table J3**). While our total number of majors has held a steady average over the past ten years, with increased instructional support (lecturers and TAs), we have decreased our time to degree averages for both freshman and transfer admits. This indicates that more support is needed if we are to continue serving our students well.

Instructional support is provided through the instructional operating budget, EEB requested 122 TAs (20.87 FTE), which are almost exclusively allocated to PhD or Masters students in the department. The UCLA Graduate Division supplies an average of $439,952 each year between the 14-15 FY and 23-24 FY. in unrestricted graduate student support that may be applied to fellowship stipends and fees. The Graduate Division also supplies the Holmes O. Miller Fellowships, Dissertation Year Fellowships, and Edwin J. Pauley Fellowships.

Table J2. Use of Lecturers in Undergraduate Program 2014-2024.

YEAR	FTE	Courses Taught by Lect.	Total Courses Taught*	% by Lecturers
15-16	3.42	20.5	66	31%
16-17	3.58	21.5	61	35%
17-18	6.23	27	63	43%
18-19	5.42	31	61	51%
19-20	4.5	27	76	36%
20-21	5.33	32	67	48%
21-22	4.49	44	73	60%
22-23	4.58	28.5	57	50%
23-24	4.5	28	59	47%

*Total Courses Taught includes only those courses which could be taught by a lecturer.

Table J3. TA Appointments 2014-2024.

Year	FTE	# of Appointments
15-16	19.21	113
16-17	18	108
17-18	23.03	135
18-19	27.83	169
19-20	23.29	137
20-21	24.5	147
21-22	24.47	144
22-23	21.7	130
23-24	21.34	128

2. Research Funding

The funding level in our department is very good for our discipline. Extramural funding has been consistent since the last review as noted by the numbers below:

	FY18	FY19	FY20	FY21	FY22	FY23	FY24
Direct	$ 3,145,539.00	$ 3,036,803.00	$ 7,426,442.00	$ 8,653,763.00	$ 8,523,809.00	$ 3,116,077.00	$ 3,357,386.00
Indirect	$ 633,966.00	$ 894,096.00	$ 1,299,683.00	$ 1,738,559.00	$ 857,237.00	$ 954,339.00	$ 841,765.00

3. Endowments and donor funds for scholarships or other activities

Locally, the department has three major endowments that provide research or dissertation writing fellowships in several fields: Lida Scott Brown for avian biology, Vavra for organismal plant biology and Bartholomew for field biology.

EEB Fund Review - As of June 30, 2024

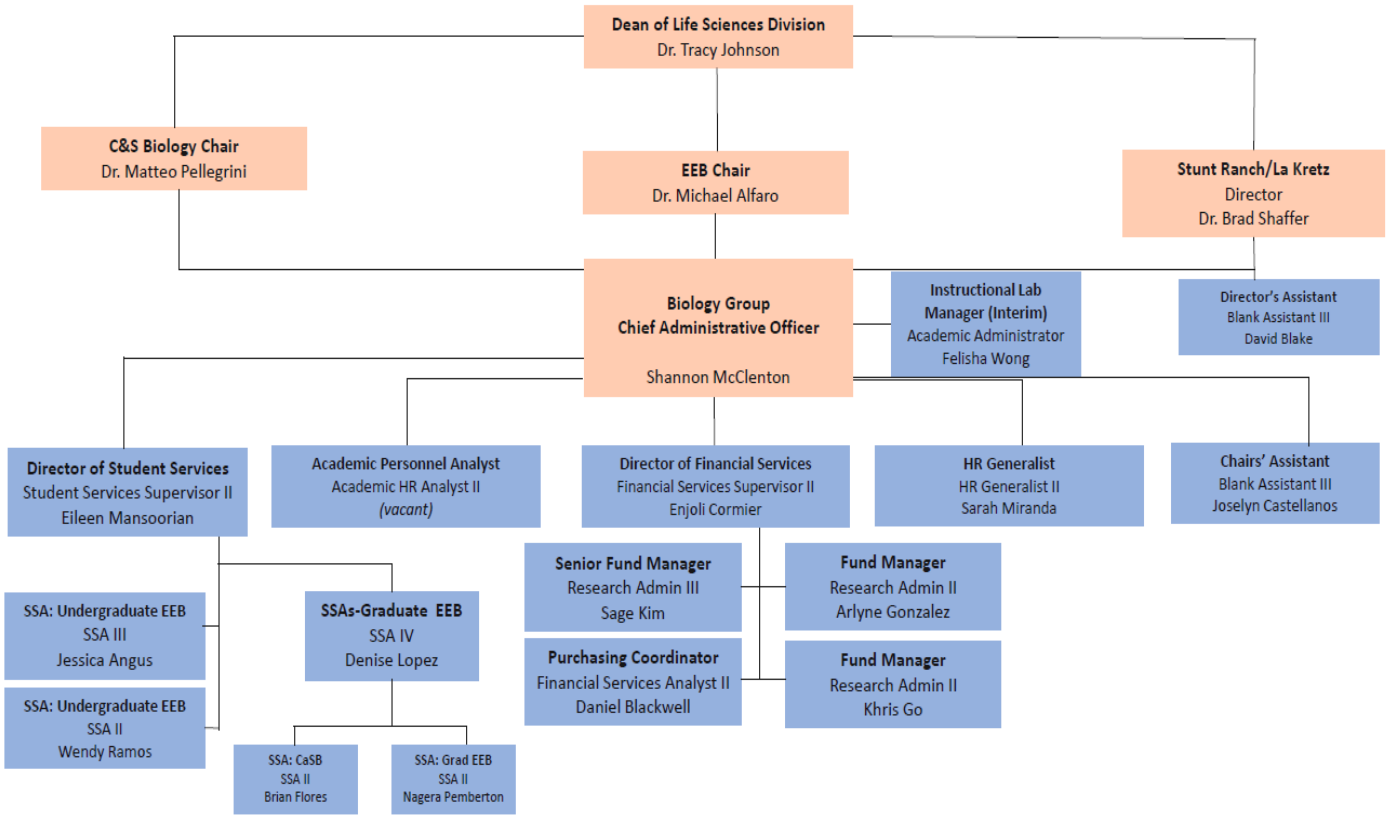
Endowment Foundation

Fund	Fund Title	Available Balance	Market Value	Projected Income
18740E / 45997	Congo Basin Institute Endowment	$ 25.65	$ 733,244.10	$ 30,929.63
82128E / 46259	Park S. Nobel Plant Biology Undergraduate Scholarship En	$ 30,852.92	$ 193,084.36	$ 8,144.67
82189E / 50645	Donald R. Dickey Chair in Vertebrate Biology	$ 91,299.21	$ 2,308,181.25	$ 97,363.47
80096E / 53043	Organismic Animal Biology Fund	$ 34,689.90	$ 82,281.86	$ 3,470.81
80543E / 53215	George A. Bartholomew Fund	$ 37,796.49	$ 266,144.66	$ 11,226.49
90113E / 56299	Biology Endowment	$ 10,642.57	$ 162,039.16	$ 6,835.12

Endowment Regental

Fund	Fund Title	Available Balance	Market Value	Projected Income
04428	BROWN LIDA SCOTT FUND	$ 148,046.93	$ 2,808,482.63	$ 104,894.53
08631	VAVRA STEPHEN A ENDOWMENT	$ 48,654.61	$ 1,671,172.17	$ 62,416.92
14532	REICH, JOSEPHINE FUND	$ 8,000.00	$ 793,172.15	$ 29,624.32

4. Roles of key staff



Section K. Goals and Plans

1. Short and Long Term Academic Goals

A. Goals for the Education Program

Lecturers. Given the new Bruin Budget model and reduced faculty size, it is critical to invest in hiring lecturers with security of employment (LPSOE), so that we have teaching *faculty* who can take on the focused teaching needs of our department. However, we need to do this strategically, hiring LPSOEs with experience in developing and implementing evidence-based pedagogy, and who—in addition to serving larger numbers of students—can use their knowledge and experience to serve as a resource to help other EEB faculty elevate their teaching, particularly with respect to inclusive teaching practices.

Curricular reform. Biological knowledge has advanced exponentially over the past 30 years, yet our approach, broadly, to the biology curriculum is based on how the discipline has been taught for 100s of years. As the amount of information that a modern biology student needs to know has grown, resulting in increases in the amount of information and courses students are required to know for graduation, little attention has been focused on reducing other areas of the curriculum to offset new areas of growth. It is essential to rethink and re-envision what we are asking our students to learn and master. In particular, our students may no longer be served by a traditional STEM curriculum where they take a year or more each of math, chemistry and physics. The Life Sciences took the bold step to replace the standard year-long Calculus for Life Sciences series with a novel, problem-based 2-quarter Mathematics for Life Sciences series that teaches students to use Python as a computational tool to solve biological problems. Assessment of this course showed significantly better outcomes in subsequent Physics classes compared to students in the traditional calculus series, highlighting that when done thoughtfully, reducing and refocusing our preparatory courses can serve the educational needs of our students better. Given that the largest course performance gaps are seen in the introductory Life Science, Chemistry, and Physics curriculum, it is time to revisit these requirements and reform our curriculum to best serve the needs of our students.

Growing the capstone majors. Field training has long been a hallmark of EEB's "specialty" majors, **Marine Biology** and **Ecology, Behavior, and Evolution (EBE)**. Students frequently cite the immersive, quarter-long field quarters as the highlight of their undergraduate experience, while instructors and TAs report high levels of personal and professional satisfaction from facilitating these transformative programs. However, as the university faces increasing pressure to accommodate a larger student body, scaling these resource-intensive field experiences has become a significant challenge, as we have insufficient faculty to offer the capstone offerings required to to grow these majors. This constraint has limited the growth of these capstone majors in recent years. At the same time, the field of ecology and evolutionary biology has diversified tremendously, with emerging career paths requiring training in genetics,

genomics, informatics, data science, and modeling, among other areas. Recognizing this evolution, EEB faculty have identified the following priorities for the future of capstone experiences within the specialty majors:

1. Preserve field training as a capstone option, maintaining the transformative quarter-long immersive experiences while also developing additional field training options that are more scalable and accessible.
2. Expand capstone offerings to include alternative experiences, such as intensive lab, genomic, computational, or data science courses, that provide students with rigorous, career-relevant skills while being less resource-intensive.
3. Explore a coursework-only option for the majors, which would offer students a path to gain diverse EEB experiences, including some field exposure, without requiring an intensive research-based capstone. In this model, the capstone experience would transition to an **honors option**, preserving quarter-long immersive research or field training opportunities for honors students who seek advanced preparation.

To achieve these goals, the department plans to diversify and restructure the capstone experiences within the specialty majors over the next 2–5 years. This approach would preserve the quarter-long immersive field training experiences as a cornerstone option while introducing additional field training courses that offer hands-on exposure to field methods and experimental design without requiring students to spend an entire quarter away. These scalable field training options could leverage the department's Natural Reserve System (NRS) sites and other local resources to provide high-quality, accessible experiences. In addition, alternative capstone options will broaden the types of career preparation available to students. These alternatives might include intensive lab-based research, genomic or informatics projects, or computational modeling courses, ensuring that the majors reflect the breadth of skills and expertise needed for contemporary careers in ecology, evolutionary biology, and marine science. By introducing a coursework-only option and reserving intensive research experiences as an honors capstone, the department can expand access to the specialty majors and accommodate the growing student population without sacrificing educational quality. This multifaceted approach will preserve the department's commitment to immersive, high-impact learning while enabling the specialty majors to grow sustainably. By offering a variety of capstone options and pathways, EEB can continue to meet the needs of an expanding and increasingly diverse student body while maintaining its reputation for excellence in undergraduate training.

B. Goals for the Research Program

The Department of Ecology and Evolutionary Biology (EEB) pursues fundamental inquiry into the processes that shape the origin, growth, maintenance, and loss of biological diversity and species interactions through space and time. We apply this knowledge to address the escalating challenges of anthropogenic change, leveraging departmental strengths in conservation biology, genomics, behavior, physiology, quantitative analysis, phylogenetics, community ecology, and field- and lab-based experimental research.

EEB's faculty are united by conceptual thinking and a collaborative culture, bridging diverse fields and scales of inquiry. These synergies underpin the department's intellectual vitality and success. However, maintaining this culture and achieving our ambitious goals require addressing an urgent structural challenge: faculty numbers have declined significantly. EEB currently operates with 25 FTE, down from a high of 31 during the review period, despite continued growth in student enrollment and expanding research demands.

The Critical Need for Faculty Growth

This contraction in faculty size has far-reaching consequences. The Biology major, the largest in the department, has approached 2,000 enrolled students during the review period, and the department anticipates further growth in this major, as well as in the specialized Marine Science and Ecology, Behavior, and Evolution majors, as additional capstone research experiences are developed (see Section K.1.A). However, EEB is under-resourced in terms of faculty to student ratios relative to other Life Science departments. For example, MCDB has 31 faculty for 944 majors, or a 30:1 student to faculty ratio. Similarly, IBP has 31 faculty for 813 majors, or a 26:1 student to faculty ratio. In comparison, ***EEB has 26 faculty for 1812 majors, or a 69:1 student to faculty ratio***. The current shortage of FTE hinders the department's ability to meet these demands and exacerbates already high student-to-faculty ratios.

Furthermore, critical areas such as Marine Science, Conservation Biology, and Evolutionary Medicine are at risk of being underserved due to impending faculty retirements. To maintain program excellence, sustain student success, and support research in these and other key areas (e.g., quantitative biology, plant biology, behavior), the department requires robust faculty growth. Thus, a core priority for EEB is to not only restore faculty numbers to the previous peak of 31 but to expand to **35 FTE** to meet future needs. This expansion will also address recent and anticipated retirements of highly accomplished senior faculty, whose leadership and expertise have been integral to EEB's success and mission.

Strategic Areas for Growth

To align with emerging scientific challenges and capitalize on departmental strengths, EEB has identified the following priority areas for strategic hiring:

1. **Marine Biology and Conservation Biology:** These disciplines are central to the department's mission and are essential for addressing the biodiversity and ecosystem crises of the Anthropocene. Faculty retirements in these areas present an acute challenge. We request **four faculty FTEs (two junior and two senior hires)** to sustain and grow these fields.
2. **Quantitative and Predictive Biology:** Modern ecology and evolution increasingly rely on advanced quantitative methods. To meet these challenges, EEB envisions hiring **three faculty** with expertise in Bayesian modeling, machine learning, and predictive biology. These hires align with UCLA's DataX initiative and will enhance EEB's capacity for cutting-edge research.
3. **Evolutionary Medicine:** This interdisciplinary field holds tremendous promise but lacks dedicated faculty to support its programs. Adding **two faculty** with expertise in evolutionary medicine will stabilize the minor and master's program, enabling consistent

course offerings and strengthening connections with the School of Medicine and Public Health.

Support for Strategic Planning and Hiring
In addition to expanding FTE, the department requests university support for strategic planning to guide its hiring initiatives. A formalized yet agile hiring plan will allow EEB to address critical areas of need while aligning with broader university goals, including the Hispanic-Serving Institution Initiative and DEI efforts. Collaborative hires and mentor professor searches focused on inclusivity will also ensure that EEB attracts diverse and innovative candidates.

Aligning Resources with Goals
Finally, achieving these ambitious goals requires adequate infrastructure and administrative support. Renovations to existing lab spaces and additional teaching and research facilities will be necessary to accommodate new hires and support departmental growth. Improved resources for grant preparation will further enhance EEB's ability to secure external funding and sustain its leadership in ecological and evolutionary research.

Conclusion
EEB's request to grow to **35 FTE** reflects a clear, actionable need to address current faculty shortages, meet rising student demand, and position the department for sustained excellence. With targeted investments in faculty hiring, strategic planning, and infrastructure, EEB can continue to fulfill its mission as a vital interdisciplinary hub and leader in addressing the environmental and biological challenges of the 21st century.

2. Needed Resources

In addition to the need for additional FTE to support the research and teaching missions (described above) EEB has identified the following resources needed to support the department.

A. Increased staffing

As mentioned in Section 7, EEB would benefit from additional funding to support the addition of FTE to our administrative team. In combination with the decentralization of the LSSA administrative core, the implementation of new operational platforms such as UCPath and Bruinbuy Plus have increased the volume of work for the current staff. The turnaround time to complete any transaction with these new systems has significantly increased leading to a culture of mistrust intrust in staffing capabilities. Faculty have felt the quality of services has left critical gaps in support for departmental academic and research missions. Frequent turnovers, loss of institutional knowledge, difficulties with recruitment and long periods of vacant positions have led to low morale, and strained relationships between faculty, students, and staff, impacting the overall functionality of the department. Subsequently, ***EEB urgently requires two additional full-time equivalents (FTE) to adequately support the research and teaching missions of the department***. The immediate needs are with the finance team (purchaser) and academic personnel. These areas require very specific skill sets and when any

vacancies in these areas leave the department in compromised situations. As noted in the staffing section, these critical areas have a single staff member, which leads to a single point of failure and no coverage. These additions are critical for meeting the increasing operational demands of the department, reducing stress on current staff, and ensuring timely support for faculty research and teaching. ***Hiring more experienced staff at higher classification levels would better reflect the department's complexity and help build an administrative team that matches the excellence and ambition of EEB's faculty***.

B. Classroom and Laboratory Modernization

Modern Teaching Spaces
EEB urgently requires modernized teaching spaces to fulfill its educational mission. Current lecture and discussion rooms lack essential AV and lecture recording technology, forcing faculty to adopt time-consuming alternatives that increase workloads and limit student access to review tools. Traditional classroom designs also hinder active learning strategies like think-pair-share, which are central to modern pedagogy. Peer institutions, such as UCI, have invested in flexible, technology-enhanced spaces that support best practices in teaching, underscoring the need for similar upgrades at UCLA.

Modern Laboratory Facilities
EEB's laboratory spaces are similarly outdated and insufficient for meeting the needs of its majors, particularly in marine science. The department's new seawater facility presents a unique opportunity to develop adjacent marine teaching labs that leverage this state-of-the-art infrastructure. Renovating nearby spaces in the Life Sciences Building (LSB) to create marine-rated laboratories would support the growing marine science major and provide flexible instructional spaces for other departmental needs. Marine-rated labs are critical for handling seawater and conducting specialized experiments, whereas traditional labs with unsealed floors are unsuitable for such purposes. This renovation aligns with past capital programs that identified potential funding mechanisms for marine lab development (see Appendix D12).

C. Seminar Space to Support Community and Collaboration

EEB's seminar series is a cornerstone of the department's intellectual community, fostering collaboration and engagement among faculty, students, and visitors. Over the last two years, the series has grown significantly, regularly exceeding the capacity of Terasaki 1100. Faculty and students who arrive late must resort to watching seminars on Zoom, limiting their ability to fully engage with speakers and peers. While the department has tried using Hershey Hall for seminars, its size, poor acoustics, and lack of seminar-specific design make it unsuitable.

To enhance the seminar experience and foster a stronger sense of community, EEB requests support for a dedicated seminar space larger than Terasaki 1100, designed specifically for speaker presentations. Alternatively, renovations to Hershey Hall to improve its acoustics and design could provide a viable solution. Such a space would not only support the seminar series but also contribute to improving morale and strengthening departmental connections.

D. Facilities to Support Marine Sciences

Dive Training Space
The current scientific diving facilities at UCLA are inadequate, creating barriers for EEB's largest user group. The lack of general assignment pool space increases costs and reduces accessibility for marine biology students, for whom diving skills are essential. To ensure continued research and education, EEB requires predictable pool allocations and a funding model that reduces financial burdens on undergraduate students.

Dedicated Marine Facilities
EEB also requires expanded marine-focused facilities to advance research and training. Developing marine culture facilities near the seawater system would support studies on aquaculture, climate proxies, and biotic responses to ocean warming and acidification. This facility could be integrated into the startup costs for new faculty or funded through recharge mechanisms to serve a broader research community. Such investments would enhance EEB's marine science program and bolster recruitment of top marine biology faculty.

E. Research Lab Space Needs

Renovating Unusable Labs
The Life Sciences Building (LSB) includes multiple lab spaces that remain unusable due to EH&S compliance issues. Renovating these spaces is essential to alleviate the department's lab space bottleneck and support its growth. Without these renovations—or the allocation of additional wet lab spaces—EEB will face constraints that could limit faculty recruitment to candidates with minimal lab needs.

Individual PI Lab Spaces
For faculty working in whole-organism biology, ecology, and physiology, shared lab spaces present significant challenges. Shifting to a model that includes individual PI labs will enable uninterrupted, specialized research and improve EEB's ability to recruit and retain researchers with unique needs.

F. Strategic Planning

The department has undergone significant transitions since the pandemic, including new faculty hires and anticipated retirements in core areas. To align these changes with long-term goals, EEB requests support for a strategic planning exercise. This process would begin with discussions on the department's research, teaching, and training missions, followed by structured retreats to develop actionable steps for achieving its objectives. University support for these discussions and retreats would be invaluable in ensuring EEB's continued success and growth.

Section L. Special Circumstances

The Department of Ecology and Evolutionary Biology has faced numerous challenges in transitioning back to in-person activities following the disruptions of the pandemic. The societal upheaval during this period, including the profound impact of social injustices brought into sharp focus by the murder of George Floyd, deeply affected the entire community and exacerbated existing challenges. Within EEB, these transitions were further complicated by significant turnover in both faculty and staff leadership, leading to uncertainty and disrupting continuity.

Moreover, there has been instability in departmental leadership, requiring an external chair search and 3 interim chairs since 2017. The appointment of Michael Alfaro as Department Chair in 2022 marked an important step toward stability, as it brought EEB's senior leadership team back to being entirely internal for the first time in several years. This internal leadership has been key to reestablishing trust, fostering community, and creating continuity within the department. Over the past two years, the department has focused on improving its climate and inclusivity, as evidenced by the revitalized EEB seminar series, which has strengthened the sense of academic community; the recruitment of transformative new faculty, including Stepfanie Aguillon, Chuliang Song, and Monica Medina; the establishment of the CARE committee; and the development of a mentoring plan for graduate students. These initiatives reflect EEB's dedication to fostering an anti-racist, inclusive, and supportive environment for all members.

While these efforts have achieved meaningful progress, EEB requires continued strategic support from the university to sustain and expand upon them. Investments in faculty hiring, infrastructure improvements, and additional departmental resources are essential to fully realizing the department's potential. By addressing these needs, the university can ensure that EEB remains a leader in tackling the pressing ecological and evolutionary challenges of the future.

Section M. Appendices

Description	Section	Link
B1. Faculty Research Interests	B (General)	Appendix B1 EEB Faculty Research Interests
B2. Faculty CVs	B(General)	Appendix B2. Faculty CVs
D1. Revised guidelines for Biology and Marine Biology majors	D (Undergrad Programs)	Appendix D1 Revisions to majors since 2014.docx
D2. Requirements for Biology Major & summary of changes	D (Undergrad Programs)	Appendix D2 Requirements for Biology BS .docx
D3. Requirements for Ecology, Behavior, and Evolution Major & summary of changes	D (Undergrad Programs)	Appendix D3 EBE requirements.docx
D4. Requirements for Marine Biology Major & summary of changes	D (Undergrad Programs)	Appendix D4 Requirements for Marine Biology Major.docx
D5. Description of courses developed during review period	D (Undergrad Programs)	Appendix D5 New Course Data_Table.xlsx
D6. Publications co-authored by undergrads	D (Undergrad Programs)	Appendix D6 Publications co-authored by Undergrads.docx
D7. Requirements Conservation Biology minor	D (Undergrad Programs)	Appendix D7 Requirements of Conservation Biology minor.docx
D8. Requirements Evolutionary Medicine minor	D (Undergrad Programs)	Appendix D8 Requirements Evolutionary Medicine minor.docx
D9. Course Enrollment	D (Undergrad	Appendix D9 Course Enrollment with Detail.xlsx

Data	Programs)	
D10. EEB survey results of undergrads since 2014	D (Undergrad Programs)	Appendix D10 Undergrad Surveys Summary.pdf
D11. Assessment Plan for Learning Outcomes for majors, 2014	D (Undergrad Programs)	Appendix D11 Undergraduate Learning Outcomes and Assessment Plan 2014.docx
D12. Marine Sciences Lab Plan	D (Undergrad Programs)	Appendix D12 Marine facility planning scheme.docx
D13. Biology Major Requirements from UCLA General Catalog	D (Undergrad Programs)	Appendix D13 Biology BS General Catalog Sheet.pdf
D14. Ecology, Behavior, and Evolution Requirements from UCLA General Catalog	D (Undergrad Programs)	Appendix D14 Ecology, Behavior, and Evolution BS General Catalog Sheet.pdf
D15. Marine Biology Major Requirements from UCLA General Catalog	D (Undergrad Programs)	Appendix D15 Marine Biology BS General Catalog Sheet.pdf
E1. DGE Program Report on EEB Ph.D.	E (Graduate Programs)	Appendix E1 DGE Report Biology PhD.pdf
E2. EEB Grad Admissions SOP	E (Graduate Programs)	Appendix E2. EEB Grad Program Admissions SOP 2025.pdf
E3. EEB First-Year Course Plan Document	E (Graduate Programs)	Appendix E3 23-24 PhD First Year Course Plan.pdf
E4. EEB Ph.D. Annual Committee Meeting Summary Form	E (Graduate Programs)	Appendix E4 Annual Committee Meeting Summary.pdf
E5. EEB Grad Publications	E (Graduate Programs)	Appendix E5 Grad Publications.docx
E6. EEB Grad Research Awards and	E (Graduate Programs)	Appendix E6 Research Awards and Extramural Fellowships.docx

Extramural Fellowships		
E7. DGE Data Packet on EEB Grad Programs	E (Graduate Programs)	Appendix E7 Graduate Division Data Packet.pdf
E8. EEB Grad Program Requirements	E (Graduate Programs)	Appendix E8 EEB Grad Course Requirements.pdf
E9. EEB Grad Courses	E (Graduate Programs)	Appendix E9 EEB Grad Courses.pdf
F1. Postdoctoral Scholars during review period	F (Postdoctoral Scholars)	Appendix F1 Postdoctoral Researchers during review period
F2. EEB Postdoc Publications	F (Postdoctoral Scholars)	Appendix F2 Postdoc Publications.docx
G1. Committee for Anti-Racism and Equity (CARE) SOP	G (Diversity)	Appendix G1
G2. CARE 2022-2023 Annual Report	G (diversity)	Appendix G2
G3. CARE 2023-2024 Annual Report	G (diversity)	Appendix G3
G4. CARE 2023 Recommendation Tracking/ Chair Response Form	G (diversity)	Appendix G4
G5. EEB 2022 Retreat Summary	G (diversity)	Appendix G5
G6. EEB 2023 Retreat Summary	G (Diversity)	Appendix G6
G7. EEB Grad Climate Survey Report (2021 survey by grads)	G (Diversity)	Appendix G7

G8. Winter 2024 Grad Survey instrument	G (Diversity)	Appendix G8
I1. EEB Task Force Recommendations	I (Comparison to Prior Reviews)	Appendix I1. Task Force Recommendations
I2. EEB Anti-Racism Implementation Committee Recommendations	I (Comparison to Prior Reviews)	Appendix I2. Implementation Committee Recommendations